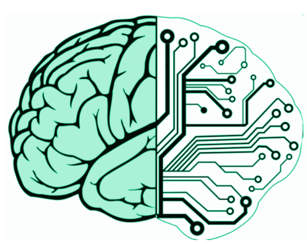
Quotation:

"Though this be madness, yet there is method in't."
Shakespeare: Hamlet: Act 2 Scene 2

The Hodgkin-Huxley neuron model V2.0

The true physics and physiology behind neuronal operation



János Végh

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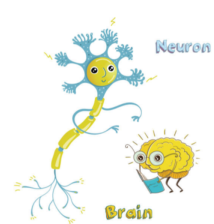

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Contents

Foreword

The importance of Hodgkin and Huxley's classic (V1.0) "model", which fundamentally changed our ideas about the physiology of neurons and which has served as the solid fundamentum for describing the functioning of neurons for several decades, can hardly be overestimated; it is truly the crown jewel of neuroscience. However, its authors, quite realistically, assessed that the physical picture behind their observations they mathematically formulated could be partially incomplete and incorrect. Already in their original communication, they drew attention to the fact that their equations did not fully describe their observations. Later they added that there exists observed (thermodynamical) evidence that does not fit into their picture at all. When making the first (and really enormous) step on terra incognita, lacking most of the required structural and operational knowledge, they were forced to formulate ad hoc assumptions and they also admitted that the formalism of the equations could describe more than one physical process. Some of the assumptions proved to be a brilliant vision, but in quite a few cases it turned out that they did not always put the right physical background behind their observations which were generally correctly described mathematically, and that their model designed for mechanical computers had obvious limitations. Already at the time of the introduction of the model, doubts arose as to whether the purely electrical model could fully describe the observations known up to that time, and later, as the accuracy, directionality and detailedness of the dedicated observations increased, it became necessary to introduce more and more ad hoc assumptions in order to maintain the consistency of the model assuming purely electrical operation.

In the past decades, numerous (V1.0x) models have been created, modifying and supplementing this purely electrical, disciplinary, original model with more or less success (which ultimately became a model due to the activities of Hodgkin and Huxley (HH)'s followers). There have also been (V1.y) models that try to model the neuron within the framework of an entirely different discipline, mainly interpreting its thermodynamic, mechanical, optical, etc. phenomena. The common feature of all these models is that they cannot provide a complete model (describing all observations without contradiction). There are also models that do not even strive to describe a physical operating mechanism; they are satisfied with providing a mathematical formalism that shows a behavior that is consistent with experience in some context; without even trying to answer the questions 'why' and 'by what mechanism'.

Due to the lack of the needed expertize in physics, biophysics introduced the idea of the ”whatnot” [1] protein mechanisms for the unexplainable forces that totally misguided physiology and neuroscience. The speculative hypotheses hyde that science can describe living matter, but needs a different approach. Claims such as ”pumps [by using protein mechanisms] ... transport ions *against their electrical and chemical gradients*” [2], page 101, or ”the flux of ions through ion channels is passive, requiring *no expenditure of metabolic energy* by the channels.” [2], page 107, are as scientific as in the pre-Newtonian age, there was a notion related to Aristotelian philosophy. That concept gave credence to the “theory” that the celestial spheres and, later on, the planets were pushed or moved by celestial movers, celestial intelligences, or angels against physical forces, without investing energy. Inventing the universal gravitational law made the complex interplay of those spheres obsolete, although, for the first look, it seemed to be unrelated.

Quotation:

” *We stand on the verge of a great journey into the unknown—the interior terrain of thinking, feeling, perceiving, learning, deciding, and acting to achieve our goals—that is the special province of the human brain* ” — from the preamble to “BRAIN 2025: A Scientific Vision” [3] @2015

Describing life science is still in the pre-Newtonian age: it has a lot of ”spheres” described by complicated theories that contradict each other, and does not have its laws of motion, in the sense as Newton’s Laws in physics. What is worse, it did not invent and test the laws describing living matter. Instead, it borrowed the disciplinary laws of physics, without recognizing that its different discipline has different conditions and requires different approximations; that is, similar, but differently used laws. Without questioning the merits of the V1.0 model, the experimental experience accumulated in recent decades, the dizzying development of experimental technology and instrumental measurements, and the ever-increasing demand for the results of neuroscience have made it necessary to develop the (V2.0) model. The model fuzes and unifies the concepts of the former V1.x models, purifies the wrong conceps and synthesizes the rigth ones. The change in the major version number symbolizes that it is a leap-like development (a drastic change in the physics concepts) instead of an incremental one (a slight improvement by introducing a new ad-hoc hypothesis).

The V2.0 model is necessarily cross-disciplinary; no single physical discipline alone allows for a complete and consistent interpretation of the phenomena of the living organism. The audience educated in the ’old school’ will find the concepts strange. In physics is is not strange at all that the concepts that are valid for the observations made in everyfay life became invalid in the world of too small or too large masses, too small or too large distances, and so on. Up to now, invalid scientific concepts have been applied to the living matter, and because of the wrong concepts, many wrong conclusions have been made. To fix those mistakes, on must strictly define the concepts, borrowed from science, for the case of living matter, because for today the wrong concepts have grown into major obstacles toward understanding the extremely complex operation of the brain, from its elementary physical processes up to its information processing and higher-level functionality.

Chapter 1

Introduction

Quotation:

"The Human Brain Project should lay the technical foundation for a **new model** of ICT-based brain research, driving *integration between data and knowledge from different disciplines*, and catalysing a community effort to achieve a **new understanding of the brain**... and *new brain-like computing technologies*."
the Human Brain Project, summarised its goal @2012

The ***dynamic*** operation of individual neurons, their connections, higher-level organizations, connections, the brain with its information processing capability, and finally, the mind with its conscience and behavior, are still among the big mysteries of science: *at which point the non-living matter becomes a living one*, *at which point the living matter becomes intelligent* and conscious; whether and how science can handle all this stuff. We really need a new understanding that we provide here. The hype in the newly launched projects is excessive and seems to lose the hope to build brain research on a firm science base. For example, the newly (at the end of 2025) launched "Brain/MINDS 2.0" program in Japan was launched using only mathematical models, *without targeting the understanding of the underlying physical processes*. In neuron models, 'dynamics' refers to how the state variables of the neuron (primarily membrane potential and ion channel properties) change and evolve over time in response to internal processes and external inputs. However, unlike in our approach, the usual processes simply use a time parameter in empirical or mathematical functions, see for an example [4]. We use a real dynamics, based on physics, in the sense as Newton introduced his laws of motion to describe response of neurons.

Quotation:

”Biological laws are much more difficult to find than physical ones” [5] @1995

1.1 Disciplinary layers

The material has multiple layers. In addition to the sections intended to be understandable by everyone (i.e. primarily readers with a suitable physiological and neuroscience background), there are also panels (typically in another chapters) for those with the relevant background, containing the necessary **mathematical 1.1.1** and **physical 1.1.1** knowledge, which require at least the appropriate level of BSc knowledge to understand, but sometimes knowledge that goes beyond this, and sometimes knowledge and explanations that make you delve deeper into the unusualness of the ideas. This method naturally leads to some redundancy, but enables the audience to read only the text his/her background qualifies for. We strive to make the material understandable without those supplementary material, and with them, the reader with the appropriate qualifications will be able to gain knowledge that encourages further thoughts.

Physics panel 1.1.1:
An example physics panel

a Physics panels contain the strict physical background.

Mathematics panel 1.1.1:
An example mathematics panel

Mathematics panels typically contain equations and their explanations.

1.2 Abstract discussion

Nature uses an infinite variety of implementing neurons. Our goal is not to discuss all infinitely complex molecular, biochemical and physiological details of neuronal operation In the Central Nervous System (CNS) they can cooperate

Numerics panel 1.1.1:
An example numerics panel

The complexity of the phenomena requires many approximations, but by using the appropriate approximations and abstractions, numerical results (sometimes only reasonable estimates) are provided in the corresponding numeric panels.

with each other, 'despite the extraordinary diversity and complexity of neuronal morphology and synaptic connectivity, *the nervous systems adopt **a number of basic principles*** for all neurons and synapses', [6] independently from the infinitely complex molecular, biochemical and physiological details. We *will base our holistic discussion on those general basic principles and create an 'abstract physical neuron'; a hypothetical element, which functions essentially like a neuron*, skipping the 'implementation details' nature uses. We essentially follow von Neumann's method in describing the foundations of computing science [7]: "we will base our considerations on a hypothetical element, which functions essentially like a neuron". We need different abstractions and approximations (furthermore, 'non-ordinary' laws of science) for describing biological processes. However, an abstraction is usable in practice only when paired with a generalization: the more abstract the assumption, the more general and widely applicable the concept or conclusion. *By abstractions, we can reduce the unbelievably detailed world into manageable pieces, and we can learn anything general.* We must show which approximations are oversimplifications and which phenomena are misunderstood; measured, or interpreted in a wrong approach. Abstraction is, in fact, everywhere, including inside each of us. It is a core element of science and cognition.

We follow von Neumann's method when he described principles of technical computing [7]. "The ideal procedure would be, to take up the specific parts in some definite order, to treat each one of them exhaustively, and go on to the next one only after the predecessor is completely disposed of. However, this seems hardly feasible. The desirable features of the various parts, and the decisions based on them, emerge only after a somewhat zigzagging discussion. It is, therefore, necessary to take up one part first, pass after an incomplete discussion to a second part, return after an equally incomplete discussion of the latter with the combined results to the first part, extend the discussion of the first part without yet concluding it, then possibly go on to a third part, etc. Furthermore, these discussions of specific parts will be mixed with discussions of general principles, of the elements to be used, etc." For example, basic concepts such as the membrane potential and its regulation in resting and transiens states, belong to chapter Physics, and their detailed scientific, physics-based, discussion is given there, but their corresponding aspects must be discussed in chapters 'Abstract neuron' and less abstract 'Physiology' as well. The contents are inherently intertwinned, and the form must follow the contents. We make this zigzag reading more accessible by using hyperlinks and cross-references throughout the document.

1.3 Electrical neuron model in a nutshell

Despite that we attach mechanical and other phenomena to the electrical ones, our motto is similar to the frequently cited saying 'Cherchez la femme': *Look for the charge*, meaning "*no matter what the problem, some charge is often the root cause*" (however, we consider also effects of the thermodynamics on charge, given that the charges are ions instead of electrons. Furthermore, we shortly consider the mechanical, optical, etc. consequences.). The expression comes from the novel *The Mohicans of Paris* by Alexandre Dumas and is frequently used in detective novels in the sense that there are always complications and unusual situations, but "no matter what the problem, a woman is often the root cause". Although, in most cases, the cause is seemingly different.

Similarly, the charge is always the primary quantity when speaking about neural operations. We follow this line and explain how processes, that traditionally belong to different classical science disciplines, redistribute charges and create electrical currents and voltages, then discuss how the cell handles the electrical charge. We note, however, that the different charge carrier (ions instead of electrons) needs care: the "slow current" shows unusual behavior and the needs using known laws in more or less different form. Also, we make hints that in most cases, the classical disciplines, alone, are not qualified to describe phenomena of living matter. The details are described in Chapter 5, where the fundamental differences caused by the charge carrier are discussed.

We show that, alhough the thermodynamic effects are inseparable from the electrical ones, an appropriate combination of the disciplines satisfactorily describe the observations. We consider that neurotransmitters, receptors and specialized membrane proteins *only implement a kind of (time and energy-consuming) chemical/enzymatic decoupling of the signal transmission mechanism.* The idea resembles opto-coupling in electronics: makes *the signal transmission independent from the local potential value.* Suppose neurons use galvanic coupling when the resting potential of one of the neurons equals the extracellular space. In that case, all connected neurons' resting potentials are equal to that of the extracellular space. Without this decoupling, the death of one neuron would immediately lead to the death of the entire neural network. Furthermore, the neurons could not make independent signal processing.

Chapter 2

Modeling for neurons

Quotation:

"What I cannot create, I do not understand" R. P. Feynman

Quotation:

"Biological laws are much more difficult to find than physical ones" [5] @1995

2.1 Modeling methods

"There is a widely accepted distinction between merely modeling a mechanism's behavior and explaining it. ... Some models are data summaries. Some models sketch explanations but leave crucial details unspecified or hidden behind filler terms. Some models are used to conjecture a how-possibly explanation without regard to whether it is a how-actually explanation."[8].

2.1.1 Disciplinary modeling

As Feynman R. P [9] wrote, "the separation of fields [of science] ... is merely a human convenience, and an unnatural thing. Nature is not interested in our separations, and *many of the interesting phenomena bridge the gaps between fields*". "When we go to investigate *we should not pre-decide what it is we are* *trying to do* except to find out more about it. ... The first principle is that you must not fool yourself, and you are the easiest person to fool.” Physiology, despite HH’s and Feynman’s warning, pre-decided that only the science discipline ’electricity’ is authorized to describe neurons’ operation. However, electricity has no place for the mass and speed of its charge carriers. After seven decades, it should be realized, admitted, and fixed.

2.2 The V2.0 modell in a nutshell

Quotation:

”Everything should be made as simple as possible, but not simpler.” Attributed to Albert Einstein.

The principle, suggested by A. Einstein, means that while one should strip away unnecessary clutter, extra features, or overly complex jargon to achieve clarity, one must be careful not to remove so much that the core message or solution becomes meaningless or inaccurate. For neurons (living matter), on the one side, it actually means that one should not reject quantities from the set of observed quantities just because they are not present in a classical discipline of the science, which once opon the time someone hypothesized (pre-decided?) to describe neurons. Furthermore, one must include the new discoveries (effects and structures) made in the past eight decades and the intentionally omitted observations which do not fit into the pre-authorized disciplinary theory. On the other side, one must remove the foreign and fake ad hoc mechanisms and hypotheses (which contradict fundamental principles of science and elementary logics). All those issues were required to obscure that the current understanding does not match the experimental facts.

In our non-disciplinary science-based model (the ’abstract physical neuron’, as we call it), a neuron (highly simplified) is an elastic sphere with two lipid layers (membrane), a similarly elastic insulating output tube (axon), and an incompressible, partially ionized electrolyte fluid with different compositions and concentrations inside and outside. They contain ions, simple chemical molecules, and complex biological formations. These components may have gradients and move at continuously changing speeds that can vary by several orders of magnitude during operation. The lipid layer is partially permeable (controlled or uncontrolled) for the components above (channels, mainly ion channels).

Between the membrane and the axon, there is a large amount of uncontrolled ion channels (Axon Initial Segment (AIS)). The elastic wall of the membrane is covered with ion layers, and therefore, a potential difference is created there (the resting potential). The membrane represents *a confined space in which ions behave differently from in an infinite space*; furthermore, it significantly alters the properties of the few-nanometer-thick layer near the membrane. In this layer, the concentration and potential change significantly with the distance from the membrane due to interactions between the membrane's ion layers and the electrolyte. Ions sent by other neurons can enter the internal volume through controlled inputs (synapse).

Under a small external influence, a neuron remains in its resting state and regulates itself through non-controlled ion channels in its membrane. If the external influence exceeds a threshold, the controlled channels suddenly release a large amount of ions into the internal space, putting the neuron into a transient state. The membrane potential changes suddenly (within a few dozen nanoseconds), and, due to the mutual repulsion of ions, the pressure acting on the membrane wall and the ion concentration in the layer near the membrane also increase significantly and suddenly. At this point, the disciplines of electricity and thermodynamics can be combined: the pressure and the electric field are proportional to the number of ions in the volume. The forces exerted on ions, calculated in a disciplinary way, are falsified in the sense that the other discipline also contributes inseparably; in a strict sense, neither of the two disciplines alone can describe ions' motion. The detailed analysis shows that the elastic energy during the action potential is about two orders of magnitude higher than the electrical energy, but the amount of ions is about three oders of magnitude lower than that of the non-dissociated molecules. It that situation, the two effects are of the same order of magnitude, so that they can modulate each other. The electric charge on the membrane's wall would produce a simple discharge current. However, its charge carriers are delivered by the damped oscillation (a soliton) generated by the membrane. The amplitude of the soliton changes the number of the charge carriers delivered, so physiology measures an oscillating discharge current (that generates a potential wave known as Action Potential (AP)). No significant mass transfer takes place.

Since *the same physical action generates the voltage gradient and the concentration gradient* (and, consequently, the pressure), they are inseparable and proportional to each other. Generating a voltage change (clamping, direct current, magnetic pulse) or a pressure change (ultrasound, mechanical, shock waves) on the membrane mutually cause the other to change. Inputting ions (synaptic input) generates both changes. The control circuit tries to restore the resting state, during which the mentioned changes decrease proportionally. The transient state of the neuron can, in principle, be described by any of the mentioned quantities, but, as a practical matter, *the effects of changes in the other entities must also be taken into account*: *no single discipline, alone, can describe the changes*. The elastic membrane implements an almost critically damped vibration, but the "vibrating mass" and the "spring force" are constantly changing. In an electric resonant circuit, the capacitor's voltage and capacitance change due to the finite amount of charge. *The pressure wave caused by a force shock and the voltage wave caused by a voltage shock describe the same effect.* In both cases, the other effect must be taken into account to some extent: an infinitesimal movement of the ion also changes both the pressure and the voltage. Changes in pressure and charge alter the membrane's thickness (and therefore its capacity), and both voltage and pressure can cause a structural change (phase change, "melting") in the lipid chains that make up the membrane.

Here, the difference comes to light: the carrier of the electromagnetic (EM) interaction has no mass, while the thermodynamic one does. In the case of the pressure wave, the speed is naturally finite (mass must be moved). For the electric wave, there are laws regarding only instantaneous interaction; so, they must be adapted to *slow ion currents* based on physical assumptions. It is simpler to describe the generation of the action potential as an electrical oscillation, and its propagation as a pressure wave (although this only represents the wave front well: the AP lasts as long as there is a non-equilibrium charge in the neuron: the repulsion between the excess ions pushes the ions out of the closed space, so the intensity of the wave continuously decreases). Hyperpolarization is described in the electrical view as a capacitive current in the sequential oscillator circuit, in the thermodynamic view as a suction effect occurring in the opposite phase of the membrane. In both cases, however, it must be taken into account that charge and mass are inseparable in ions, and that their cross-disciplinary effects significantly reduce the applicability of the disciplinary laws. Furthermore, as our force analysis suggests, the ions' motion results an interference between the longitudinal thermodynamic oscillation of elastic origin and a discharge process of electrical origin.

In simple words, a neuron combines different disciplines; it works like a two-tact biological internal combustion engine, and can be described theoretically as a special Carnot-type thermodynamic engine, see section **??**. The synaptic input currents operate the ignition. The influx of Na^+ ions acts like an explosion, producing a large pressure impulse due to electrical repulsion between ions; see section **??**. The neuron's volume remains practically unchanged (representing an iso-volume process), and the pressure wave vibrates the particles (including dissociated ions) as a longitudinal wave, and the resulting offset potential moves the ions out of the neuron through the only available output channel: the axon. The pressure wave is similar to a sound wave, with negligible material transport. The force analysis shows that the pressure wave causes a longitudinal vibration, while the changed potential adds a potential-dependent longitudinal force component for moving the ions.

In the first stroke, the pressure impulse invests energy into compressing the elastic membrane and starting a compression wave (a soliton). In the second stroke, the elastic membrane pumps out some electrolyte (the driving force is a combined thermoelectrical/mechanical force rather than some magic battery) through the ion channels in the AIS, which "resists" it, thereby generating a potential wave known as AP; well described by electricity. The force analysis shows that the pressure wave causes a longitudinal vibration, while the changed potential adds a potential-dependent longitudinal force component for moving the ions. The compression produces heat, as evidenced by a rise in temperature. The expansion consumes heat, as evidenced by a decrease in temperature. These are well-understood processes in thermodynamics. However, they are still unexplained in biophysics, although they were observed seven decades ago [10]. The elastic membrane functions as a damped oscillator, which, in its negative amplitude phase, reverses the direction of electrolyte flow (and, consequently, the direction of the current carried by the ions). In the discipline of electricity (when using the serial RC model), it can be interpreted as a capacitive current, which in physiology is observed as hyperpolarization that inverts the polarity of the output voltage. The ions represent both mass and charge simultaneously, and, correspondingly, one can observe, in addition to the electrical potential wave, mechanical, optical, and other features changing. In different disciplines, pressure and voltage represent the same phenomenon, using concepts from various disciplines, although the speeds of those waves differ significantly.

Today, two major branches of disciplinary theories compete for being applied to describe the neuronal operation [11, 12, 13, 14, 15, 16]. Those cited references also compare those theories. The 2-decade-old Heimburg-Jackson thermodynamic theory is trying to break in alongside or displace the 7-decade-old all-electric Hodgkin-Huxley theory. The two theories apply one of the scientific disciplines developed for inanimate nature, unchanged, to living nature. There are unexplained observations for each of the two theories, and neither can describe the biological neuron's functioning without contradictions. In a disciplinary view, there is no chance of the two theories fusing. Although they can describe a wider range of phenomena together, the conditions under which they are applicable exclude their joint use. Moreover, to conceal contradictions and fill gaps, biophysics assumes (typically unspecified protein) mechanisms that contradict the fundamental principles of physics, and some of them even the disciplinary laws.

The electrical view essentially stems from the (Nobel Prize-winning) work of Hodgkin and Huxley [17]. They could not make a perfect job, mainly due to the lack of discoveries made several decades later, including ours, about handling the finite speed of the biological ionic currents (and, due to that, the dynamic features), which drastically change their conclusions. In contrast with their *empirical description*, which delivered mathematically formulated measured observations, our discussion, although it follows essentially the same principles, reinterprets and pinpoints the used fundamental terms, sets up a physical model, and *explains* the physically underpinned processes. "However, the theory could not explain the physical phenomena such as reversible heat changes, density changes, and geometrical changes observed in the experiments" [14]. The thermodynamic view roots in the (possibly Nobel-prize-winning) idea of Heimburg and Jackson [18]. They proposed that the action potential is essentially a sound wave (a soliton). "However, there are several other questions that this has to answer like ion flow involvement in nerve signal propagation as stated by the HH model and also the faster propagation in myelinated nerves than in unmyelinated" [14]. If we consider that the ion flow means charge propagation in a membrane tube where the specific capacity depends on the thickness (of the myelin sheath) of the wall [19], the faster propagation is not mysterious anymore. Those apparent contradictions are simply consequences of ions' non-disciplinary features. By using the correct (non-disciplinary) discussion of the phenomena, we can explain all those observations in more detail in section **??**.

The electrical view essentially stems from the (Nobel Prize-winning) work of Hodgkin and Huxley [17]. They could not make a perfect job, mainly due to the lack of discoveries made several decades later, including ours, about handling the finite speed of the biological ionic currents (and, due to that, the dynamic features), which drastically change their conclusions. In contrast with their *empirical description*, which delivered *mathematically formulated measured observations*, our discussion, although it follows essentially the same principles, reinterprets and pinpoints the used fundamental terms, sets up a physical model, and *explains* the physically underpinned processes. "However, the theory could not explain the physical phenomena such as reversible heat changes, density changes, and geometrical changes observed in the experiments" [14]. Given that our (non-disciplinary) model natively connects charge and mass of ions [20], it can explain the missing phenomena. The thermodynamic view roots in the (possibly Nobel-prize-winning) idea of Heimburg and Jackson [18]. They proposed that the action potential is essentially a pressure wave (a soliton). "However, there are several other questions that this has to answer like ion flow involvement in nerve signal propagation as stated by the HH model and also the faster propagation in myelinated nerves than in unmyelinated" [14]. If we consider that the presence of ions in the neuron's closed volume means an unbalanced force component exerting only on ions (but not on the vibrating neutral molecules) and the ion flow means charge propagation in a membrane tube where the tube's specific capacity depends on the thickness of the axonal membrane's myelin sheath (aka the distance across the membrane surfaces), the faster propagation is not mysterious anymore.

The excellent textbook's statement that "the resting membrane potential results from the separation of charge across the cell membrane" [2] is only half the truth; we tell the second half in section **??**. We refute their statement that "resting ion channels establish and maintain the resting potential" [2], page 126. The ion channels are passive players. The interplay of the lipid condenser and the electrolytes on both sides of the permeable membrane establishes the resting potential, and ionic gradients maintain it. The textbook separates the neuron membrane's states into resting and transient states. It introduces resting and transient ion channels, but does not introduce that the different physical processes in those states need different models. As we show in section **??**, *two different physical mechanisms operate in the resting and the transient states*, so two different models are needed for describing the same components.

A primary reason neurophysiology has been misguided is that it has in mind a static picture of the dynamic life of neurons. It started from the static inanimate anatomic microscopic pictures, continued with the static clamping investigations (where the negative feedback of an external electric circuit eliminates the natural gradients) and underpinning the theory by analyzing frozen (inanimate) samples by electron microscopes. Those investigations revealed a tremendous amount of crucial details, but they obscured the idea that those samples are just snapshots of a permanently changing system. Neuroscience projected the mechanisms of the static resting state to the dynamic transient state and attempted to understand the transient state as perturbations [21] to the (entirely different) resting state. Of course, that attempt required more and more ad hoc non-scientific assumptions. We agree that "Understanding and predicting molecular responses in single cells upon chemical, genetic or mechanical perturbations is a core question in biology." [22] However, it is not possible without understanding that the transient state is a state on its own right, with different physical mechanisms, rather than a perturbation of the entirely different resting state.

Due to the lack of the needed expertize in physics, biophysics introduced the idea of the "whatnot" [1] protein mechanisms. Claims such as "pumps [by using protein mechanisms] ... transport ions *against their electrical and chemical gradients*" [2], page 101, or "the flux of ions through ion channels is passive, requiring *no expenditure of metabolic energy* by the channels." [2], page 107, are as scientific as in the pre-Newtonian age, there was a notion related to Aristotelian philosophy. This concept gave credence to the "theory" that the celestial spheres and, later on, the planets were pushed or moved by celestial movers, celestial intelligences, or angels against physical forces. Life science does not have its laws of motion, in the sense as Newton's Laws in physics, at least at cellular level.

The book [2] asks the central questions, "How do ionic [i.e., electrical and chemical] gradients contribute to the resting membrane potential? What prevents the ionic gradients from dissipating by diffusion of ions across the membrane through the resting channels?" However, it leaves them essentially open by giving only a qualitative answer, discussing only membrane permeability, without explaining how the resting potential is created. We demonstrate in section **??** that classical methods of electricity enable us to calculate the membrane's potential resulting from charge separation and polarization, why and how its resting value is created, furthermore, how it is regulated. We make some physically plausible assumptions to provide numerical figures.

Our results underpin that "the crucial system in biology isn't a molecule or a molecular class whatsoever, but the interface created by biomolecules in water" [23]. We add that mainly physical processes at various speeds (instead of molecular classes such as proteins) and ionic gradients (instead of protein mechanisms) control the processes. More precisely, inseparable thermodynamic and electrical processes set up and maintain the potential established by the electrical and thermodynamic backbone defined by the lipid/protein structure of the cell and the ionic solutions it contains. It is one of the frequent cases when one effect implements a functionality (the backbone closely defines the frames of the operation), and the other (in this case, combined thermodynamics and electricity) corrects it when it implements a complex operational (dynamical) functionality: "stimulated phase transitions enable the phase-dependent processes to replace each other ... one process to build and the other to correct" [24].

2.2.1 Essential differences to the V1.0 modell

Chapter 3

Resting Potential

Chapter 4

Transient potential

Chapter 5

Physics for neurons

5.1 Introduction

The roles of time, space and matter are subjects of endless debates in science. Considering finite interaction speeds is against using a "nice and classical physics" with its nice mathematical formulas, but omitting the different speeds misled and may mislead research in several fields. Biology produces situations where the complexity of phenomena and the needed carefulness meets the ones needed in cosmology. The difference is that, in biology, the consequences of phenomena are immediate and they can be studied experimentally.

To describe the related phenomena, we must scrutinize, case by case, which interactions are significant and which interaction(s) can be omitted; instead of setting up ad-hoc models for living matter that contradict each other and fundamental scientific principles when used outside their narrow range of validity. To provide their correct physics-based description, we must understand the corresponding behavior of living material, including that it works with slow ion currents, electrically active, non-isotropic, structured materials, and consequently, its temporal behavior (the speeds of interactions) matters. We must consider macroscopic and microscopic phenomena (continuous and discrete features) at the same time, different science fields, and their interplay (or mixing). "Living complex systems in particular create high-ordered functionalities by pairing up low-ordered complementary processes, e.g., one process to build and the other to correct". [24] We must check the validity of our abstractions.

First of all, we introduce (and, in this aspect, correct the current common understanding) that neurons' "output product" [25] *is created and transmitted by thermoelectric processes instead of net electric phenomena or net thermodinamic.* Feynman was right, also in this point: "*nature is not interested in our separations, and many of the interesting phenomena bridge the gaps between fields.*" [9] We derive the correct mathematical and physical handling processes, not considering the disciplinary boundaries, pointing to where the disciplinary omissions and approximations are not valid, deriving the required correct handling of the physical processes together with with the mathematics required to handle them. We use the theory developed in section **??**, where (among others) the time derivatives of the processes are introduced, in this way enabling the quantitative mathematical description of the neurons' abstract operation, that serves as a basis for discussing quantitatively the neural computing processes (as we discuss in section **??**, these derivatives are the laws of motion of ions) and their information handling. However, these corrections must be carried out in disciplinarily separated chapters.

As Schrödinger and Feynman implicitly suggested, we revisit the approximations that led to classical physics (where we derived the well-known 'ordinary' laws). We scrutinize the "construction" and "working" of living matter in a cross-disciplinary way, using non-ordinary approximations and abstractions. Then, laws based on the same first principles in a different approximation can describe living matter. We show that nature and its observable phenomena are complex; it is pointless to dispute which of the scientific disciplines describes neuronal behavior. The non-disciplinary approach shows that the firm and fast electric interaction sufficiently well describes neuronal operation after introducing the equivalent "thermodynamic electrical potential".

5.2 Schrödinger on life

In this chapter we reinterpret some concepts of physics to show that science can describe living matter. However, we must derive the appropriate approximations from the first principles of science and construct the correct form of laws for the conditions of living matter, instead of taking over the laws constructed by using different approximations from the first principles for the inanimate matter.

Quotation:

"Is life based on the laws of physics?"…"*New laws to be expected in the organism*"…E. Schrödinger: What is life?[1] @1944

5.3 Addendum

In this section, we add some comments, pinpointings and fixes to the chapters of book [6], needed mainly to make enhanced abstractions for biology, with the goal to derive the needed 'non-ordinary' laws of physics. The quotations are from that book, without citing the book per quotation, providing only page number. Our fixes and notations are typeset in blue, and confine themselves

Physics panel 5.2.1:
Schrödinger on 'non-ordinary laws'

a "From all we have learnt about the structure of living matter, we must be prepared to find it working in a manner that cannot be reduced to the ***ordinary*** **laws of physics**. And *that not on the ground that there is any 'new force' or what not*, directing the behavior of the single atoms within a living organism, *but because the construction is different from anything we have yet tested in the physical laboratory.*"..."How can the *events* in *space and time* which take place *within the spatial boundary* of a *living organism* be *accounted for by physics* and chemistry?" ..."living matter, while not eluding the 'laws of physics' as established up to date, is likely to involve 'other laws of physics' hitherto unknown, which, however, once they have been revealed, will form just as integral a part of this science as the former." (for the interpretation of those points, see **Physics panel 5.2.2**) E. Schrödinger: What is life?[1] @1944

to the most needed text, so the best way or reading is to read the two texts simultaneously.

Correct neuroscience textbooks such as [26] call the attention to that the analogy (from the point of the used equations) between the equivalent electrical circuits and biological neurons is not complete .

5.3.1 Ion movement in excitable cells

Page 9 "In excitable cells, movement of ions across the plasma membrane results in changes of electrical potential across the membrane, and these potential changes are the primary signals that convey biological messages from one part of the cell to another part of the cell" Given that those messages are delived by ions, the conveyed biological messages also mean changes of concentration. After crossing the membrane, the ions (as slow currents) also physically move in the layer on the surface of the plasma membrane. To consider the process and the dynamically formed component is essential in understanding why and how APs form, as well as how those signals are processed and transmitted. Their effect is significant. As shown in Page 12, the total number of ions in the spherical cell is $2 * 10^{13}$. As we estimate in section **??**, the number of ions conveyed in a single signal is about 10^7, which seems to be negligible. In line with our estimate, in the order of 10^4 pulses in quick succession are needed to drastically change the number of ions (aka concentration or potential of the bulk) in the cell. However, the number of uncompensated ions is estimated, in line with our estimate for the order of number of ions per pulse, as $4.7 * 10^7$, so it is absolutely necessary to consider the finite resources, as we do it in section 5.4.

Physics panel 5.2.2:
Schrödinger's points on living matter

a In his very accurately formulated question (see **Physics panel 5.2.1**, Schrödinger focused on (at least) these significant points

- '*events*' Unlike non-living matter, *living matter is dynamic, changing autonomously by its internal laws*; we must think differently about it, including making hypotheses and testing them in the labs. Those laws are non-ordinary 'because the construction is different', but its principles must not differ from the ones we already know.

- '*space and time*' Those characteristic points are significant changes resulting from processes that have material carriers, which change their positions with finite speed, so (unlike in classical mechanics) the events also have the characteristics 'time' in addition to their 'position'. In biology, the spatiotemporal behavior is implemented by *slow ion currents.* In other words, instead of 'moments' we must consider 'periods' and define the 'events' that define the beginning and the end of those periods.

- '*within the spatial boundary*' Laws of physics are usually derived for stand-alone systems, in the sense that the considered system is infinitely far from the rest of the world; also, in the sense that the changes we observe do not significantly change the external world, so its idealized disturbing effect will not change it. The 'construction' also results in constraint forces. Furthermore, we must consider the issue of finite resources.

- '*accounted for by physics*'[by extraordinary laws] We are accustomed to abstracting and testing a static attribute, and we derive the 'ordinary' laws of motion for the 'net' interactions. In the case of physiology, nature prevents us from testing 'net' interactions. We must understand that some interactions are non-separable, and we must derive 'extraordinary' laws. The forces are not unknown, but the known 'ordinary' laws of motion of physics are about single-speed interactions.

- '*living matter*' To describe its dynamic behavior, we must introduce a dynamic description. Processes happen inside it, and we can observe some characteristic points. The internal processes also manifest in changing physical quantities which are inseparable from the ones the experimenter intended to change.

- '*yet tested in the physical laboratory*'[including physiological ones] We test those 'constructions' in laboratories, in their actual environment, and in 'working state'. The measurement represents an intervention into their internal processes that falsifies the measurement results. As we did with inanimate matter, we must develop and gradually refine the testing methods as well as the hypotheses. Not only in measuring them (including measurement design and evaluation) but also in handling them computationally, we need more or less different thinking and algorithms.

5.3.2 Physical laws that dictate ion movement

Page 10 "The first two laws concern two processes: diffusion of particles caused by concentration differences and drift of ions caused by potential differences." Actually, there are three processes: it remains outside the scope of the HH theory that the mutual repulsion of ions generates a shock wave in the electrolyte. The generated wave mechanically moves the ions that per definitonem means a current. As our force analyzis shows, that mechanical force can dominate the ion movement in the transient state. The third law concerns the relationship between the proportional coefficients of the first two processes, the diffusion coefficient D and the drift mobility μ." Unfortunately, here the reciprocal relations, which we discuss in section **??**, between the processes is missing, as well as the role of the finite resources, which we discuss in section 5.4. This is a fundamentally wrong abstraction. The worst context of this abstraction is that it hides that the speeds of diffusion coefficients and speeds differ by several orders of magnitude, as it refers to the Nernst-Planck equation at zero flux, where, really, both speeds are zero. The equation, in addition, does not consider that by moving ions, we change the potential and concentration simultanously and inseparably; that is, handling the case with partial derivatives is not allowed.

Page 15"The negative sign indicates that I flows in the opposite direction as $\frac{\partial V}{\partial x}$ and in the opposite (same) direction as $\frac{\partial [C]}{\partial x}$ if z is positive (negative). This equation describes the passive behavior of ions in biological systems." Unfortunately, there are doubts here if partial derivation can be applied at all: when changing one of the abstract entities the other changes simultanously, given that the ion represents mass and charge simultanously.

5.3.3 The Nernst equation

Page 15"The NPE gives the explicit expression of ionic current in terms of concentration and electric potential gradients." As we discuss in section **??**, NPE is valid only in the 'mean-field' approximation, i.e., when we assume that the diffusion and electrical interactions have the same speed. Furthermore, although it was derived for equilibrium state, it implies that for the short time of forming an AP, the concentration also changes, given that the potential changes. According to their Table 2.1, for a short period, up to two orders of magnitude change in the concentration can take place. See also section 6.1.6.

5.3.4 Ion distribution and gradient maintenance

Page 17 "These ionic concentration gradients across the cell membrane constitute the driving forces (or chemical potentials) for ionic currents flowing through open channels in the membrane. In other words, the ionic concentration gradients act like DC batteries for cross-membrane currents." First, the relation is mutual; it could be considered that there are chemical batteries. Second, the ionic currents flow not only through the membrane; they can flow also through the extracellular space. Furthermore, one must tell also when it flows. Third, the driving force of those DC batteries changes during operation.

5.3.5 Membrane permeability

Page 24 "Within the membrane, if one assumes that $[C]$ drops linearly with respect to x"Since the ions are simultanously also charges, the potential also drops linearly, as explicitly said on page 26. Ions must accelerate across the membrane surfaces.

5.3.6 Space-charge neutrality

"In a given volume, the total charges of cations is approximately equal to the total charge of anions ... The principle of space-charge neutrality therefore holds in any volume in the biological system except within the plasma membrane, where the electric field is nonzero." The paper correctly states that the statement is valid only approximately, but misses the point that the volume must be choses appropriately and that the essence of life is the temporaly inbalance of charges; either the dynamic "charge pumping" processes in the resting state, or the transition state leading to issuing an AP.

5.4 Science concepts

Science has created abstract concepts such as space and time, mass and charge. It based its laws on abstractions/idealizations that were concluded by approximations and contained *only one* of these concepts. Inventing non-independence of space and time led to creating the theory of relativity. Under certain circumstances one must treat particles as a kind of continuous wave, and continuous waves as a particle. All those inventions had far-reaching consequences. These micro-objects interact with their environment, including our measuring instruments. With a measurement method suitable for particle detection, we see them as particles, and with an instrument suitable for wave measurement, we see them as waves. (How one micro-object sees the other remained cryptic.) This kind of micro-level behavior is inherent in nature; our methods of description and measurement must be adapted to nature and not the other way round. We understood that the Newtonian approach has its limitations; finite-velocity interactions can only be described by using a different kind of mathematics: by introducing the concept of space-time. In another approach, mass is also related to space and time; taking this into account, we can describe nature with curved space-time. We can also say that according to classical physics, we can describe phenomena with sufficient accuracy using only the aforementioned concepts (the zeroth derivatives). According to modern physics, however, for a more accurate description, we must take into account the first derivative with respect to time, and even the second derivative with respect to time for a general description.

Science's first principles could serve as a firm base for all its disciplines. As we discuss, *its disciplines use abstractions based on limited-validity approximations* based on the same first principles. However, *the approximations are different even for science's different disciplines, and even more for biology and physics.* In physics, most processes we observe are fast enough so that we can use the abstraction that they are essentially jumps between states. In most cases, the approach can be –more or less– successful. Notice, however, that in the case of a jump we neglect the need to explain why and how the transition happens. For the slower, well-observable processes, we have the laws of motion that describe how the processes happen under the effect of some driving force. We also experienced that nature is not necessarily linear (in the sense that it depends only on space coordinates but not on their derivatives), which we can describe by "nice" mathematical formulas. A century ago, A. Einstein invented that the approximations I. Newton introduced two centuries earlier are not sufficiently accurate for describing the movement of bodies at high speeds. In other words, a new paradigm, the constancy of the speed of light (today: the maximum speed of interactions), must have been introduced that caused a revolution in physics and led to the birth of "modern physics".

Classical vs modern science concepts

Science, unfortunately, is separated into classical and modern science based on whether the theoretical description assumes infinitely fast interaction (the Newtonian model) and continuous quantities or acknowledges the finite interaction speed (the Einsteinian model) or acknowledges the discrete nature of some quantities or connects the discrete and continuous views. Experience shows that the generated forces, regardless of their origin, can be summed, and one can apply Newton's laws of motion. *The finite interaction speed is erroneously associated with the speed of light and frames of reference moving relative to each other with speeds approaching the speed of light.* Assuming that the interaction speed is finite is sufficient to build up the special theory of relativity [27] (using the speed of light as the value for its external parameter). Still, the Minkowski-mathematics [28] behind the special theory of relativity works with any speed parameter c. The same mathematics describes technical [29] and biological [30] computing systems, where there are no moving reference frames, but the finite interaction speed has noticeable effects on the operation of the system. Furthermore, connecting the discrete and continuous views needs care in the case of biological objects and processes.

Quotation:

"*In thinking about the physical basis of the action potential perhaps the most important thing to do at the present moment is to consider whether there are any **unexplained observations** which have been neglected in an attempt to make the experiments fit into a tidy pattern*" [31], page 70.

The so-called modern physics was born when experiments began to contradict the idea that all phenomena of nature could be described in such a simple way. Neuroscience has similarly accumulated a number of experiences that contradict theory, although mainstream neuroscience has either ignored them or explained them with ad-hoc flawed and unfounded hypotheses. It happens despite Hodgkin's opinion about the unexplained observations. In this field (and in the physical foundation of biology in general), a similar modern approach is needed, which first requires reviewing the fundamental assumptions of the underlying physics.

Science disciplinarity

Science disciplines attempt to discover the infinitely complex nature in a disciplinary approach. To describe a *well defined* range of phenomena, we use approximations and omissions, and we create abstractions which can then be described by known laws using the universal language of mathematics. Newton's laws deal with masses only. Coulomb's laws deal with charges only. We use the abstraction "charge" and "charge carrier" for electrons, protons, ions, etc., and we can describe the electricity-related abstract features of the carriers. The charge has no mass, but the charge carrier does. We must not forget, however, that *those laws have been derived based on approximations and omissions*, and so they also *have their range of validity*. To apply laws from different fields of science, we must scrutinize whether all laws we use are applied within their range of validity. "Biological laws are much more difficult to find than physical ones" [5]. We add that mainly because those laws cannot be uniquely mapped to the disciplines of physics. The charge and mass can be separated conceptually, but not physically.

Classical physics classified phenomena into disciplines that largely described nature well. Modern physics has revealed new relationships, but they have also organically integrated into the fabric of science as a whole. To understand the living organism, in the spirit of R.P. Feynman, we must make "excursions" between disciplines. E. Schrödinger expressed his conviction that no new force or unknown interaction is emerging; only a previously unknown regularity concept must be found, then *the non-ordinary laws organically integrate into the fabric of science, together with the already known laws.* In this chapter, we make an attempt to find such a concept in the hope that scrutinizing the features of ions leads to a solution.

Regarding ions, separately, we can handle their charge and mass well, we know the relevant laws. Their properties are analogous to the particle-wave duality: depending on the interaction (i.e. the measurement method), we see an ion as either a charge or a mass. *We see the same object, but we measure it based on the concepts of different disciplines.* Therefore, we must connect charge and mass, just like space and time in the case of relativity. They do not exist independently of each other and their combined behavior differs from what we expect based on the classical disciplinary approach. Connecting two abstract notions generally result in counter-intuitive experiences. Interestingly, similarly to relativity, their velocity establishes the connection between the two quantities here.

Furthermore, we must introduce a connection for the currents represented by ions, such as the one introduced by statistical mechanics when connecting the discrete and the continuum approaches. However, we base the correspondence on geometrical considerations and, due to the unseparable connection between the two properties, we cannot apply statistical laws based on one property in an unchanged form. Moreover, in most biological cases, the number of particles involved are so small that their statistical interpretation is severely limited. *The fundamental approximation of physics that the force field acts on the particle, but the particle's effect on the field is negligible, is not met.* In some cases, closed spaces give rise to counterforces that fundamentally change the processes that take place. *Ignoring these properties of living systems has led to the erroneous statement that science cannot describe living matter and its processes.* Describing living matter is undoubtedly more complicated than describing inanimate nature and requires a revision of several approximations. However, it is possible. *Physics is much more than a collection of formulas concluded for inanimate matter and valid within the frames of a discipline that can be applied without thinking about whether they can be applied to the conditions of living matter.*

Locality and finite resources

Quotation:

> "How can *the events in space and time* which take place ***within the spatial boundary*** of a living organism be accounted for by physics and chemistry?" [1]

The closed volume of the biological objects and the finite speed of the charge carriers is also a problem of finite resources. In biology, we cannot use the fundamental assumption of physics that, although a field acts on the ion in question, the ion does not affect the field (the other ions generate). The material transport represents simultaneously mass and charge, so the transport itself gradually changes the gradients (the field itself). The transferred ions decrease the field in the volume they departed from and increase it in the volume where they arrived (a problem of finite resources); furthermore, the concept of 'field' is slightly different from the classical one, see below. We must consider that the resources are finite. A main source of confusion is that the phenomena happen in a limited and closed region of space (*within the spatial boundary*), and we study processes (in a period instead of a moment) where the environment is not "infinitely far" from the studied object (*the studied process interacts with its environment*), and the forces exerted on it belong to different disciplines. The boundaries of a closed volume may exert constraint forces and fundamentally affect the processes inside the volume.

Due to the field-dependent speed within the electrolyte, we must consider the autonomous change between the microvolumes where the ion traverses. This process keeps the entire volume of the electrolyte in a (more or less intense) continuous change, which makes life possible. The more distant parts of the biological cell will see any change in the local values of the state variables with a delay. Furthermore, biological objects inside the cell can absorb ions and charge up. With their potential, they alter local gradients, accelerating or decelerating ions. Not to mention that biological objects can be active in the sense that (depending on the environmental conditions) they can let ions from one separated volume part into the other, thus changing the number of charge carriers in the volume under study. Co-existing processes may affect each other by changing the available resources: changing one quantity changes the other. Mathematics descibes such processes by linked differential equations, such as the Lotka-Volterra (or "predator-prey") equations. For details, see section **??**.

These processes are what E. Schrödinger coined as "*the construction is different from anything we have yet tested in the physical laboratory*" [1]. Consequently, measurements must be designed and carried out with care; the routine methods used for measuring objects from inanimate nature, cannot surely be applied in unchanged form to living objects.

Fields in science

The concept of force fields is fundamental for science since I. Newton introduced that remote objects can interact with each other. Traditionally, a field is defined by the force a test charge (or mass) would experience at a specific point in space, even if no charge/mass is currently there (assuming an infinitely small charge or mass, despite that we know their discrete nature). R. P. Feynman questioned whether fields were fundamental physical entities or simply convenient mathematical descriptions of the "condition of space". He cautioned against "mistaking the map for the territory," suggesting that while (electric and magnetic) fields are useful, they might not be the ultimate reality. Anyhow, the fields are created by nearby ions which can change their positions during the interaction, furthermore, the ions under test interact with them and this way change their environment. Although the fundamental interactions forces and laws are the same that we are used to when testing objects from the inanimate nature, the "construction" is entirely different. This difference needs much attention because the laws concluded for objects of inanimate nature are not valid for objects in living nature in unchanged form.

Mass and charge may have the corresponding fields given that they have well-known remote interactions. However, thermodynamics does not consider either gravitational or electrical fields: the interaction are implemented by short-range interaction (direct collisions). Although there are theories for both fields where (typically virtual) particles represent the action of the field, those particles are not identical with the sources of that field. To extend the concept of field to thermodynamics, one must extend the idea of deterministic forces with stochastic forces; so the resulting 'field' is of stochastic nature. In the case of ions in electrolytes (particularly in structured biological matter), they move under the effect of electric and thermodynamic fields. In their laws of motion, one has to add electrical and thermodynamic fields, i.e., a deterministic and a stochastic field. The result is stochastic, anyhow. The case is further complicated by the fact that the propagation speeds are different for the two cases. One can tell only stochastically valid statements and must derive special laws of motion (say, for the selectivity of ion channels; see section **??**).

5.4.1 Dichotomies in modeling

For centuries, science developed its methods for deriving abstract concepts by reducing the features of a real object to an abstract (idealized) one that cannot be reduced further, such as mass or charge. Ions are an exception when using the concepts of classical disciplinary physics: *the ions are charge and mass simultaneously, without a further possibility of reduction.* The consequences of this item are listed in Table 1 of paper [12] as the "Electrical vs Mechanical" dichotomy. Science has derived laws for the forces acting on those abstract objects, such as Newton's universal law of gravitation and Coulomb's law of electricity. Then one could apply Newton's laws of motion. Experience shows that the generated forces, regardless of their origin, can be summed, and one can apply the laws of motion using the resultant force (see below for a discussion of the resultant force and its application in biology). However, notice that those experiences are based on the Newtonian 'instant interaction', i.e., that the interactions' speeds are equal.

Paper [12], which, in the context of this section, tends to be the precursor of our ideas, sheds light on some remarkable dichotomies, scrutinizing which can result in "a sound basis for unification of the physics of nerve impulses". By deeply agreeing with that they have a "potential impact on our understanding of (the physical nature of) neuronal signaling", we list and uncover more dichotomies; furthermore, we uncover their fundamental, partly philosophical reasons.

5.4.2 Complete measurement

In physics, measurement means quantifying something relevant to the process under study. A 'complete' measurement measures all relevant quantities. *The* *different disciplines of physics restrict the measured quantities to the ones which are, in general, that the discipline studies. The remaining quantities remain outside the scope of the discipline.* The fundamental physical quantities of mechanics are length, mass, and time, which form the basis for defining all other quantities like velocity, force, and energy in that field. The fundamental physical quantities in thermodynamics are Temperature, Energy (Internal Energy, Heat, Work), and Entropy, which characterize systems at equilibrium and describe energy transformations; Pressure and Volume are also key variables. Those of electricity include electrical charge (Coulomb), the basis of all electrical phenomena; electrical current (Ampere), the rate of charge flow; voltage (Volt), the potential difference driving current; resistance (Ohm), opposition to current; power (Watt), the rate of energy transfer; and energy (Joule), the capacity to do work. As seen, there is little overlap of the studied quantities between the disciplines.

When making a measurement, one must keep in mind that measuring devices are designed for the purposes of determining a quantity interpreted in one of the disciplines of science. It is the task of the experimenter to figure out which physical quantities (that maybe interpreted in different disciplines) play a role in the process and to define the set of quantities that describe the process, instead of pre-deciding which discipline's set of inanimate science will be used for biology which is not one of the disciplines of inanimate science. In biology, that set does not necessarily matches any of the sets used in one of the disciplines. *A disciplinary study (such as the electrical and thermodynamic ones in physiology) provides an incomplete set of measured quantities.* None of the disciplines alone can describe electrolytes, since they consider different and incomplete sets of physical quantities: thermodynamics misses charge, electricity misses mass. As discussed in connection with the Onsager's relations, a disciplinary (incomplete) measurement does not discover that an unexpected change (an unexplainable "miracle") in the value of one of the quantities belonging to another discipline is accompanied by a change in the value of the other quantity. This change is clearly the case with the measurements performed in the spirit of HH: the values the electrical instruments provide are accompanied by the values of mechanical/thermodynamic quantities which were not measured, partly due to the obvious measuring difficulties and because they are outside the scope of the discipline electricity. The theory, however, should not remain disciplinary. If one attempts to reduce ion-related phenomena to a single abstraction (see the mentioned theoretical descriptions), one experiences that some quantity (the charge or the mass) changes in an uncontrolled way. The fundamental reason of the incompatibility on the disciplinary theories is that they consider an incomplete set of physical quantities. The disciplinary separation in classical science does not apply to the study of living matter. *We must make a complete measurement and interpret them in a cross-disciplinary way.*

Thermodynamics provides a framework for handling ions and determining their thermodynamic properties. However, as good thermodynamic textbooks (including the one on the thermodynamics of the membrane [32]) emphasize, *thermodynamics derives its concepts for non-interacting particles*, so one cannot expect its validity for ionic solutions [20] (Boltzmann assumed that, in the absence of long-range interaction between the particles, the sizes of cells in the phase space do not change). In addition, he required the presence of a vast number of particles in a homogeneous, isotropic, infinite volume, which is typically not the case in biology.

Electrical vs Mechanical

In the electrical abstraction, no mass is present, so one can use the equations assuming 'instant interaction' (aka an infinitely large interaction speed, and, consequently, an infinitely large acceleration), which leads to non-physical explanations of the physiological observations (such as 'delayed current'). In the mechanical abstraction, mass is to be moved, making the finite-speed interaction evident. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively and simultaneously, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [20]. This item is listed in Table 1 in [12] as the "Electrical vs Mechanical" dichotomy.

To deliver a current, one needs moving charged particles that need exerting of some external (electrical, magnetic, or chemical) force or a mixture of forces. We have speeds of EM interaction, thermal motion of the charge carriers, macroscopic current, drift speed, and their mixing, simultaneously, in the same phenomenon. In the theoretical description of processes, instant interaction (i.e., the abstraction of non-physical, infinitely large interaction speed) is used in most cases, which is a good approximation only if the charge carriers are electrons. In the cases when absolutely needed, the generic notion of "speed" is used without specifying which one of the mentioned speeds it means.

The low speed of ions in electrolytes introduces further problems. Any change in the local value of the state variables will be seen by the farther parts of the cell with a delay, The material transport represents simultaneously mass and charge, so the transport itself changes the gradient. This process keeps the entire volume of the electrolyte in (more or less intensive) continuous change. Furthermore, the biological objects inside the cell can absorb ions and charge up. With their potentials, they change the local gradients, accelerating or decelerating the ions. Not to mention that biological objects can be active in the sense that (depending on the environmental conditions) they can let ions from one separated volume part into the other. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [20].

Atomicity

Although science is aware that the apparently continuous matter in nature cannot be divided infinitely, even though it knows that the abstracted discrete 'material points' (as A. Einstein coined) have well-defined discrete values, it occurs rarely that both views must be applied in describing a single phenomenon. The continuous and discrete approaches (also called macroscopic and microscopic views) seem to be independent of each other. Connecting those views was one of the tasks performed by thermodynamics for particles with no long-range interactions. Inventing discreteness of energy led to quantum mechanics. Connecting the features of the microscopic discretete constituent objects to the macroscopic continuous features led to thermodynamics. However, there is no similar discipline for electricity, although the behavior of charge carriers is similar to that of neutral discrete particles. It is not evident for electrons, but it is significant for ions. This item is listed in Table 1 of paper [12] as the "Macroscopic vs Microscopic" dichotomy.

Reversibility

The third dichotomy is "Reversible vs Irreversible". A reversible process can proceed in either direction in time. Cellular operation seems to be cyclic (i.e., mostly reversible) in the short run, while in the long run, spanning from birth to death, it is irreversible. Experience suggests that a mixture of reversible and irreversible processes describes neuronal operation. As reviewed in [12], in contrast with the electricity-centered disciplinary conception HH model [17], all other disciplinary models discussing phenomena suggest reversible operation.

Reversibility seems to be closely related to the dichotomy "Electrical vs Mechanical" and to the nature of the exclusively used electrical HH theory. In the framework of that theory, Ohmic currents flow through resistors that irreversibly dissipate heat due to friction, no matter in which direction ($W = I^2 \times R$) the ion currents flow. By introducing the concept of 'delayed current' and the hypothesis that some hidden power controls the operation of neurons by altering their conductance, physiology gave rise to the fallacy that science and life sciences are almost exclusive fields. Furthermore, their (unintended) model provokes questions (for a review see [15]) whether it is a model at all, and what controversies it delivers. As Hodgkin wrote [31]: "Hill and his colleagues found [10] that an initial phase [of the action potential] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." Since that discovery, experimental evidence from the missing "leakage current" [33] to the conversion between elastic and kinetic energies [34] to demonstrating the pressure-wave-like behavior of AP witnesses that at least part of the phenomenon is of thermodynamic origin; that is, it requires a cross-disciplinary explanation. For discovering the reciprocal relations in thermodynamics [35], also in electrolytes, Lars Onsager was awarded the 1968 Nobel Prize in Chemistry. Those relations, together with the theoretical understanding, manifesting in the cross-disciplinary Nernst-Planck relation, paved the way to a combined theory. However, no thermodynamic explanation has been given so far. Our discussion provides, in line with the experimental results, a complex unified electrical/thermodynamic description of neuronal operation.

5.4.3 Speed

The role of speed and time, particularly in the context of an object's changing location over time, has long held a mystique in the realm of scientific discovery (and recently returned to be mystic again in cosmology). This intrigue can be traced back to historical debates, such as Zeno's paradoxes. The acknowledgment that an object's movement speed can influence our observations is a topic that has sparked significant scientific discourse over the years. In this section, we aim to draw parallels between the historical debate on the finite speed of light and its contemporary implications in various scientific disciplines, such as the finite speed of ionic currents in biology.

It has been a long-standing mystery that interactions with different speeds play their role *simultaneously*. The issue forces researchers to give non-scientific explanations to everyday phenomena only because *they routinely assume that the interactions have the same speed, and they use the laws about strictly pairwise interactions*. They have no choice: there is no formalism to handle non-equal speeds.

We need different abstractions (finite-speed interaction in modern physics) for different phenomena that require different mathematical handling, which is not as simple and friendly. The *speeds of observation* and propagation of electric fields remain the same in biology, and it is easy to extrapolate, mistakenly, that *all* interactions have infinitely large interaction speeds. However, also *slow* interaction speeds exist, furthermore, *different interaction speeds can intermix in the same phenomenon. Neglecting that effect introduces the need to assume fake mechanisms and effects* for explaining some details; which are naturally explained by assuming finite interaction speeds and their combinations. More discussion follows in section **??**.

Instant Interaction

The above dichotomies are rooted in deeper layers of science. Notice that in the electrical abstraction, no mass is present, so one can use the equations assuming 'instant interaction', which in biology led to using non-physical explanations of the observations (such as 'delayed current'). In the mechanical/thermodynamic abstraction, mass is to be moved, making the finite-speed interaction evident. Classical physics is based on the Newtonian idea that space and time are absolute, so everything happens simultaneously. Consequently, when nature's objects interact, it must be instantaneous; in other words, their interaction speed is infinitely large. Furthermore, electromagnetic waves with the same high (logically, infinitely high) speed inform the observer. This self-consistent abstraction enables us to provide a "nice" mathematical description of nature in various phenomena: the classical science.

We learned that the idea resulted in "nice" reciprocal square dependencies, Kepler's and Coulomb's Laws. We discussed that the macroscopic phenomenon "current" is implemented at the microscopic level by transferring (in different forms) discrete charges; furthermore, that solids show a macroscopic behavior "resistance" against forwarding microscopic charges. We also learned that *without charge (and, without atomic charge carriers), neither potential nor current exists*. We did not learn, however, that *also thermodynamic forces can move the ions*, given that they have inseparable mass.

The low speed of ions in electrolytes introduces problems. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [20]. (Historically, Onsager's work in 1931 referred to thermoelectricity and transport phenomena in electrolytes, thus connecting experimentally two separate disciplines.)

Physics notoriously suffers from the lack of handling different simultaneous interactions; facing such a case leads to misunderstandings, debates and causality problems. Such a famous case is the interaction speed of entanglement. In that time, E. Schrödinger introduced his famous Law Of Motion into quantum-mechanics entirely analogously as I. Newton introduced his Laws of Motion. Similarly to the Newtonian 'absolute time', the quantum mechanical interaction is supposed to be 'instant' (this is the price for having 'nice' equations in classical and quantum mechanics), i.e., its speed is supposed to be infinitely high. However, at that time was already known that the electrical interaction (propagation of electromagnetic waves) is finite, so if an object has quantum mechanical interaction (aka entanglement) and electrical interaction, the corresponding forces start at the same time, but arrive at the other object at different times. The entanglement arrives instantly, the electromagnetic effect arrives at the time we can calculate from the interaction speed and the spatial distance of the objects. This leads to causality problems: the effect of the two interactions of photons entangled earlier in an exploded supernova should be measured at two different times; meaning a "spooky remote interaction" as A. Einstein coined, and leads to contradictions such as the Einstein–Podolsky–Rosen paradox. Actually, the issue roots in the improper handling of mixing interaction speeds: the Schrödinger-equation introduces the infinitely large interaction speed, while the EM interaction has finite speed.

The confusion and question marks in connection with describing the life by science mostly arise from the interpretation of notion 'speed' in physics. When discussing the underlying physical laws, we go back to the very basic physical notions instead of taking over the approximations and abstractions used in the *classical physics for non-biological matter* and less complex interactions. As we emphasized many times, we construct laws and conclusions based on somewhat simplified abstractions about nature, in all fields of science. The notions and laws depend on the circle of phenomena we know and want to describe. The Newtonian and Einsteinian worlds are basically distinguished by considering their *speed dependence* that actually means *explicite time dependence*. Interesting consequences are that in the Einsteinian world, the mass is not constant, the time and space are not absolute, and so on. We can be prepared to some similar counter-intuitive experiences in physiology: "we must be prepared to find it working in a manner that cannot be reduced to the **ordinary** laws of physics"[1]. Here we scrutinize the basic notions and discover some differences between physics and biology as consequences of the required different abstractions and approximations.

During our college studies, we mentioned that light is an electromagnetic wave with a vast but finite propagation speed. Still, we forgot to highlight that, at the same time, it is the propagation speed of the electric (, and magnetic, and gravitational) interaction force fields as well. The effect of "Retarded-Time Potential" is also known in physics and communication engineering. Algorithms "marching-on-in time" and "Analytical Retarded-Time Potential Expressions" are derived to handle the problem; for a discussion, see [36]. The Telegrapher's equations (unfortunately, also used to describe biological signal transfer) explicitly assume a finite propagation speed millions of times slower than the (implicitly assumed) EM interaction's. The issue is not confined to large distances: designers of micro-electronic devices also must consider the effect: they introduced clock time domains and clock distribution trees; see, for example [37, 29].

Science uses 'instant' in the sense that one interaction is much faster than the process under study; we consider the faster interaction as instant. The approach of classical science is based on the oversimplified approximation that the interaction speed is *always* much higher than the speed of changes it causes and that the processes can *always* be described by a single stage. In our approach, for biology, we put together a *series of stages* to describe the observed complex phenomena, where the stages provide input and output for each other, involve more than one interaction speed, and use per-stage-valid approximations. We simplify the approximations by omitting the less significant interactions and introduce ideas for accounting for the different interaction speeds. This way, we reduce the problem to a case that science can describe mathematically. *This procedure is fundamentally different from applying some mathematical equations derived for an abstracted case of science to a complex biological phenomenon without validating that we use the appropriate formalism.*

Speed of light

In 1676, the Danish astronomer Ole Rømer was making meticulous observations of Jupiter's moon Io and concluded not only that the speed of light is finite, but he measured its value with a sufficient accuracy. Rømer never published a formal description of his method, possibly because of the opposition of his bosses, Cassini and Picard, to his ideas. Cassini knew Rømer's idea and the measurement data. However, instead of accepting the finite value of the speed of observation, he made periodic corrections to the tables of eclipses of Io to take account of its irregular orbital motion: *periodically resetting the clock* .

The speed of light must remain infinitely large.

However, the theory of finite speed quickly gained support among other natural philosophers of the period, such as Christiaan Huygens and Isaac Newton [38]. Although Newton surely knew that the observation speed was finite, in his "Philosophiae Naturalis Principiai Mathematica" [39], published in 1687, he decided to refer to observations that they happen "at the same time" despite knowing that what we observe at the same times, happen at different times. Using instant interaction results in "nice'" mathematical laws and enables us to describe most of nature's experiences with sufficient accuracy. Einstein, in 1905, discovered [40] that the speed of observation (in moving reference frames) may play a decisive role in interpreting scientific phenomena. The results he derived using Minkowski-coordinates [28] were counter-intuitive, with many unexpected consequences. Instead of introducing improvement(s) or correction(s) to the existing classic principles and methods, he introduced a new principle: the finite (limiting) interaction speed. The *disciplinary analysis of the reception of Minkowski's Cologne lecture reveals an overwhelmingly positive response on the part of mathematicians, and a decidedly mixed reaction on the part of physicists* [41] has turned to the exact opposite. Today, physics generally accepts the description, that is, the existence of finite interaction speed (resulting in the birth of a series of modern science disciplines). However, other science disciplines, including biology and computing science, refute (or at least do not use) it; despite that its effects are evident.

Speed in neuroscience

Helmholtz, in 1850, sent a short report off to the Academy [42] "I have found that a measurable time passes when the stimulus exerted by a momentary electrical current on the hip plexus (Hüftgeflecht) of a frog propagates itself to the nerves of the thigh and enters the calf muscle." His teacher "had thought that the speed of nervous conduction might be in excess of the speed of light and could probably never be measured. Helmholtz's father, on hearing of the experiment and the surprisingly slow measured speed, wrote to his son that he would as soon believe this result as that one can see the light of a star that burned out a million years ago" [43].

With the development of measurement technology, it became evident that finite speed is a general feature of the "nervous connection". (Somehow, "the speed of nervous conduction" has been renamed to "conduction velocity", neglecting the clear distinction that physics makes between the two wording.) With the dawn of instrumental electronics and computing, the McCulloch-Pitts model [44] introduced the picture that the brain can be modeled by a network of simple perceptron nodes connected by wires; that is, it comprises a two-state equipotential membrane connected with perfect wires. The experimental research also quickly (re-)discovered that those wires forward signals in a particular way; the speed of the potential wave is finite. Furthermore, *the axons are not equipotential during transmission.* Although its structure is practically identical with axons, *biology assumes that, unlike an axon, the membrane remains* *equipotential during its operation, although the evidence shows the opposite*: 'the action potential spreads as a traveling wave from the initial site of depolarization to involve the entire plasma membrane' [45].

When seeing that assuming an equipotential membrane was wrong and a single equipotential surface (in other words, classical physics' instant interaction) cannot describe neurons adequately, multi-compartment models (typically comprising equipotential cylinders with different potentials) have been introduced [46]. (Notice that it is a consequence of the wrong oscillator model hypothesized by Hodgkin an Huxley: the membrane is modeled as a series of resistors and capacitors.) Then (forgetting that Ohm's Law is valid only for the classical physics's 'instant interaction', furthermore, that no external potential is connected to either of the compartments, and no charge is present at the beginning, except at the input of the first compartment), the individually equipotential compartment pieces were connected by individual resistors. This model shows that the more compartments, the better the agreement (accuracy) with experiments. It happens because the shorter is the size of the compartment (approaches a differential equation), the less noticeable is the deviation from the true non-equipotential surface. This conclusion means that charging the capacitance attached to the compartment takes time, resulting in a delayed distributed current. Using infinitely many compartments, we would arrive at the differential equations describing a delayed distributed current on the surface of the non-equidistant membrane. However, biology did entirely not give up its position. It admitted that membrane current exists, but only between compartments, and its speed must be infinite (or, at least, the speed of EM interaction). However, at least the compartment pieces must be equipotential. Instead of fixing the wrong hypothesis, biology is "periodically resetting the clock". Instead of accepting that the charged ions represent a "slow current" (compared to the "fast current" represented by electrons and their charge transmission method), biology introduced changing conductance, delayed current, rectifying current, and so on; effectively blocking the understanding of the neural operation.

Finite-speed interactions

When speaking about speed, especially about the speed of charged objects inside biological objects, one needs to consider microscopic and macroscopic levels of understanding. On the boundaries of the two levels, we need to make distinctions between different kinds of speeds, among others (in units of m/s), the propagation speed 10^8 of the electric field (aka potential gradient), the speed 10^5 of thermal motion and potential-accelerated motion, the apparent speed of current (potential-assisted speed of a macroscopic stream, both in metals and electrolytes; mainly due to the repulsion of nearby ions in the stream) 10^1, for ion current inside a neuron (see Fig. 1 in [47]) 10^{-2}; diffusion speed of electrons in a wire 10^{-4}, drift speed of the individual carriers in aqueous solutions 10^{-7}; and of ions moving in a narrow tube filled with viscous liquid 10^{-8}. Fortunately, in most *but not all* cases, different mechanisms (such as the Grotthuss mechanism or free electron model; for a review, see [48]) at the level of microscopic structure help to create the illusion of a high macroscopic propagation speed (million times higher than the speed of its microscopic carriers). *The same carrier can have macroscopic speeds differing by orders of magnitude, depending on the context*; see a biological example at ion channels. When more than one of those speeds plays a role in the phenomenon we study, we must carefully consider its context and prepare for handling *fast* and *slow* effects, furthermore, their mixing.

When an object can interact with another in a way abstracted by science as more than one interaction type, we need to find the relation (the 'extraordinary' law) between them. Such a famous case is electricity and magnetism. Their interrelation is defined by the Maxwell equations: how an electrical field creates a magnetic one and vice versa (notice that the law is about their *space derivatives, aka space gradients* instead of the entities themselves). While we understand that the speeds of electromagnetic and gravitational interactions are finite, we can use the 'instant interaction' approximation in classical physics because one effect of the first particle reaches the second particle simultaneously with the other effect, leading to the absence of a time-dependent term in the mathematical formulation.

An apparently similar case is found in electrodiffusion, where ions can be abstracted as mass and charge, one belonging to thermodynamics and the other to electricity. There is, however, an essential difference between those cases: the interaction speeds are the same in the first case (moreover, in the spirit of classical physics, the interactions are instant) and differ by several orders of magnitude in the second one. Of course, the Maxwell equations can be nicely solved and modeled also for biology if one introduces [49] that the axial currents have the same speed (BTW: which was measured as 20 m/s) as the electrical and magnetic waves, furthermore the longitudinal current is (?)defined(?) to have no attenuation. Furthermore, it is likely also defined that current needs no driving force and this is why the positive and negative ions flow in the same direction. It is really a novel paradigm leading to "(mis)understanding cell interactions", but definitely describes some alternative nature.

Processes vs states

The finite speed introduces something unusual into science: processes, rather than jumps between states must be considered. The processes connect states and we need the corresponding laws of motion to connect them.

Speed in laws of science

Actually, the famous Coulomb's Law is expressed as

$$\frac{F(t)}{Q_1} = k\frac{Q_2}{\overrightarrow{r}^2} \tag{5.1}$$

$\overrightarrow{r}$ is a space-time distance. That is, in the Newtonian approximation, time is identical at all places, so we used to omit it. However, physics knows also the notion of *retarded time*. Considering the finite field propagation speed requires revisiting the fundamental physical laws. (in a Lorentz-transformed form) should be written as

$$\frac{F(t)}{Q_1} = k\frac{Q_2}{r^2}(t-\frac{r}{c}) \tag{5.2}$$

It is an implicit equation usually solved numerically or using perturbation methods for moving sources.

The electrostatic field that charge Q_1 experiences due to the finite propagation speed c of the electric field (or interaction) corresponds to that Q_2 at a distance r *generated* $\frac{r}{c}$ *time ago* (k is the constant describing the electrical interaction). This term has no role if the two charges do not change their position; similarly to that in the special theory of relativity, only the relative movement leads to complications. If the distance changes, its effect is so tiny that the term can usually be omitted. So, our college knowledge can serve as a good first approximation.

This speed term pops another law from classical physics into our minds: Kirchhoff's junction rule. The law is perfect in the approximation 'instant interaction' that classical physics uses, but not for biology. First, because it expresses charge conservation, *it is invalid when charges are "created" inside biological objects* (ions diffuse into the junction; see the role of ion channels in the wall of membranes). Second, it is not valid for input currents arriving with finite speeds into finite-size space regions, but it is valid for a single point in space-time (in other words, in differential equation form). As we discuss in section **??**, using a wrong definition of current means assuming 'instant interaction', that is, that neural signals propagate with the speed of light. The currents (and the voltages), measured at two different points in space-time, are different. Consequently, for extended objects (such as a line-like finite-size neuron), it is valid only with a time delay

$$I_{out}(t) = -I_{in}(t - \Delta t) \tag{5.3}$$

The time delay in biology is in the 1 *msec* range. *We must not describe the axon or the membrane with the non-differential form of the Kirchoff equation: the input and the output currents flow at different times (the charge carriers need time to reach from input to output); only the differential equation form expresses charge conservation* (furthermore, in the case of "producing" ions, even by the differential form is invalid). For its exact interpretation, see sections on axonal charge delivery and on the true membrane current, Fig. **??**, and the text around it.

Studying electrical phenomena on structured media, such as biological cells, needs much care. We must not apply laws derived from entirely different conditions (mainly metals). The "ghost image" can be the "zeroth order" approach to understanding the AP: we assume that a"slow current" causing to appear as Post-Synaptic Potential (PSP), flows out through AIS some Δt time later than in flowed in "at the other end" of the neuron. The time delay is needed for the slow current to reach from the entry point to the exit. In fact, the figure displays Kirchhoff's junction rule for the slow currents in biology: the rush-in current triggered by exceeding the threshold leaves the junction a few tenth of millisecond later. The difference between the numerical solution and the simple difference is mainly due to the distributed nature of the input current. As the conduction speed increases, the difference current decreases, and at the speed of "instant interaction", results in Kirchhoff's junction rule as known in the technical electricity.

5.4.4 Spatiotemporal

Classical physics is based on the Newtonian idea that space and time are absolute so everything happens simultaneously. Moreover, their observationis also instantaneous. Consequently, when their objects interact, it must be instantaneous; in other words, all interactions have the same (and, logically: infinitely high) speed. Furthermore, electromagnetic waves with the same high (logically, infinitely high) speed inform the observer. This self-consistent abstraction enables us to provide a "nice" mathematical description of nature in various phenomena: the classic science. In the first year of college, we learned that the idea resulted in "nice" reciprocal square dependencies, Kepler's and Coulomb's laws. We discussed that the macroscopic phenomenon "current" is implemented at the microscopic level by transferring (in different forms) discrete charge, and that movement of charges has no effect on the environment. Furthermore, that *without charge (and, without atomic charge carriers), neither potential nor current exists.* In the next semester, we learned that the speed of light is finite and that solids show a macroscopic behavior "resistance" against forwarding microscopic charges. One semester later, we learned that nature behaves differently at high speeds, at low sizes and energies; furthermore, that the transition between the continuous and discrete views needs new concepts.

In biology, it was evident that the transfer (conduction) time must be considered together with the computing (synaptic) time (in this sense, presynaptic to postsynaptic transmission time). The name "spatiotemporal" and a (separated) time dependence is commonly used [21], in the sense that Precise Firing Sequence (PFS) "*tended to be correlated with the animal's behavior*"; furthermore, that "*the results suggest that relevant information is carried by the fine temporal structure of cortical activity*" [50]. The "*neural dynamics*" was studied and "*spatiotemporal spreading of population activity was mapped*" [51] by methods used to describe the *static computing methods*: interspike intervals histograms, auto-correlation and cross-correlation. Because of the peculiarities of this information handling, *there are severe doubts whether the notions of the classic neural information theory are valid for biological computing systems* [52]. The correct method of describing biological computation is still missing, given that the significant item of the computing is missed: *the time and position are connected through the information transfer speed* (called conduction velocity).

Will be based on [30, 53, 54]

In Fig. 5.1, for visibility, two spatial and one temporal coordinate are shown. In the following figures, some illustration enable to omit one more spatial dimension, i.e., effectively to draw the events as a two-dimensional diagram.

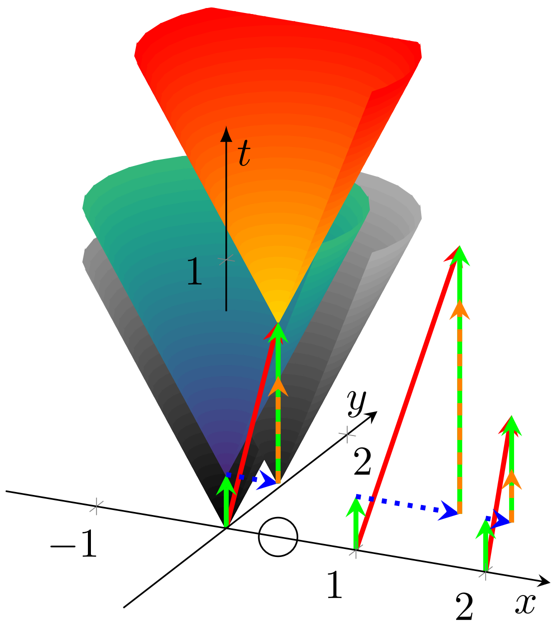


Figure 5.1: The temporal diagram, i.e., the way of calculation to combine the spatial distance (transmission time, blue arrows) and computing time (green arrows) illustrated in the time-space coordinate system. The orange-green vertical arrow shows that the second computing unit must idly wait until the transmitted result reaches its position, because of the finite transmission time. The axes x and y refer to space coordinates (transformed to time using the conduction velocity), the axis t refers to the time itself. The arrows starting from points 0, 1 and 2 on the x axis illustrate timing for three different propagation speeds. The red vector points from the beginning to the end of the process. Its length may serve as a statistical entity to describe temporal distance of the units.

5.4.5 Mathematics

Galilei said, "Mathematics is the language in which God has written the universe". However, it is not sure that when we attempt to read a piece of the universe written in that language, we use the right piece and dialect of the language, and even that humans already invented the needed piece. Not mentioning whether one understands the given piece correctly. For example, mathematical calculus (integral and differential) was invented mainly for the practical needs of analyzing spatial motion of celestial bodies. Similarly, Minkowski's mathematics theory proliferated widely [55] only after inventing the special theory of relativity. Although the mathematical description was developed earlier, there was no practical need to apply it. The classical laws of motion were valid only until more meticulous observations required to consider speed and acceleration (time derivatives of position) dependence in addition to position dependence. Newton's *static* laws remained valid, but for the *dynamic* description we must revisit the second law of motion.

Also, we must not forget that "mathematics is not just a language. Mathematics is a language plus reasoning. It's like a language plus logic. Mathematics is a tool for reasoning." (Richard P. Feynman) Mathematical formulas work with numbers but math theorems and statements begin with "If ... then". They have their range of validity, even when they describe nature. Using mathematics to describe the classical equations of motion, to calculate forces and times that speed up bodies above the speed of light is possible, but in that case mathematics is applied to an inappropriate approximation to nature. When approaching the speed of light, different physical approximations (that calls for a different mathematical handling) are to be used. *A mathematical formula, without naming which interactions it describes and naming under which conditions and approximations the formula can be applied, are just numbers without meaning. It surely describes something but only eventually describes what we studied.* Galilei made measurements with objects having friction, but his careful analysis extrapolated his results to the abstraction that no friction was present. *We know his name because of making meticulous abstractions and omissions (and mainly: recognizing the need to do so!)* instead of publishing a vast amount of half-understood measured data.

The discrete-continuous dichotomy raises the interesting question: how long quantities describing matter remain differentiable? At macroscopic quantities, for example, in technical electricity, surely they are: for a large number of charge carriers flowing through a cross section, the statistical fluctuation of the number of charge carriers obviously is sufficiently small to operate with the approximation that the fluctuation is zero, so the definition that, at a cross section, $I = \frac{dQ}{dt}$ defines the current has sense. Even one can approximate the current by saying that dQ cannot be smaller than the elementary charge (recall what a revolution happened due to assuming the the energy of a radiation cannot be arbitrarily small). Mathematically, one can say that the time course of a single charge carrier is a δ-function, which is surely not differentiable. Then, is the rush-in current with its 10^7 charge carriers differentiable? If it happens in fractions of a nanosecond, over a surface of the membrane? The same question for the single ion channels? Or, when that amount of rush-in current is distributed over a several millisecond range? In a microsecond, in average, about thousand electrons represents the current; at the beginning and the end of an AP, only a few dozens. These are reasonable questions when measuring voltage gradient that such a low current current generates on a resistor represented by some ion channel.

The interdependent behavior of charge and mass has an interesting, entirely mathematical consequence as well. The definition of a partial derivative of a function of several variables is its derivative with respect to one of those variables, *with the others held constant.* In the case of ions, changing the function with respect to mass or charge means simultaneously changing also the other;

that is, the partial derivatives depend on each other. In other words, one cannot calculate the partial derivatives; only the total derivative. As a consequence, equations that use the partial derivatives of concentration and electrical potential are either incorrect or only more or less good approximations. Classical mathematics cannot be applied. Consequently, in the case of ions and electrolytes, one must not use the well-established concepts (enthalpy, entropy, etc) of thermodynamics in unchanged form. (BTW: the discrete nature of charged particles leaves the question open, under which conditions remains the ion current differentiable.)

Experience shows that thermodynamics describes correctly the macroscopic behavior of physical media when an enormous number of microscopic objects are present in a macroscopic volume. Are the biological objects large enough to neglect the statistical fluctations of the objects they contain? The Nernst-Planck equation (and the corresponding transport equations) relates concentration and voltage gradients. Are those laws valid at the size of ion channels, where it is hard to interpret those derivatives? Do those statistical laws remain valid for quasi-continuous infinite volumes, for non-infinite closed volumes, or for individual ions? Recall that thermodynamics follows a hybrid path that calculates the *probable number* in a volume in the phase space and works that as if it were the *actual number* of particles. Mathematics and physics do not have methods behind the pairwise, instant interactions.

Science has a well-established method for deriving its laws. It groups resemblant phenomena to major disciplines and they branch along even more resemblance to sub-disciplines. The hierarchy is not strict, given that "nature is not interested in our separations, and many of the interesting phenomena bridge the gaps between fields." [9]. Mathematics is a discipline on its own right. It abstracted its notions from nature's concrete objects and phenomena, its abstract methods serve as a solid base for the fellow sciences and are frequently used to describe new discoveries. It is commonly accepted that physics is the very basic tool to study nature at some elementary level. Chemistry, at another level of abstracting nature, is based upon its laws, and constructs its laws for the phenomena it researches. The laws of physics are borrowed and respected, and the laws of chemistry cannot conflict with the laws of chemistry. Biology is the discipline studying living matter, and is considered to be an even higher level abstraction, with its own laws. These laws, as expected, cannot conflict with the laws of chemisty and physics. Howevever, it looks like some laws, mainly the ones belonging to thermodynamics, are are not perfect for describing biological phenomena. Especially, the life itself seems to contradict the second law of thermodynamics. E. Schrödinger suspected that some non-ordinary laws are to be derived to describe life.

Classical physics is simplifying the things until it can attribute a single substance (such as mass or charge) to the tested object and in its laboratories derives accurately the laws and determines the attibutes of the substances. As E. Schrödinger has drawn attention to, "the construction [of living matter] is different from anything we have yet tested in the physical laboratory". Science experienced many times that the used approximations do not describe some phenomena accurately. At that times, scrutinizing the fundamental assumptions (such as whether the Galillei's relativity principle remain valid when approaching the speed of light, or the matter remains continuous when decreasing extremely the tested sample s size) has led to establishing new scientific fields, such as theories of relativity or theory of quantum mechanics. When describing living matter, one needs some more approximations.

5.4.6 Biology

The above items are what E. Schrödinger coined as "*the construction is different from anything we have yet tested in the physical laboratory*" [1]. Consequently, measurements must be designed and carried out with care; the routine methods used for measuring objects from inanimate nature cannot surely be applied to living objects without changing them. We discuss some of the relevant terms and notions of physics, differentiating which approximation is appropriate only for physics (mainly electricity and thermodynamics) and, instead, which approximation should be used for biology. *Biophysics simply translated the corresponding major terminus technicus words from the theory and practice of physics' major disciplines, mainly from electricity, which were worked out for homogeneous, isotropic, structureless metals, and for strictly pair-wise interactions with a single (actually, 'instant') interaction speed; to the structured, non-homogeneous, non-isotropic, material mixtures and for multiple interaction speeds.* Those notions do not always have unchanged meaning, and how much they do, depends on the actual conditions. The precise meaning needs a case-by-case analysis.

A common fallacy in biology is that physics cannot underpin the operation of living matter, citing E. Schrödinger. However, the claim falsifies his opinion by omitting the most essential word 'ordinary'. Schrödinger wanted to emphasize the opposite: there is no new force (no unknown new interaction), only studying living matter needs different testing methods in the physical laboratory. He suggested to answer the question "Is life based on the laws of physics?" affirmatively, but expected to discover the appropriate forms of physical laws describing the 'non-ordinary' (in our reading: non-disciplinary) behavior of living matter. No doubt, the basic notions and terms must be interpreted precisely for living matter, much beyond the level we used to at college level. However, after that reinterpretation, we can interpret features of living matter, although we need a more careful, many-disciplinary analysis to do so. We must use the appropriate abstractions and approximations for the phenomena, depending on the level needed in the given cooperation of objects and interactions.

The physical models consider infinitely large volumes, surfaces, distances; furthermore, and most importantly: instant interactions. Is the cell large enough to consider it infinitely large (at least on the scale using ion's size); that is to apply laws of science derived for the abstraction 'infinitely large'? When working with charge, we know that charge is quantized, while the macroscopic quantities voltage and current are continuous (derivable). Do cells contain a sufficient number of charge carriers to apply macroscopic notions? Do the thousands of times smaller ion channels transfer enough charge to speak about well-defined current? When a couple of ions are transferred through an ion channel, do they change significantly the potential that accelerates them?

Life, including the brain's operation, is dynamic. As Schrödinger formulated, the "construction of living matter" differs from the one science used to test in its labs. The scientific abstraction based on "states" (i.e., on instant changes) fails for the case of biology, where "processes" happen (i.e., the changes are obviously much slower). The commonly used measuring methods such as clamping, patching, and freezing, reduce the life to states (and correspondingly, the related theories describe states with perturbance [21]). On the one side, this technology fixes the cell at some well-defined static state and enables one to observe a static anatomic picture of the cell. On the other, it eliminates the dynamic processes, i.e., *hides for forever the essence of the life that the cell exists in a continuous change governed by laws of motion.* Those methods stop the processes for studying them, in this way killing their dynamicity to be measured. It was forgotten that using feedback for stabilizing an autonomously working electrical system means introducing foreign currents and this way falsifying its operation.

Furthermore, it is hazardous to introduce technically (and incorrectly) derived and misinterpreted macroscopic features and interpret them as fundamental electrical notions. In general, instead of understanding and developing the correct scientific base of the operation, saying that science cannot describe it. The idea of conductance has been introduced to neurophysiology almost a century ago. It was taken from physics, where the notion was derived for metals (conductors, instead of electrolytes). Since then, its original interpretation has been forgotten, and today (in contrast with physics, fundamentally due to the wrong concept of 'equivalent circuit') it has become a primary entity for describing electrical characteristics of biological cells. We explain how the right physics background enables us to discover wrong physical models and misinterpreted notions of physics in neurophysiology; furthermore, how the right interpretation opens the way to the correct interpretation of neuronal information. We set up an abstract electrical model of neuronal operation.

We derive the needed 'non-ordinary laws', which are derived by using the same first principles as the 'ordinary laws', but are abstracted for the approximations valid for living matter. As we discuss, those 'ordinary' laws were derived for strictly pair-wise interactions at very high speeds. In biology, we can observe interactions at million times smaller speed, in inhomogeneous, non-isotropic, structured material. *Biology does not have the conditions for which the ordinary laws of physics were derived.* We show that the ordinary laws are also the result of approximations (including omissions) and by using the appropriate approximations for the biological cases we can derive those 'non-ordinary' laws of physics. Which laws are more complex to derive and need joining different disciplines, furthermore, we must use several stages (with approximations changing from stage by stage) instead of one single stage in the case of the 'ordinary' laws. However, *all laws follow the same principles.*

Biology, and especially neuronal operation, produces examples where wrong omissions in complex processes results in absolutely wrong results. In those cases some initial resemblance between our theoretical predictions and our phenomena exist, but the success in simple cases provides no guarantee that the model was appropriate: "the success of the equations is no evidence in favour of the mechanism" [17]. Finally, all laws are approximations and the accuracy of verifying their predictions is limited. Several theories can describe the same phenomenon with the required accuracy. We also show in the section 5.4.3 about the finite interaction speeds that the mostly known laws (from Newton, Coulomb, Kirchoff, etc.) are also approximations. They have their range of validity, although it is often forgotten.

One such neuralgic point of omissions and approximations is the vastly different interaction speeds; furthermore, that where the speed is considered at all, *the same speed is assumed for all interactions.* The laws are abstract also in the sense that, say, the objects in the laws of physics have either mass or electric charge, but not both. It is the researcher's task to decide which combination of laws can be applied to the given condition. For example, one can assume in most cases that the speeds sum up linearly, except at very high speeds. Biology provides excellent case studies where different interactions shape the phenomenon and special care must be exercised. We give a short review of history and kinds of interaction speeds.

Another point is that science started with the assumption that the non-living matter is continuous, although it was early discovered that there are smallest pieces of that matter. When we approached that size, we experienced that different subsets of science laws describe that matter and the atoms they contain; it is one of the hardest tasks to establish relations between those subsets. Again, we used abstractions that the continuous matter is infinitely large and that the isolated atoms are infinitely far from each other and from the external world. We also experienced the semi-infinite cases, and studied the behavior of surfaces and interfaces; which, again is different from both that of the atoms and their large masses. Given that biological objects are between those microscopic and macroscopic sizes, and they are surrounded by surfaces, we must be prepared that no simple rules describe their behavior.

Neuronal operation is at the boundary where sometimes, in the same phenomenon, one interaction can be interpreted at macroscopic level, some another must already be interpreted at microscopic level. Furthermore, a series of stages (instead of a single state) and processes (instead of stages) describe the subject under study. Given the vital role of charge and current in neuronal operation, we give their precise interpretation. Furthermore, we must consider that the processes happen in a finite volume, "within the spatial boundary". Special emphasis is given to the true interpretation of conductance, one of the central terms in biology.

Chapter 6

Electricity for neurons

The excellent textbook [2] (of course, having HH V1.0 in mind) separates the neuron membrane's states into resting and transient states. The statement that "the resting membrane potential results from the separation of charge across the cell membrane" is only half the truth; we tell the second half in section 6.1. We refine their statement "by discussing how resting ion channels establish and maintain the resting potential" [2], page 126. As we show in section 6.2, two different physical mechanisms operate in the resting and the transient states. A primary reason for misguiding neurophysiology was projecting the mechanism of the resting state to the transient state.

The book [2] asks the central questions, "How do ionic [i.e., electrical and chemical] gradients contribute to the resting membrane potential? What prevents the ionic gradients from dissipating by diffusion of ions across the membrane through the resting channels?" However, it leaves them essentially open by giving only a qualitative answer, discussing only membrane permeability without explaining how the resting potential is created and regulated through charging and polarization. We demonstrate in section 6.2 that classical methods of electricity enable us to calculate the membrane's potential. Our results underpin that "the crucial system in biology isn't a molecule or a molecular class whatsoever, but the interface created by biomolecules in water" [23]. However, we add that mainly physical processes and speed gradients control the processes. More precisely, inseparable thermodynamic and electrical processes keep the potential set by the inseparable electrical and thermodynamic backbone defined by the lipid/protein structure of the cell. It is one of the frequent cases when one effect implements a functionality (the backbone closely defines the frames of the operation), and the other (in this case, combined thermodynamics and electricity) corrects it when it implements a complex operational (dynamical) functionality: "stimulated phase transitions enable the phase-dependent processes to replace each other ... one process to build and the other to correct" [24].

6.1 Membrane's electricity

The importance of the subject was correctly estimated decades ago [56]: "For the past forty years, our understanding and our methods of studying the biophysics of excitable membranes have been significantly influenced by the landmark work of HH. Their work has had a far-reaching impact on many different life science sub-disciplines where concepts of cell biology have come to be important. The ionic currents and electrical signals generated by neuronal membranes are of obvious importance in the nervous system." To fix the flawed and incomplete hypotheses has a similar importance. Recall that if you have the HH model in mind, and measure with a voltmeter, you will surely measure voltage and current, and nothing more (no other quantity). This way the accompanied other changes can remain hidden.

Concerning the origin of the membrane's resting potential, we disagree that "the resting membrane potential is the result of the [low intensity] passive flux of individual ion species through several classes of resting channels" [2]. This claim cannot be true: the flux is not passive (to move the ions, energy required), furthermore, no setpoint is defined, so some (low) current flows due to the interplay of the electrostatically set setpoint and the reversal potential of the ions inside and outside. The claim was already question-marked whether the membrane permeability is a primary property for the membrane potential generation, for a review, see [57]. That interpretation leads to causality issues [15].

It was already stated that the membrane's resting potential is, of course, of electrostatic origin [58, 59, 60, 57, 61]. Paper [62] stated correctly "that separation of charges produces a potential" and derived the thermodynamic part in two different ways, but did not account for the electrical potential. However, the classical electrical potential is only a small part of the actual potential; the rest is contributed by the polarization of the electrolyte's dipoles. Moreover, *the closed volume of the neurons drastically differs from the infinite space. The charges attract and repel each other, but since they experience the counterforce of the limiting surface (the membrane), a static mechanical pressure (furthermore, a shock wave, if the amount of charge on the membrane suddenly changes) are also created, which naturally connect the electrical and mechanical effects, as well as the electrical, mechanical, and thermodynamic theories of operation.* That is, *the 'net electrical' model implies the effects from other disciplines, but does not provide a way to calculate them.* We must discuss them in a cross-disciplinary way; furthermore, we must cross the borderline between particle-based and continuous points of view of nature. Moreover, we must be aware of that the features of living matter may change while being measured in physics laboratories as usual: "the construction is different from anything we have yet tested in the physical laboratory" [1].

When the balanced state is perturbed, the system attempts to find a new balanced state, using temporal processes. In this scenario, the 'carrier' - the ion - can be influenced by two interactions. We discussed elsewhere that those phenomena in living matter need deriving and applying 'non-ordinary' laws of science [63], and the time course of those processes can be described by the laws of motion of biology. Fundamentally, due to the mixing of interaction speeds, *considering only one of the interactions (the disciplinary way) leads to incorrect conclusions and theoretical interpretations.*

6.1.1 Condenser

Cell biochemistry sufficiently well describes the lipid bilayers that constitute the membrane [64, 65]. The negative and positive ends of the lipids that constitute the membrane, in an electrolyte solution, attract and bind ions, forming a charged layer on the membrane surfaces. In this way, the membrane in electrolytes acts as a condenser, comprising two charged sheets on its surfaces, with ill-defined parameters. Their boundaries, thicknesses, and compositions depend on the operating state (unlike conducting plates with well-defined sizes and boundaries). The condenser is not ideal: the charged sheet comprises ions with attached aqueous constituents from the electrolyte, so its thickness is far from the ideal zero thickness of classical electricity. Furthermore, in its wall, it inseparably comprises ion channels (from an electrical point of view: resistors), and the condenser plates are in polarizable electrolyte solutions.

Cell science does not establish the membrane's electrical phenomena; instead, it attempts to elucidate complex protein activities that explain charge transfer, resting and transient potentials, and explains the entire *biological operation using mathematical formulas without a realistic physical background.* Electrophysiology combines currents of electrons and ions, and *uses Ohm's Law for its admittedly non-Ohmic systems.* It introduces foreign (clamping) currents into biological systems, attributing their effects to changes in membrane conductance without explaining how *voltage and current can be independent of charge carriers.*

6.1.2 Resistor

Ions differ from electrons we used to in physical electricity in many features. Their conduction mechanisms are entirely different, and they are not necessarily present in the discussed medium (say, when the medium is surrounded by a charged surface, the field moves free ions out of the volume). They may be produced by biological mechanisms, and they are slow (have a couple of m/s speed, creating the illusion of 'delayed current'). Both disciplines provide an atomic level description, but unfortunately they use the same words for their concepts, although their meaning can be entirely different.

In solids, as the Drude model describes, electrons constantly bounce among heavier, stationary crystal ions, that make up the structure of the material. With each collision, though, the electron is deflected in a random direction *with a speed that is much larger than the speed gained by the electric field.* The net result is that electrons take a zigzag path due to the collisions, but generally drift in a direction along the electric field. Important, that *the collisions are inelastic*, so the electrons lose (most of) their energy. The solid's gridpoints absorb the energy, which later is released in form of heat. The process is irreversible.

In biological matter, ions convey electricity. So called ion channels are used for the transfer. Here diffusion and electromigration takes place, by orders of magnitude different speeds. The process is mostly reversible. The resistance is interpreted as the interaction of charge carriers with the particles in the material that are transported by a driving force (that can comprise also a force of chemical origin). In solid state, the volume considered is "infinitely large" (that is, the field is homogeneous and the propagation is isotropic), and the charge carriers are uniform that is only one type of interaction is present. In biological objects, the volume is more or less limited (is finite), there may be charged objects present that make the propagation non-isotropic. Furthermore, the charge carriers are non-uniform that introduces thermodynamic interaction in addition to the electrical interaction. Ions, under the electrical/thermodynamic field (and maybe mechanical constraints) accelerate to the Stokes-Einstein speed (see Eq.(6.28)), move against a friction in the electrolyte. The ions suffer (essentially elastic) collisions with neutral molecules and transfer their momentum (and so: energy) to them. From the point of view of ions, the process is energy loss and momentum loss. From the point of neutral molecules, the process is energy gain and momentum gain. In a free space, the energy and momentum quickly distribute in the entire volume, so the process in finally irreversible.

If the acceleration occurs in a narrow tube such as an ion channel, the wall prevents gaining momentum in the direction perpendicular to the direction of acceleration. Given that the collisions are elastic, no energy dissipates; so, in principle, the process is reversible. Strangely enough, this way the electrical force accelerates also the neutral molecules; furthermore, their momentum (using an elastic membrane) can fully contribute to the operation of neuron (forms the AP).

In a narrow tube (such as an ion channel) the momentum exchange is limited to one direction. If the ions are accelerated through the channel, due to the frequent collisions, they can entirely transfer their energy to a large number of neutral molecules. Those accelerated molecules having the same speed may represent a significant "impact force", sufficient to cause significant mechanical changes throughout biological systems. The momentum transfer by collisions is a two-way process, it is reversible also in the practice. If a small amount of ionized particles is present in the tube, under an external electrical field, the charged particles can accelerate the neutral ones. Vice versa, if the bulk of the particles are moved mechanically (say, when an ultrasound wave presses the neuron or a rush-in process transfers electric energy to the membrane), the one-directionally moving neutral molecules transfer momentum to the charged particles, this way producing measurable electrical waves (such as an AP).

The secondary electric quantities can be derived from the primary quantity: the bulk of the undissociated molecules 'resist' transferring ions under the effect of an electric field (the liquid must be pressed through the tube). One can measure potential and current along the ion channel (and so, to calculate resistance), but the behavior of the resistance/conductance is drastically different. One has a pneumatic system instead of a voltage generator and the resistance originates in viscosity. Most of the energy is converted to mechanical energy instead of heat, making the process (mostly) reversible.

The AP initially produces an impulse force that compresses the membrane. The compressed membrane, while performing a damped oscillation, pushes and pulls the incompressible electrolyte that contains a small amount of dissociated ions, this way (per defitionem) creating an observable current and voltage wave. In the positive and negative half periods of the elastic oscillation, due to opposite elastic/pneumatic forces, the current and voltage change their sign. Given that the charges move at their Stokes-Einstein speed, the voltage-, current- and pressure waves are synchronized, as observed. The different experimental observations measure the same effect by using their disciplinary measuring devices.

The energy of the ions cannot dissipate; instead, the neutral and charged particles share it, and convert between kinetic and elastic energies (the electric energy is much smaller). That is, the neuron can produce heat in the positive phase of the elastic oscillation (the electrolyte is compressed), and can reabsorb heat in the negative phase (decompression), in full accordance with the correctly interpreted laws of physics. This way, by introducing the correct interpretation of "resistance" for ion channels, the seven-decades-old contradiction [10] between the electrical and thermodynamic observations are finally resolved, furthermore, the conflicting theories can be naturally unified.

6.1.3 Electrolytes

Specific chemical substances can naturally hold either a positive or negative electrical charge and react to their micro-, macro-, and electrical/mechanical/thermodynamic environments. A molecule has internal electrical forces that keep its ions in place, so it has two charge centers (dipoles). When another dipole or a macroscopic external electrical field (which can be of electrical, magnetic, mechanical, or chemical origin) appears near the molecule, its perturbing effect can affect the relation of the ions to each other. The two charge centers initially increase their distance (the molecule polarizes). When that disturbance is strong enough, the ions can entirely separate (the molecule ionizes). The local electrical field fluctuates, so their state is dynamic: molecules dissociate, and free ions recombine. Furthermore, the process also plays an important role in the energy supply of neurons: the strong electric field near the membrane hydrolyzes Adenozine Triphosphate (ATP) molecules, and moves ions to the "condenser plates", thereby accumulating "potential energy". (The claim that the ions pass through ion channels by using the "downhill method" does not use energy, is wrong: they are accelerated by the condenser's field and consume energy in that way.) Depending on their environment, substances can exist in base, ionized, and polarized states (and exhibit corresponding behaviors).

(In physics, the term 'polarization' is reserved for the redistribution of charges in multiply charged systems without generating an external current. Changes in the membrane's potential difference are a consequence of complete charge separation (and the movement of slow charges), even though, unfortunately, physiology refers to it as polarization. As discussed below, true polarization is also present, which confuses the description. External currents contribute to the changes of potential.)

In biological cells, in a resting state, a small (approximately 10^{-3}) fraction of the molecules dissociates (i.e., ions separate from their counterparts with opposite charges) and moves freely within the volume. In other words, an electrolyte liquid can conduct electricity (under the effect of an electrical or thermodynamic field) because the positive and negative ions are mobile. The remaining molecules can be more or less polarized, allowing for the possibility of generating an internal electrical field in the solution. In a transition state, all components (including ions, molecules, and ATP) have gradients, and they move towards their balanced state.

6.1.4 Deriving the "thermodynamic electrical field"

In a segmented electrolyte, electrical charges are typically globally balanced; however, they may become locally unbalanced due to physical factors. **Physics panel 6.1.1** describes how the chemical and electrical spatial gradients depend on each other, **Physics panel 6.1.2** shows a practical approximation for calculating it, with the calculated figures in **Numerics panel 6.1.1**.

Physics panel 6.1.1:
The Nernst-Plack law

a The Nernst-Planck electrodiffusion equation in a balanced state

$$\frac{d}{dz}V_m(z) = -\frac{RT}{q*F}\frac{1}{C_k(z)}\frac{d}{dz}C_k(z) \tag{6.1}$$

describes in one dimension the mutual dependence of the *spatial gradients* of the electrical and thermodynamic fields on each other. In good textbooks (see, for example, [66], Eq (11.28)), its derivation is exhaustively detailed. In the equation, z is the spatial variable across the direction of the changed invasion parameter, R is the gas constant, F is the Faraday's constant, T is the temperature, q the valence of the ion, $V_m(z)$ the potential, and $C_k(z)$ the concentration of the chemical ion. The two sides are the derivatives of the equation

$$V_m = \frac{RT}{q*F}\ln\left(\frac{C_k^{ext}}{C_k^{int}}\right) \tag{6.2}$$

known as *Nernst equation.* In other words, **Eqs.(6.1) and Eq.(6.2) are the differential and integral formulations of the same knowledge**. The limits of the integration are chosen arbitrarily (by choosing the reference concentration C_k^{int}).

Physics panel 6.1.2:
The membrane's eqivalent electrical themodynamic field

a We assume that, in a balanced state, the concentrations on the two sides of the membrane are C_k^{ext} and C_k^{in}, respectively; furthermore we assume that

$$\frac{dC_k}{dz} \approx \frac{C_k^{ext} - C_k^{in}}{d}$$

To simplify the math writing, we neglect the change caused by considering the difference (maybe we should apply a near-unity factor); we use the larger concentration value instead. Using the constants and $T = 300$, we can derive the "equivalent thermodynamic electrical field" across the membrane for the case of a permeable ion channel in the membrane

$$E_{thermal}^{C_k} = -\frac{RT}{q * F}\frac{1}{C_k^{ext}}\frac{C_k^{ext} - C_k^{in}}{d} \approx -\frac{RT}{q * F}\frac{1}{d} \tag{6.3}$$

The equivalent thermodynamic electric field does not depend on concentration, but rather is linear in temperature and inversely proportional to membrane thickness (given in nm). We provide the numerical values in **Numerics panel 6.1.1** for the case of ion channels in the membrane's wall.

In simple words, the Nernst-Planck equation states that the changes in concentrations of ions create changes in the electrical field (and vice versa), and in a stationary state, they remain unchanged. Notice that in the case of an ion mixture, *joint charge-generated electrical fields exist, and they are equal, per ion, to the concentration-generated "electrical fields"*, see section 6.2.1.

In the steady state, some other forces also contribute to the mentioned forces. To discuss how nature restores the steady state when a microscopic change happens in a balanced state of a biological solution, we write the well-known Nernst-Planck equation (see Eq.(6.1)) in a slightly extended form:

$$0 = \underbrace{e_{el} * E_{Gap}^{Total}(c, \Delta z)}_{Electrical\ force} + \underbrace{e_{el} * E_{thermal}^{C_k}(d)}_{Thermodynamic\ force} \underbrace{\big(+F_{ext}(z)\big)}_{Constraint} \underbrace{\big(+F_{Transp}(z)\big)}_{Transport\ force} \tag{6.6}$$

We multiplied the usual two terms by the elementary charge, so its terms are expressed as forces, plus we added a transport-related external force (its role is discussed below). Furthermore, changing one force triggers a corresponding counterforce and/or causes the ion to move in a viscous fluid. (An excellent summary of the possible biology-related forces is given in [68], with a lot of cited references.) This equation, expressing Newton's first law, has no place for

Numerics panel 6.1.1:
Neuron's equivalent thermodynamic electrical field

It is valid separately for all C_k

$$E^{C_k}_{thermal}(d) = -2.585 * 10^{-2}\frac{1}{d} \quad \left[\frac{V}{m}\right] \tag{6.4}$$

Figure 6.1 depicts the field's dependence on the membrane's width and how it compares to the charge-generated electrical field. The figure illustrates the setpoint(see section 6.2.2), the boundaries of the changes, and the interesting phenomena that occur within the shaded cross-section. The figure is an illustration, but its figures are not far from the true ones. For a 5 nm membrane thickness, it results in $5.17 * 10^6\ \frac{V}{m}$, which compares well to the value provided by Eq. (6.18). (Notice that the membrane's thickness may change during the action potential, see [67]: the rush-in extra charge compresses the membrane. Similarly, the concentration of the electrolyte in proximity to the membrane also changes.) Correspondingly, by using that $U = E * d$,

$$U^{C_k}_{thermal}(5) = -25.85 \quad \left[mV\right] \tag{6.5}$$

effects such as "Pumps that maintain ion gradients ... transport ions *against their electrical and chemical gradients*" [2], page 101. It implies some *new force* that moves ions, confronting Schrödinger's science-based opinion: "And that not on the ground that there is *any 'new force'* or whatnot, directing the behaviour of the single atoms within a living organism"[1]. Biophysics still did not provide either the origin of the 'new force' or the reason why Newton's laws are not valid for ions in living matter, which undermines the credibility of those protein-based mechanisms.

Expressed differently

$$-F_{constraint} = Eq.(6.18) + Eq.(6.4) + Eq.(6.30) + External \tag{6.7}$$

That means when describing an ionic transfer process, *we must not separate the electrical current from the mass transfer* (in other words, to discuss electrical or thermodynamic models in a disciplinary way): they happen simultaneously and mutually trigger each other. Notice that *the thermodynamic term is ion-specific while the electrical term is not.* To be entirely balanced, the system must be balanced to all elements. This way, changing one concentration implicitly changes all other concentrations and the electrical field.

Notice the special role of the constraint force by the closed volume represented by the membrane's wall. The thermodynamic contribution also means a pressure from the colliding particles, and the electrical repulsion exerts a force

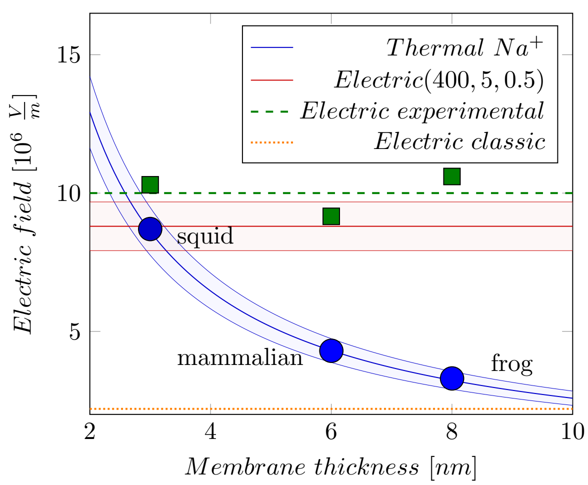


Figure 6.1: The thermodynamic "equivalent electrical field" depends on the membrane's thickness; see Eq.(6.4). The electrical gradient depends on the charged layer's concentration and thickness; see Eq.(6.18). The dots mark measured data shown in Table 6.1.

on the membrane's wall. Those disciplinary forces act simultaneously and inseparably.

6.1.5 One segment

For the discussion below, we assume that the segment has a two-dimensional surface boundary, and we discuss the gradients along a line perpendicular to that plane's surface. We neglect the electrokinetic and surface-related electrical effects connected to membranes and nanopores [59] and focus on the balanced state of membranes in ionic solutions. We compose the segments from such layers (sheets), (x, y) parallel plates, and describe the gradients in the direction of z. We introduce the concept of a 'thin physical layer' parallel to the membrane, characterized by a finite width. The detailed calculations are shown in **Numeric panel 6.1.2**.

6.1.6 Two segments

Now, we split the volume into two segments and compose them from 'thin physical layers' (sheets) with different potentials (and concentrations). We consider

Physics panel 6.1.3:
The effect of Nernst-Plack law of the GHK equation

a

Warning The potential derived in **Physics panel 6.1.1** inherently comprises an additive term (a potential difference), so they must not be compared directly if they use a different reference C_k^{int} (furthermore, it is nonsense to combine quantities from the intracellular and extracellular sides additively, whether concentrations, mobilities, or Nernst voltages, as it happens in the Goldman-Hodgkin-Katz (GHK) equation (see section 6.2.3).
The derivatives are spatial derivatives; the temporal derivatives (needed for describing the time course) of the concentration(s) and voltage are derived in [63]. In 'ordinary' physics, where the charge and mass are independent, changing one side changes the other in the opposite direction. In 'non-ordinary' physics [20], valid for electrolytes, the two differentials must change in the same direction, given that ions' charge and mass cannot be separated (BTW: this is why Eq. (6.2) results in a negative potential difference). (The derivation in 'ordinary physics', such as [66], Eq.(11.30), is wrong, at least because of two reasons: the same speed is assumed for mass and charge transport; furthermore, the used partial derivatives are not independent; the details are discussed throughout the paper.)

the immediate environment of the neuron's membrane as adjacent parallel (x, y) plates and find the electrical field's z component of those plates in the points as shown in Fig. 6.2. **Physics panel 6.1.4** discusses the difference between the classical and biological condenser. Furthermore, **Physics panel 6.1.5** discusses the issues with the ill-defined biological boundaries.

Charge layers in segments

As we show, the finite thickness will result in a lack of balance (introducing inhomogeneity and creating a voltage and concentration gradient) proximal to the surfaces of the membrane, even if the concentrations on the two sides are the same. Changing the bulk concentration or potential in one of the segments creates a corresponding gradient across the separating membrane, increasing the inhomogeneity proximal to the membrane. The ions will experience an extra force due to the gradient; however, the mechanical counterforce of the membrane will keep them back in the segment. The *concentration and potential, inseparably and having the same time course*, will change across the two sides

Numerics panel 6.1.2:
The electric field of a physical charged sheet

In the calculations below, we require the 'surface charge density' (interpreted as an 'infinitely thin layer' in physics, expressed in $C*m^{-2}$). We can derive (assuming singly-charged ions) the *volume charge density*

$$\sigma_V(c) = c*N_A*e_{el} = c*6.023*10^{20}*1.602*10^{-19} = 96.5*c \quad \left[\frac{C}{mM*m^3}\right] \tag{6.8}$$

where N_A is the number of ions in a *millimol* and e_{el} is the elementary charge. Notice that all ions contribute to the charge density, so simply $c = \sum_k C_k$.

Here, we run into a conflict between the 'infinitely thin' layer of physics and the biologically implemented (finitely) 'thin physical layer', so we use a thickness parameter Δz. By assuming an arbitrary 'physical layer thickness' Δz, we can calculate the *surface charge density*

$$\sigma_A(c, \Delta z) = 96.5*c*\Delta z \quad \left[\frac{C}{m^2}\frac{1}{mM*nm}\right] \tag{6.9}$$

We interpreted classical physics' notions from the theory of 'continuous' electricity, interpreted for 'infinite' cases, for the finite world of biological objects and the quantized atomic world. Our idea is similar to Boltzmann's for connecting the continuous medium to particles. When ions are present in an electrolyte having concentration c in a 'conducting layer' of thickness Δz on an isolating surface, they produce a field

$$E_z(c, \Delta z) = \frac{\sigma_A(c, \Delta z)}{\epsilon_o} = 10.89*10^3*c*\Delta z \quad \left[\frac{V}{m}\frac{1}{nm*mM}\right] \tag{6.10}$$

of the membrane, just because of the gap's physical features the membrane represents.

We consider that the segment is composed of electrically conducting discs (the ions are free to move on the surface), and the charged discs' contribution to the electrical field at point P (see Fig. 6.2) can be calculated as known from the theory of electricity. Due to the symmetry in the z direction, in a homogeneous solution, the resulting electrical field in the plane perpendicular to the z direction is zero, as we have contributions of equal magnitude with opposite signs.

Notice that here we used the abstraction of an infinitely thin "charged sheet", and there is a step-like gradient in the electrical field in the gap. The physical reality is that the charge is carried by ions, which simultaneously carry mass. In equilibrium, the forces due to the voltage and concentration gradients must

Physics panel 6.1.4:
The classical condenser

a In classical electricity, the charge remains on the surface of the conducting layer, and the distribution is homogeneous (although the charge is concentrated on the surface and the electrical field is zero inside the conducting layer). Hence, the electrical field contributions cancel each other. In electrolyte segments, (a tiny fraction of) the molecules decompose into an ionized state (dissociate), and the created ions interact with a bounding membrane using a not entirely understood mechanism [69]. The rest of the dipole molecules become polarized but do not dissociate, forming virtual charges that allow the electrolytes to contain electrical fields. Although not explicitly stated, we assume that electrical double layers are present near the membrane, and ions from the dispersion medium are adsorbed onto the particle's surface, depending on the medium's chemical properties.
Correspondingly, the theory non-equivocally explains the potential near the membrane surface; see the discussion and review in [70], mainly their Figure 1. That model considers a symmetrical condenser, that is, both plates contain charges of the same sign. The Donnan function is a (discontinuous) step-like potential. Its continuous version introduces a mathematically smooth, non-physical function at the boundary of the charged layer. One must describe a physically discontinuous layer (electrolyte, ion layer on the membrane's surface, and isolating membrane) with a single continuous function and gradients (potential and concentration) with physical meaning.

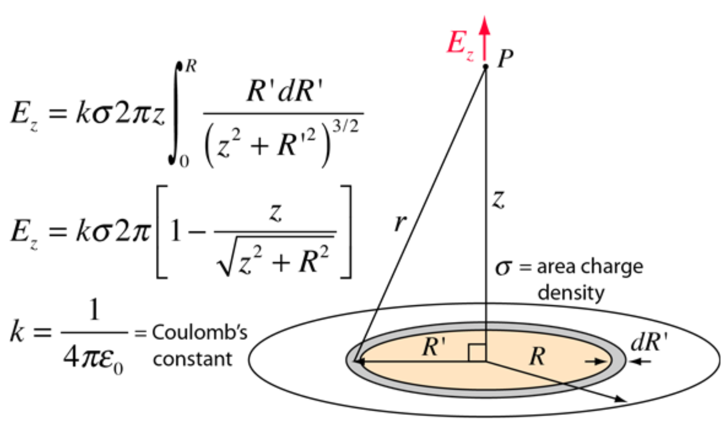


Figure 6.2: The electrical field of a disc of charge can be found by superposing the point charge fields of infinitesimal charge elements. This field can be facilitated by summing the fields of charged rings. "From Hyperphysics by Rod Nave, Georgia State University". ©hyperphysics.phy-astr.gsu.edu/hbase/

Physics panel 6.1.5:
The boundary problem of a physics condenser

a The behavior near the boundary layers is problematic, primarily because the layers themselves are ill-defined. Physically, they are not ideally plain and well-defined surfaces; the layer thicknesses are limited by the size of the ions/molecules and their associated ions; the layers do not necessarily satisfy the ideal requirements of the respective physical laws. Mathematically, a single 'universal' function that spans all regions surely cannot accurately describe the behavior of the physical model. So, we chose a piecewise description where the pieces seamlessly fit at the boundaries. We assume that between the surfaces of the membrane, classical electricity describes the electrical field; the charged layer and the electrolyte jointly produce the electrical field; we have a uniformly charged layer from separated charges on the surface and a polarized electrolyte outside them.
Unlike in the classical theory, the electrolyte segment next to the surface layer contains dipole molecules that have more or less balanced charges (the polarization also depends on the external electrical field), so they have much less mobility than the dissociated ions: their size and mass are bigger, and their charge is a fraction of that of the ions. Their virtual charge becomes apparent only in the appropriate environment, yet it generates an electrical field similar to that of a real charge. The final reasons for creating virtual charges are real charges and/or external fields, so an electrical field due to virtual charges can also exist inside dielectric matter. We assume the membrane is transparent for the electrical interaction (the electrical field affects the ions in the other segment on the other side of the membrane) but not for their masses (mechanically separate the segments). The membrane does not need to generate any counterforce (except for collisions).

be counterbalanced by an external force.

Simple conducting sheet (a physical condenser)

First, we consider a condenser that obeys the laws of classical electricity and has the geometrical size of a biological neuron. We apply the physics terms to that pseudo-biological object and test whether the values we derive are consistent with physiological phenomena. A tiny (according to [6], page 12, about $2*10^{-3}$) portion of the positive ions leaves their negative counterions behind and forms a thin positively charged ion layer on the surface of the membrane (of course, the same happens on the opposite side with negative ions). Experience shows that – also in biology – two skinny layers of ions are formed on the two sides of the membrane of finite width, see the caption of figure (Fig. 11.22 in [45]):

"The ions that give rise to the membrane potential lie in a thin (< 1 nm) surface layer close to the membrane" (As shown below, these ions give rise only to part of the potential, and the dipoles in the electrolyte contribute the rest). A sub-nanometer-thick conducting layer covers the two surfaces at the top of a 5 nm-thick insulator. So, we model the cell with two finite-width "conductive sheet" layers. These layers represent an electrical condenser (so we can calculate the internal electrical field between the plates) and counterbalance each other's electrical field outside the condenser. We may assume that the solid surface provides the needed counterforce to keep the charges at rest.

We cannot expect geometrically flat surfaces, as some biological objects of up to 1 nm in size are present on the surface. So, we must assume a physical 'uniformly charged sheet' with an average thickness $\Delta z = 0.5$ nm on the surface. For the sake of simplicity, we assume a step-like function for the electrical field. It is zero in the segment outside the "conducting sheets" (the 'bulk' portions) and inside the sheets. Between the sheets, it jumps to the value of the electrical field of a charged sheet with surface density σ_A (see Eq.(6.9)). (In this picture, we see a jump in the value of the electrical field on the two sides of the physical layer. The physical layer emulates the well-defined boundary on the side opposite to that proximal to the membrane. Here, we still assume that a mechanical counterforce due to the surface's roughness keeps the charges inside the charged layer.)

Numerics panel 6.1.3:
The voltage in a classical neuronal condenser

We assume that all unbalanced (dissociated) ions are in the layer, and the rest contains no dissociated ions; furthermore, the ion concentration is unchanged in the segment. In the case of a 400 mM solution and a 0.5 nm 'thin physical layer' Eq.(6.10) evaluates to an electrical field

$$E^{Gap}_{Classic}(400, 0.5) = 0.22 * 10^7 \left[\frac{V}{m}\right]$$

and at a 5 nm membrane thickness, the voltage across the plates becomes

$$U^{Gap}_{Classic}(400, 0.5, 5) = E^{Gap}_{Classic}(400, 0.5) * 10^{-3} * d = 10.84 \left[mV\right] \quad (6.11)$$

The number of uncompensated ions needed for the cell, using the method in [6], is $0.88 * 10^7$. The calculated values are about a factor of 5 lower than the experimentally derived values. "An electrical potential difference about $50 - 100$ mV ... exists across a plasma membrane only about 5 nm thick, so that the resulting voltage gradient is about $100,000$ V/cm" [45]. "The number of uncompensated ions needed for the cell is $4.7 * 10^7$" [6].

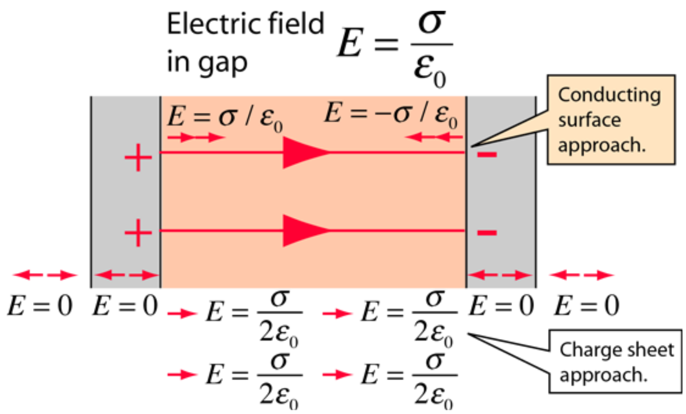


Figure 6.3: The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then one can use Gauss's law to calculate the electrical field between the plates. Presuming the plates are at equilibrium with zero electrical fields inside the conductors, the result from a charged conducting surface can be used. "from Hyperphysics by Rod Nave, Georgia State University". ©http://hyperphysics.phy-astr.gsu.edu/hbase/

Based on the calculations using the classical model, the numerical figures **Numerical panel 6.1.3** suggest our simple model to be a strong oversimplification. We are in the correct order of magnitude, but we arbitrarily assumed a layer thickness, a non-dielectric medium, and a physically unreasonable step-like electrical field. The deviation from the expected values suggests that dipoles in the electrolyte bulk significantly alter the achievable electrical field and potential across the plates.

A usual parallel plate condenser (shown in Fig. 6.3) can be derived with such a model. We have two infinite conducting sheets and an insulator layer between them. The magnitude of the opposite charges on the two plates must be the same as in the classical picture. *Given that the amount of surface charge densities is concluded from the ionic concentration in the segment (in a balanced state), the sum of ionic concentrations of the neuron must be the same in the segments.* A constant electrical field is present inside the membrane (across the plates of the condenser), and there is zero electrical field inside the parallel conducting plates and outside the condenser. In this ideal picture, the charges are aligned along the boundary between two (infinitely thin) conducting layers and cannot move. The attractive force between them and the opposite charges on the other plate keeps them fixed in the direction of z, and the repulsive force between the charges with the same sign, the surface density in the (x, y) plane remains uniform: the infinitely thin plates are equipotential. This picture illustrates the equilibrium state of charges, consisting of infinitely small non-dissociating charge carriers and perfectly smooth surfaces.

Condenser in dielectric material

In its integral form, Gauss's flux theorem states that the flux of the electrical field out of an arbitrary closed surface is proportional to the electrical charge enclosed by the surface, irrespective of how that charge is distributed. The surface layer represents a steep potential and concentration gradient. Above, we assumed that the counterforce that keeps the charges in their place against their electrical attraction on the membrane's surface is a mechanical force: the ions cannot pass through the membrane. However, such a counterforce does not exist on the side toward the 'bulk' part of the segment. The charges on the plates do not generate an electrical field toward the bulk, but the concentration they represent does, according to the Nernst-Planck equation, provided there are charge carriers in the bulk. We hypothesize that virtual electrical charges exist in the electrolytes, and their field provides the missing electrical field.

When explaining the effect of dielectricity, we must explicitly consider the duality of ions that they obey laws of electricity and thermodynamics *simultaneously*; furthermore, the complexity of the electrical structure of the solution and that the charge carriers have finite size (see "electron size" vs. "dipole size"). According to the theory of electricity, the free charge carriers (the dissociated ions) are located on the surfaces where they form powerfully charged thin layers (condenser plates) in the segments separated by a membrane on the two surfaces of the membrane; two proximal charge layers on the surfaces of the membrane. We assume that those ions behave as point charges, i.e., due to the attraction from the opposite charges on the opposite plate, they do not produce an electrical field toward the side of the bulk of the electrolyte layers. In the classical picture, those layers represent step-like gradients in the electrical field along the z axis, and the constraint that ions must not enter the membrane provides the necessary counterforce.

From a thermodynamic point of view, a driving force acts as long as a concentration gradient between neighboring layers exists. From an electrical point of view, the layers are simple parallel-plate condensers that produce no electrical field outside their closed volume and are connected serially; plus, the top layer has free ions. The electrical field is proportional to the bulk concentration in the layer. Changes in the local electrical gradient may also alter the degree of dissociation and polarization, producing a graded local electrical field. These two driving forces have opposite directions and in a balanced state, the same magnitude. The counterforce acts in a way that constrains the molecules to stay in their layers; it adapts (also compensates for the different interaction speeds) while the gradients are changing. It presses the proximal layer against the membrane and the neighboring layers against each other.

Due to the membrane's finite width and surface roughness, the conducting layer is also finite. The electrical field is homogeneous within the layers. The rest of that electrical field toward the bulk layer directs the dipoles proximal to the neighboring dipole layer, and their polarization increases; in other words, they "produce" an electrical field. The final effect is that a dipole layer is attracted to the charged layer, and it forms another layer that shows a charged layer toward the bulk. The process repeats and results in a decaying electrical field based on the distance from the membrane's surface.

In our model, we build the volume of the electrolyte from thin layers of directed dipole molecules (the size of molecules limits the thinness Δz) in the volume, separated into two segments with electrolytes by a membrane with a finite thickness d (that is, a point's distance in the electrolyte from the two disks will be z and $z+d$, respectively). The contributing potential of an infinitesimal volume on the axis at point z due to the charged sheet is

$$dU(c,\Delta z,z) = \underbrace{E^{Gap}_{Classic}(c,\Delta z) * z}_{Potential\ from\ sheet} * \frac{1}{z^2} = E^{Gap}_{Classic}(c,\Delta z)\frac{1}{z}$$

The total potential due to all elements dz from the left side is (in the mathematical formulas below, we express the distance in units of $\frac{z}{d}$, so here z is dimensionless).

$$U_{left}(c,\Delta z,d) = E^{Gap}_{Classic}(c,\Delta z) * d\int_{-\infty}^{0}\frac{1}{z}dz = E^{Gap}_{Classic}(c,\Delta z) * d\Big[\ln|z|\Big]_0^{\infty} \tag{6.12}$$

In the case of a single-segment volume, within the segment, a similar potential with an opposite sign is generated by the charges on the neighboring sides of the considered charged sheet, counterbalances the potential described by Eq.(6.12). However, when a membrane with thickness d separates the segments (with no charges in the gap), the potential on the right side will be

$$U_{right}(c,\Delta z,d) = E^{Gap}_{Classic}(c,\Delta z) * d * \Big[\ln|z+1|\Big]_1^{\infty} \tag{6.13}$$

We can write the electrical field in the form

$$E(c,\Delta z,d) = E^{Gap}_{Classic}(c,\Delta z) * ln\left(\frac{z}{z+1}\right) \tag{6.14}$$

That is, the gap sets the potential difference across the membrane to

$$U(c,\Delta z,d) = E^{Gap}_{Classic}(c,\Delta z) * d * \left(\Big[\ln|z|\Big]_{-\infty}^{0} - \Big[\ln|z+1|\Big]_1^{\infty}\right) \tag{6.15}$$

We use the approximation that $\ln(\infty) \approx \ln(1+\infty)$, and we arrive at that

$$U(c,\Delta z,d) = E^{Gap}_{Classic}(c,\Delta z) * d * \left[ln\left(\frac{z}{z+1}\right)\right]_0^1 \tag{6.16}$$

describes the potential across the plates; that is, the classical potential shall be multiplied by the result of the integral. It was experimentally confirmed [71] that the potential of the membrane is proportional to the logarithm of the distance from the membrane.

Given that the same number of charged ions must be present on the two sides of the membrane, their resultant surface density σ_A must be the same. Notice that the effect is purely electrostatic, resulting in an asymmetric ion distribution; in that balanced state, no permeability is needed. If the membrane is (slightly) permeable, ions will move across the membrane until equilibrium is reached. The resulting potential difference is directly proportional to the concentration difference.

The classical condenser has an electrical field, as the blue dashed line shows in Fig. 6.4. No field is outside the condenser or inside the condenser plates (see Fig. 6.3). There is a sudden jump in the field on the conductor's surface (the armatures) and a constant field between the plates. (The charged layer can be ideally thin if electrons are sitting on the surface, but is of finite thickness on the opposite side where ions are on the surface. This latter effect is not discussed in classical electricity.)

The biological condenser behaves differently; see the red diagram line. An electrolyte (instead of an insulator) outside the condenser significantly alters the field's structure. The electrical field on the surface and between the plates changes by a factor of 3-4. Outside the plates, the polarization creates a field that changes logarithmically. An ion layer with finite thickness is built near the membrane's surface. It is uniformly charged, so the field inside it is linear. At the internal surface of the condenser, it takes the field value calculated for the gap; on the other side of the layer, it takes the logarithmic field value at that position. The dashed line represents the gap field, the continuous line represents the charged layer's field, and the dotted line represents the electrolyte's field. We observe the "correspondence principle": discrete and continuous fields join seamlessly; furthermore, as the electrolyte's polarizability decreases toward zero, the field approaches the classical value.

6.1.7 Bridging the micro- and macro views

We assumed that the function can be interpreted for regions $(-\infty, 0)$ and $(1, +\infty)$. However, we have arrived at the boundary between particles and continuous electricity. We use a trick similar to the one Boltzmann used in his famous equation, except that we calculate the number of charge carriers from geometrical instead of statistical assumptions. The minimal thickness of layers is limited, at least by the size of ions. Furthermore, the membrane's surface is not flat; there are structures (mainly proteins/lipids) with sizes of up to 1 nm, so it is likely realistic to consider a layer thickness of 0.5 nm as we did above (on our mathematical scale $0.1 = \frac{0.5\ nm}{5\ nm}$), for which a multiplier of 1.7 between the classical and the electrolytic fields exists. A similar calculation using a layer thickness of 0.1 nm (on our mathematical scale, 0.02) results in a multiplier of 3.2; so we use a multiplier of value 3 (that corresponds to a layer thickness of 0.125 nm (the radius of a Na^+ ion is 0.116 nm)). Correspondingly, we assume an electrical field

$$E^{Gap}_{Dielectric}(c, \Delta z) = 3 * E^{Gap}_{Classic}(c, \Delta z) \quad \left[\frac{V}{m}\frac{1}{nm * mM}\right] \tag{6.17}$$

due to the dielectric properties of the segment. This contribution arises from the dielectricity in the segment (for simplicity, we have neglected the relative permittivity). It is to be added to the value of the classical contribution, the field generated by the conducting plates, so the final multiplier is 4. (The experience is known in technical electricity: the electrolytic capacitors *achieve several times higher charge storage capacity by using (pseudocapacitance)*. Using "roughened anode foil" increases the electrolyte thickness, and the roughening provides the necessary mechanical support. A thicker electrolyte layer wraps the condenser plates, and the dipoles in the thicker electrolyte provide an additional charge storage facility.)

Correspondingly,

$$E^{Total}_{Gap}(400, 0.5) = 4 * E^{Gap}_{Classic}(400, 0.5) = 8.8 * 10^6 [\frac{V}{m}] \tag{6.18}$$

and at a 5 nm membrane thickness, the voltage across the plates becomes

$$U^{Total}_{Gap}(400, 0.5, 5) = E^{Total}_{Gap}(400, 0.5) * (5 * 10^{-9}) = 45 \; [mV] \tag{6.19}$$

"The plasma membrane of all cells, including nerve cells, is approximately 6 to 8 nm thick and consists of a mosaic of lipids and proteins." [2], page 71. Using such a value for the membrane's thickness may yield a thickness up to 60% higher. Furthermore, the usual concentration is also up to 10% higher. Hodgkin, in 1964, measured molarity values in squid axons for ions K^+, Na^+, and Cl^-, (400,50,40-150) inside and (20,440,560) outside, and they provided potential values $55 - 75$ mV [6]. Given that we used a plausible but ad hoc "charged layer thickness" and gap distance, we cannot expect a better agreement. However, our qualitative conclusions remain valid and can be adjusted quantitatively to the results of dedicated measurements.

We have the boundary conditions that we know the electrical field at the membrane's surface through the volume charge density, and the Nernst-Planck equation delivers the change of the electrical field due to the change in concentration. The physical constraint is that the layer thickness cannot be infinitely small. If we use a parameter for layer thickness equal to that of the surface layer, the calculated electrical field distribution will be scalable.

6.1.8 Ionic currents in a cell

On a microscopic scale, a charge generates a potential field that acts on other charges. It is a common fallacy in physiology that a "potential field travels": neither potential nor current changes without a physical movement of charges. In a conducting wire filled with ions (the solids have a different current transfer mechanism), there are free charges; their number per unit volume is given by

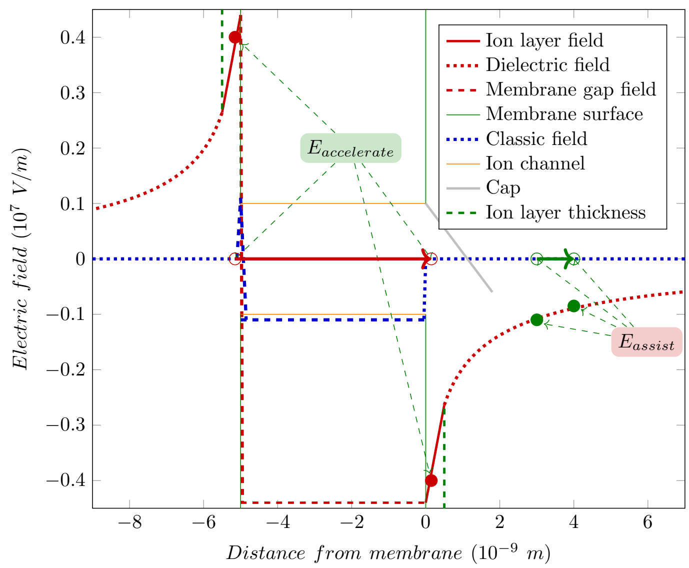


Figure 6.4: The neuronal membrane's *electrical field* in the function of the distance from the membrane's surface. The thickness of the 'atomic layers' proximal to the membrane's surfaces are also shown. The electrical field is linear inside the uniformly charged atomic ion layer on the surface and is described by Eq.(6.14) in the electrolyte. For visibility, a layer thickness 0.5 nm is assumed. For illustration, a simple capped ion channel in the membrane's wall is also displayed.

n, and q is the amount of charge on each carrier. If the conductor has a cross-section of A, in the length dx of the wire, we have charge $dQ = q * n * A * dz$. If the charges move with a macroscopic speed $v = \frac{dz}{dt}$, at the macroscopic level, we define the current I as the charge moved per unit of time as

$$I = \frac{dQ}{dt} = q * n * A * v \tag{6.20}$$

The effect was hypothesized at the very beginning: "it seems difficult to escape the conclusion that the changes in ionic permeability depend on the movement of some component of the membrane which behaves as though it had a large charge or dipole moment." "it is necessary to suppose that there are more carriers and that they react or move more slowly" [17]. The charged ions, attracted by the opposite charges on the other plate and repelled toward the single exit point AIS, really move slowly and behave as if they were a component of the membrane.

Notice that if any of the factors is zero, the macroscopic current I is zero.

Microscopic carriers must be present in the volume, and have charge, the cross section must not be zero, and the charge carriers must move with a potential-assisted speed, which needs an electrical field generated by the system, see Eq.(6.7). When there is no constraint force, the thermodynamic and electrical forces will produce an effective force that moves the ions. That is, the ions can move without an external electrical field, as experienced in biological systems. One of the fundamental mistakes by HH was neglecting the Coulomb force in ion-electrical interactions and assuming that diffusion processes or ideal electrical batteries create the electrical potential.

In biological systems, where ions represent the "current", it is either a native current (without an external potential) or an artificially injected current or potential generating a current. This way, the current is consistently accompanied by a change in concentration gradient, as the moving ions represent both mass and charge transfer simultaneously (also, the repulsion leads to a "skin effect": in electrical wires and axons, the charge travels near the outer surface). The potential and current are connected through the features of the medium (material) that hosts our measurement.

According to Stokes' Law, to move a spherical object with radius a in a fluid having dynamic viscosity η, we need a force

$$F_d = 6 * \pi * \eta * a * v \tag{6.21}$$

(drag force) acting on it. A (microscopical) electrical force field $\frac{dV}{dz}$ inside the wire would accelerate the charge carriers continuously

$$F_{electric} = k * E(z) * q \tag{6.22}$$

with a constant speed v. It is not the *drift* speed because of the electrical repulsion, it is a *potential-assisted* or *potential-accelerated* speed that can be by orders of magnitude higher. The medium in which the charge moves shows a (macroscopic, speed-dependent) counterforce F_d, which in steady state equals $F_{electric}$, that is :

$$I = \frac{k * q^2 * A}{6 * \pi * \eta * a} * n * E(z) \tag{6.23}$$

We extend the original idea by using

$$E(z) = E_{electric} + E_{thermal} + E_{external} \quad \textit{in electrolytes}$$

that is, for living matter

$$\begin{aligned} E(z) &= \frac{\Delta V}{\Delta z} & \textit{in linear potential} \quad (6.24)\\ &= \frac{dV}{dz} & \textit{in graded potential} \quad (6.25)\\ &= E_{electric} + E_{(thermal)}(+E_{(external)}) & \textit{in electrolytes} \quad (6.26) \end{aligned}$$

The amount of current in a wire is not only influenced by the electric force field (specific resistance) but also by the number of charge carriers n. While the latter is commonly considered constant and part of the former, this is not necessarily the case for biological systems with electrically active structures inside. The medium's internal structure ("the construction) introduces significant modifications. *Applying an electric field to a biological wire can generate varying amounts of current as the number of charge carriers changes due to biological effects.* For axons, we use a single-degree-of-freedom system, a viscous damping model, so the *ions will move with a field-dependent constant velocity in the electrical field*; so it takes time while they appear on the membrane. The activity of potential-controlled ion channels in its wall may change n in various ways; furthermore, that change can result in 'delayed' currents during measurement, for example, in clamping, as physiological measurements witness.

If we have a concentration $C(z)$, in the volume $A * dz$, we have $dQ = C(z) * A * dx * q$ charge, resulting in another expression for the current

$$I = \frac{C_k(z) * A * dz * q}{dt} = C_k(z) * A * q * v \tag{6.27}$$

By combining equations (6.23) and (6.27):

$$v(z) = \frac{k * q}{C_k(z) * \eta * 6 * \pi * a} * E(z) \tag{6.28}$$

The higher the resulting space derivative (gradient) and the fewer ions that can share the task of providing a current, the higher the speed. We hypothesized (it needs a detailed simulation) that in the case of this charged fluid, the electrical repulsion plays the role of 'viscosity' [63]. The higher the charge density, the stronger the force equalizing the potential; so η is the lower, the higher the charge density (proportional to C_k). For the sake of simplicity, we assume that the speed is proportional to the space gradient of the local concentration and potential. Recall that *our equations refer to local gradients only. The electrical gradient can propagate only with the speed of the concentration gradient*, given that only the chemically moved ions can mediate the electrical field [63]. *The lower interaction speed limits the other interaction speed if the interactions generate each other.*

The dependence of the diffusion coefficient on the viscosity can be modeled by the Stokes-Einstein relation:

$$D = \frac{k * T}{6 * \pi * \eta * a} \tag{6.29}$$

so we can express the speed with the diffusion coefficient D and temperature T

$$v(z) = \frac{D}{T} * \frac{q}{C_k(z)} * E(z) \tag{6.30}$$

We introduce *a time-dependent speed gradient $v(z,t)$ which plays a vital role in the processes that occur in living matter.*

It is important to remember for alternating current experiments: the different ions will move at different speeds under the effect of the same $E(z,t)$, which is the function of the local concentrations and the frequency of the alternating current. The conduction speed sensitively depends on the temperature, and through it, the shape parameters [72] of the AP and especially the length of the "relative refractory" period [73].

6.1.9 Imitating slow currents

At the dawn of finding methods for describing neuronal operation, HH published high-precision measurements [17] enabling detailed testing of theories explaining the observed physiological behavior. *Their good physical model that* "movement of any charged particle in the membrane should contribute to the total current" *lacked considering the mutual repulsion and finite speed* of those particles (they are about 50,000 times heavier and a million times slower than the electrons, and they move in a closed space segment). Furthermore, they have started from the commonly used wrong assumption that conductance is a primary electrical entity. They could "find equations which describe the conductances with reasonable accuracy and are sufficiently simple for theoretical calculation of the AP and refractory period". The "reasonable accuracy" and the need for a simple calculation method (performed on a mechanical calculator) obscured the fact that the laws for electrons are not the same as the laws derived for ions.

As discussed in connection with Eq. (6.20), the current depends linearly on the number of charge carriers n. The physical processes changing the number of charge carriers also define the intensity of the currents, providing a way to imitate slow currents by fast currents (by using current generators using special function shapes), where the spatiotemporal time course of the physical process of producing ions in the system modulates the number of charge carriers.

As an example, we discuss the case of slow current traveling in an axon. The ions travel a finite distance with a finite speed, so there must be a delay between their entry and exit times. The case of switching a clamping voltage on, to some measure, is analogous to the arrival of a spike. Initially, the axon contains no ions. In the classical picture, the axonal current flows into the membrane with capacity C_m and increases the membrane's voltage V_m with a time constant

$$\frac{dV_m}{dt} = -\frac{1}{C_m} I_{axon};\ V_m(t) = \frac{I_{wall} * (1 - e^{-\alpha * t})}{C_m} \tag{6.31}$$

that generates a change in the membrane current

$$\frac{dI_m}{dt} = \frac{1}{R_m}\frac{dV_m}{dt};\ I_m^{on}(t) = g_m(V)V_m(t) \tag{6.32}$$

where $g_m = \frac{1}{R_m}$ is the conductance of the membrane. That is, the measurable current equals the product of the conductance and the clamping voltage. Equs.*(3)-(5) in [17]* express this relation. *If we assume that the axonal current is "fast", we arrive at the wrong conclusion that the conductance is voltage- or time-dependent.* In contrast, if we assume that the axonal current is "slow", we naturally conclude that Ohm's Law is correct and valid also for biology: *the* *conductance/resistance is constant, but the number of charge carriers changes*, see section 6.1.8.

There is no voltage-dependent conductance [74]. Instead, the finite speed of ions and the wrong assumption that conductance is a primary entity mislead physiological research. *With wording that "conductance changes", one states that charge carriers appear/disappear/reappear; that is, the charge and mass conservation are not fulfilled.* Claims such as "the current cannot disappear, it has to go somewhere" [66], page 9, are wrong: *there is no conservation law for current, which is the derivative of charge with respect to time.* The correct physics background behind the early phenomenon [75] is that the number of charge carriers, at the cross section where the measurement is carried out, changes as a wave of ions with variable intensity moves slowly along the axon (they are apparently "created" for the static measurement arrangement). The changing conductance is a fallacy stemming from the idea of "fast current", which is valid for electrons but not for slow ions.

In contrast, when the clamping voltage is switched off, the axon is still filled with charge carriers. The resting potential reaches the end at the membrane "instantly" (in this situation, the conduction mechanism is similar to that in metals). The driving force disappears, the ion stream stops, and the ions discharge into the membrane. The lack and the presence of ions in the axon when switching clamping on and off, respectively, produce the difference that "*conductances [actually, currents] increase with a delay when the axon is depolarized but fall with no appreciable inflexion when it is repolarized*" [17]. The potential is equalized by the AIS current, producing a net exponential decay:

$$I_m^{off} = I_{Wall} * e^{(-\frac{\alpha}{R_m C_m} * t)} \tag{6.33}$$

During the regular operation of a neuronal membrane, after opening the ion channels, a vast amount of ions flows into the intracellular space from the extracellular space, somewhat similar to the effect of switching a clamping voltage. The essential difference is that the ions arrive through the axon to the joining point in clamping. In contrast, through the membrane's ion channels, they directly contribute to the current on the membrane's entire surface. The membrane's size is finite, so with a finite current speed, it takes time until the charges on the membrane's surface arrive at the AIS, as discussed for the axonal current. Fig. 6.5 shows the measured rising and falling edges of the clamping experiment performed by HH [17]. The continuous lines show the diagram lines fitted to the experimental data (notice again that the fitted polynomial significantly differs from the correct exponential function around the time 0.5 msec); the dashed line shows the expected result of the rising edge, calculated with the time constant of the falling edge. The number of charge carriers follows a saturating curve, which decreases the current in the axon. HH measured that their time constants differ far outside their experimental error, which would not be possible if they were the result of the charge-up and discharge of the same oscillator circuit. The measured rising edge corresponds to a charge-up process where the charge-up current is a saturating curve due to the change in the number of charge carriers in the axon; see Eq.(6.23).

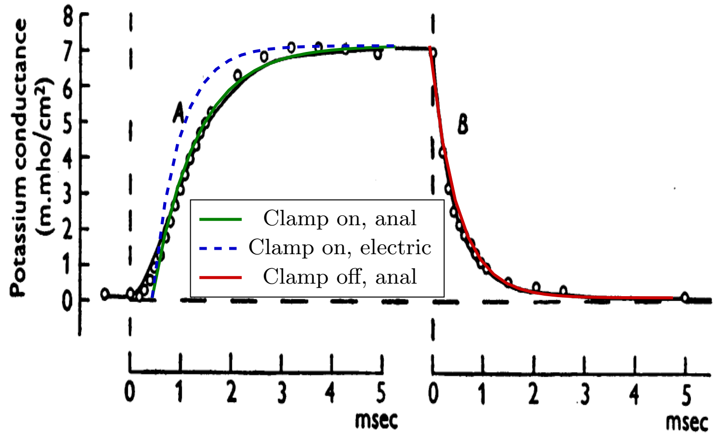


Figure 6.5: The measured current, instead of conductance, (Fig. 2 from [17]), demonstrating that two different physical effects result in the rising and falling edge of the measured current.

The *slow* input currents (although due to slightly differing physical reasons for the different current types) are described by the following analytic form valid for *fast* currents:

$$I_{in} = I_o * (1 - \exp(-\frac{1}{\alpha} * t)) * exp(-\frac{1}{\beta} * t) \tag{6.34}$$

where I_o is the current amplitude per input, and α and β are timing constants for current in- and outflow, respectively. They are (due to geometrical reasons) approximately similar for the synaptic inputs and differ from the constants valid for the rush-in current. To implement such an analog circuit with a conventional electronic circuit, voltage-gated current injectors (with time constants around 1 msec) are needed. For example, see in Fig 6.6 the PSP measured function shape published in [76], fitted with our theoretical function Eq.(6.34). The corresponding voltage derivative is as follows:

$$\frac{dV_{in}}{dt} = \frac{I_o}{C} * \left(\frac{1}{\alpha} * exp(-\frac{1}{\alpha} * t - \frac{1}{\beta} * t) - \frac{1}{\beta} * exp(-\frac{1}{\beta} * t) * exp(1 - exp(-\frac{1}{\alpha} * t))\right) \tag{6.35}$$

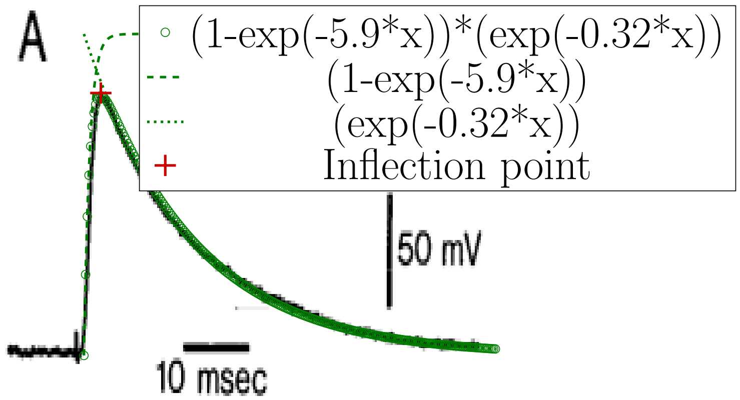


Figure 6.6: The PSP shape measured by [76] and fitted by function forms given by Eq.(6.34).

6.2 Forming membrane's potential

In this section, we model the neuron as an electrical circuit that comprises an electrolyte condenser implemented by its membranes, surrounded by two electrolyte segments. The membrane comprises low-transmission-capacity ion channels in its wall and high-transmission-capacity channels toward its axon; it also uses slow ions rather than electrons. The condenser produces a potential (that depends on the membrane's thickness and the concentrations in the two segments, as discussed above). Furthermore, the two electrolyte segments produce Nernst voltages that are combined to yield a single voltage. According to Kirchhoff's Voltage Law, in a balanced state, the sum of those voltages between the intracellular and extracellular segments must be zero. However, one must take into account the effects of the "slow current" (emulated by "fast current" with a plausible time course in the simulator). We omit biological regenerative processes (such as ATP production via hydrolysis, ion pumps and channels functioning, and so on) and discuss only the most important physical primary processes. In light of our findings, we must reconsider several key concepts in neurophysiology.

The electrical processes are inseparable from the corresponding thermodynamic processes [63], which we discuss by deriving an "equivalent thermodynamic electrical field" in section 6.1.4. The other minor effect is a low-intensity "passive flux" produced by the "resting channels". They are different from the ion channels in the AIS [77, 47, 78, 79, 80]. The "resting current" is about two orders of magnitude lower [33, 81] than HH assumed at the time of writing their papers; correspondingly, its role is much smaller in the transient state, and it even drastically changes the electrical model used in producing an AP which we discuss elsewhere [82].

6.2.1 States

Figure 6.1 shows, in the function of the membrane's thickness, the electrical field across the membrane due to electrical charges (it does not depend on the thickness of the membrane), for different physically plausible assumptions; furthermore, the thermal electrical field. The two electrical fields show significantly different dependencies on the neuron's parameters. Recall that although the parameters we used are biologically plausible, they are only estimations from non-dedicated, incomplete measurements. The qualitative conclusions, however, remain valid.

The balanced state is set where the thermodynamic and electrical "electrical field" diagram lines cross each other; around 5 nm; the value we assumed in evaluating our equations. On the other hand, we can check the electrical field's dependence on the summed-up concentration of the ions in the segment (see Fig. 6.7) (dehydration, for example, changes the setpoint through changing concentrations). Here, the thermodynamic electrical field is constant but different at different width parameters. The field of the classical condenser does not provide a reasonably balanced state (a matching line). The dielectric diagram line seems to provide a realistic estimation when using biologically reasonable parameters. One can describe the neuron as a balanced system at the intersection of the sloping and the horizontal lines. In the resting state, only minor deviations (fluctuations) from this state exist in both the electrical field and the chemical concentration.

In the transient state, if we assume that a charge transfer causes a significant potential change around 100 mV in the layer of width 10 nm, it means a $10^7\left[\frac{V}{m}\right]$ jump in the electrical field that is caused by a $100\left[mM\right]$ jump in the concentration (for only a short time and only in the mentioned layer, but not in the bulk; see also Fig. 6.9). One can infer that a neuron can minimize its offset voltage (and so: its energy) by either decreasing its membrane's width, the sum concentration of the electrolyte, the electrical field, or by a process with a temporal course that changes their combination, thereby implementing a true multi-physics phenomenon. During issuing an AP, the system finds its way back to the crossing point (the path in the phase space is not simple: concentrations on both sides of the membrane and the electrical field in the mentioned layers change, all having different speeds; electrical, thermodynamic, mechanical and so on, changes happen; the process is a relaxation (a waving), as it is seen in the shape of an AP).

We can imagine how the electrical process occurs in this controlled system (for visibility, the positions of the marks are not proportional to the mentioned numbers). The system is in a balanced state at a crossing point. The two circles connected by bent arrows represent that when an excitation begins at the point with lower concentration and higher potential (the cytoplasm side and

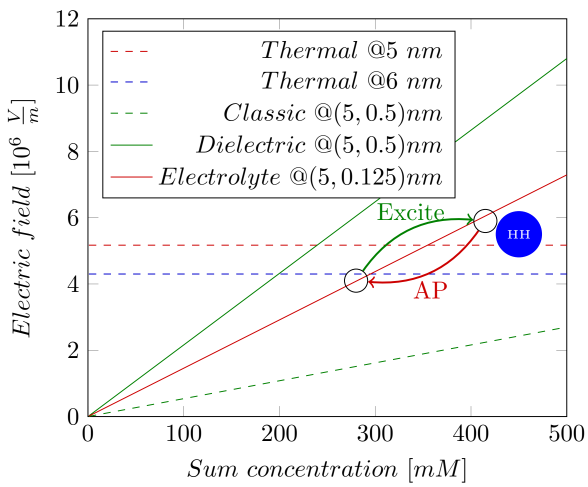


Figure 6.7: The electrical and the thermal "electrical fields" at different concentrations, see Eq.(6.18). The thermal electrical field varies with the thickness parameter and depends on the concentration; see Eq.(6.4). Balanced states are where the sloping lines cross the horizontal ones. The dot marks where HH provided measured data.

negative resting potential), ions rush into the surface layer, so the ion concentration increases, which increases the electric field; this is seen that the membrane potential suddenly increases. (We note here that, due to ions' repulsion, the pressure also increases proportionally with the potential.) The increased potential starts a current, but the current is slow, and the intensity of the current through the channels in the membrane's wall is much lower than the current of the drain (AIS) so the system issues an AP as it returns to its (balanced) starting point. Given that it receives a large amount of Na^+ ions, its primary effort is directed to get rid of them (the electrical gradient cannot distinguish the electrical charges by their chemical nature), so the output current through the AIS comprises mainly Na^+ ions. The system will move along a line, with the restriction that it is a movement in phase space (the change in concentration is not proportional to time), and the different points on the membrane's surface may have different potentials at different times (spatiotemporal behavior). Due to the capacitive current, the local potential may temporarily drop below the setpoint (as observed in physiology, this phenomenon is known as hyperpolarization). The time course of the processes is described by the laws of motion of biology [63] (the time derivatives of the Nernst-Planck equation). The simplified model operates as described in section 4 of [83] and its computational details are provided in [84].

Based on the correct physical model, we can get closer to understanding and reducing the effects of neurological diseases [85, 86], which is not possible while having a wrong model in mind, with batteries and autonomously changing resistors inside the neuron. The change in parameter values potentially leads to neurological diseases as suspected: "Structural alterations of the AIS are likely to have an impact on normal neuronal activity" [85]. When lipid debris accumulates on the surface of the neuron membrane [85], the thickening of the layer Δz reduces the electric field strength (i.e., changes the neuron's 'setpoint'). In that case, the resting potential (and the threshold potential) may change, which weakens AP generation. The elongated shape of the neuron means that the resistance of the AIS increases that changes the parameters of the AP; see Eq.(6.48). "In brain tissue from Alzheimer's disease patients ... the most common changes were a proximal shift or a lengthening of the AIS" [85]. Also, as [87] observed, the setpoint changes with temperature (the "thermodynamic electrical field" changes with the temperature, while the "electrical electrical field" does not, so their crossing point changes), reflected in the dynamics of the AP.

Resting state

In the resting state, the goal is to provide stability with slow, less intense currents. As [2] discusses, there are 'holes' (non-controlled ion channels) in the membrane where the counterforce is missing, so the difference in the electrical and thermodynamic forces could accelerate ions until the Stokes-Einstein speed (see Eq.(6.30)) is reached. Given that the deviation from the setpoint is slight, the current's intensity and speed are low. The slow ion transport delivers both charge and mass, so the gradients permanently direct the process control variable toward the reference point. Given that the 'resting channels' are scattered on the surface of the membrane and it takes time for the gradients delivered by slow ions to reach the entrance of an ion channel, some slight variations in the actual value of the membrane's potential necessarily exist. Given that the AIS comprises non-gated ion channels with much higher density than those in the membrane, the variations and the control ("leaking") are marginal, and the current flows mainly through the ion channels in the membrane's walls, as observed in experimental physiology.

In Fig. 6.8, the electrical voltage contribution is unidirectional. In contrast, the two-component thermodynamic contributions, which depend on the concentrations, are bidirectional. If choosing the zero potential at the middle of the electrical potential, we can interpret that the concentration combinations maintain their ionic concentrations (and so also thermodynamic voltage contributions) independently on both sides, starting from a zero potential level. They generate potentials autonomously in the segments proximal to the opposite membrane surfaces. The electrical charge on the membrane generates an additional voltage independently. However, that voltage must fit between the

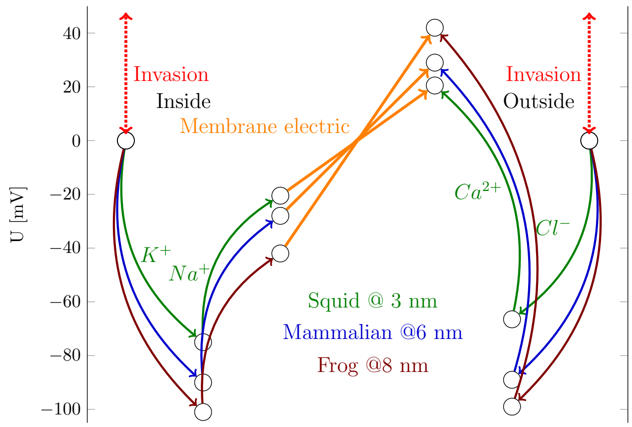


Figure 6.8: The mechanism for controlling neuronal operation is the same in all cases shown in Table 6.1. It appears that the thicker membrane produces a higher membrane electrical potential, requiring a higher thermal potential to compensate. However, it can solve the task using a lower salt concentration, since the electrical membrane potential accelerates ion flow through the membrane channels, thereby speeding up the operation. The electrical fields for the different species have similar values, as shown in Fig. 6.1. Furthermore, the total concentration decreases as the membrane thickness and potential increase.

two thermodynamic voltages. Any deviation in potential between the intracellular and extracellular sides initiates a current that drives the concentrations and voltages toward a balanced state. The process is complex: the sum of surface concentration of ions must be the same on both sides for the condenser, and the ratios on the intracellular and extracellular sides must be adjusted to satisfy the conditions

$$U^{K^+,Na^+}_{internal} = -U_{electric} \qquad +U^{K^+}_{internal} + U^{Na^+}_{internal} \qquad (+U^{invasion}_{internal}) \tag{6.36}$$
$$U^{Cl^-,Ca^{2+}}_{external} = U_{electric} \qquad -U^{Cl^-}_{external} - U^{Ca^{2+}}_{external} \qquad (+U^{invasion}_{external}) \tag{6.37}$$

In a balanced state, the left sides are zero: at the contact point between the electrolyte and the membrane, there are no potential differences, while in a perturbed state, they are non-zero and so they represent driving forces for restoring the balance. (Recall that the ions are slow and that the local electrical and concentration gradients control the local potentials.) From the derivation, it follows that

$$U_{electric} = -2 * (U^{K^+}_{internal} \quad + U^{Na^+}_{internal}) \tag{6.38}$$

$$U_{electric} = -2 * (U^{Cl^-}_{external} \quad + U^{Ca^{2+}}_{external}) \tag{6.39}$$

$$U_{electric} = U^{K^+}_{internal} \qquad + U^{Na^+}_{internal} = -(U^{Cl^-}_{external} \quad + U^{Ca^{2+}}_{external}) \tag{6.40}$$

Our statement is the precise mathematical formulation of that "Thus ions are subject to two forces driving them across the membrane: (1) a *chemical driving force*, a function of the concentration gradient across the membrane, and (2) an *electrical driving force*, a function of the electrical potential difference across the membrane." [2], page 129. Fig. 6.8 quantitatively underpins that "action potential generation in nearly all types of neurons and muscle cells is accomplished through mechanisms similar to those first detailed in the squid giant axon in early research by HH" [88]. *In all species, the chemical and thermodynamic processes keep a balance in the cells of different construction in both resting and transient states.*

According to the above derivation, we consider three characteristic segments in the neuron. In the middle, we have an electrostatically charged isolator plate with positive and negative charge carriers on its two surfaces and none inside, so it has a homogeneous electrical field. The magnitude of the field depends only on the total concentration of the corresponding ions in the segments. On both sides are ionic solutions of different compositions. If the membrane is permeable for the ions, they can move from one segment to the other as the resultant field dictates, until the fields get balanced.

The membrane's electrical potential defines the resting potential, and the "electrical" and "thermodynamic" potentials work together to control neuronal operation. The anatomically defined "electrical" membrane potential, combined with the vast number of ions in the extracellular space, provides a stable concentration that serves as a solid reference point (a setpoint). In the resting state, the membrane potential is balanced by the presence of always-open "resting ion channels". When ions flow into the membrane, the "resting ion channels" can keep the balance: the internal K^+ and Na^+ concentrations are adjusted as the Na^+ flows in.

Figure 6.8 also shows that in its native state, the neuron is electrically balanced: in the internal and external segments, the resulting thermodynamic and electrical forces are equal. Any foreign invasion into the neuron's life (chemical, mechanical, or electrical, including clamping) changes the left-side reference value to an externally set value. The neuron must adjust its Na^+ and K^+ concentrations to reach a new balance while the invasion persists. That is, any invasion ("testing in the physical laboratory") changes the concentrations and voltages inside the cell. In addition, the internal biological processes can trigger internal processes that contribute (from the measurement's point of view) foreign currents or processes, see also section 6.2.1. The feedback from the clamping methods introduces a compensating current. Even a simple conductance-measuring device is active: it applies a test voltage that generates a test current, which can sensitively affect the ionic composition of the neuron (recall that concentrations may differ by up to six orders of magnitude). Although modern electronic devices attempt to conceal this effect, they subtly influence biological operation.

From our result, it follows that for more chemical elements, a per-element coupled set of equations exists, plus the sum concentration agreed electrical fields that are valid simultaneously and define the process variable (aka membrane potential) of the neural regulatory circuit. In a more symmetrical form

$$0 = \frac{U^{Total}_{electric}(c, \Delta z, d)}{2} + \sum_k U^{C_k}_{thermal}(d) + invasion \quad for\ all\ k;\ inside\ and\ outside \tag{6.41}$$

$$c = \sum_k C^{ext}_k = \sum_k C^{int}_k \tag{6.42}$$

In the classical measurement by HH, the sum of the concentrations of K^+ and Na^+ in the cytoplasm and extracellular fluid (in mM) is 380 and 390, respectively, while the concentration of the Cl^- are 450 and 460; practically all equal, as our analysis has derived. The deviations are well within the uncertainty of the measured values.

That is, we have an unidirectional electrical potential difference and oppositely directed thermal voltages. In the balanced state, the two potentials must be equal. When changing, say, one concentration, the other concentration on the same side, and the surface charge density of the condenser, changes accordingly. The changed electrical voltage re-adjusts the thermodynamic voltage and, consequently, the concentrations in the other segment. There is no simple way to express, say, $[Na^+]_i$ in the function of $[K^+]_i$ or vice versa, and similarly for the other ions. Everything moves. The size matters: the amount of ions in the external segment is many orders of magnitude higher, and the gradients are inversely proportional to that amount.

As the figure suggests, an external invasion, typically an electrical voltage on the intracellular side, changes the balanced state, and due to the parameters being linked, changes all concentrations. When moving the system out of its balanced state, in any way, a driving force appears that moves the system towards finding a new balanced state. However, recall that the ions are slow, so *the changes are not instant.*

Table 6.1 shows our results for three different famous published cases (Table 2.1 in[6]). It looks like the calibration of the concentration of negative and positive ions needs scaling, and the measurement of Ca^{2+} comprises issues (the Ca^{2+} concentration in all cases is inaccurate and had to be corrected).

When calculating the values in Table 6.1, we used $\Delta z = 1.05$ and employed the membrane thickness data from various publications to demonstrate that our theoretical approach is correct and that using the correct data can yield even the absolute values of the membrane potential. The higher-than-expected value of Δz may suggest that the ions and the membrane's double layers in the

Table 6.1: The summary data table for equilibrium (Data from Table 2.1 in[6])

Ion	Inside [mM]	Outside [mM]	$U_{therm} \frac{RT}{zF} log(\frac{[C]_o}{[C]_i})$ [mV]	U^{theor}_{electr} [mV]	U^{exper}_{electr} [mV]	U_{membr} [mV]	Width [nm]	E_{membr} [MV/m]
	Squid (HH) T=293 K^o						**3**	14.0
K^+	400	20	$-75.5 = 58log(\frac{20}{400})$	**41.9**	**41.4**	-62.0		
Na^+	50	440	$+54.8 = 58log(\frac{440}{50})$					
Cl^-	40	560	$-66.5 = 58log(\frac{560}{40})$	**41.9**	**42.0**	63.0		
Ca^{2+}	0.27*	10	$+45.5 = 29log(\frac{10}{0.27})$					
	* used for the published value 0.4							
	Frog muscle (Conway) T=293 K^o						**8**	9.61
K^+	124	2.25	$-101 = 58log(\frac{2.25}{124})$	**88.9**	**83.6**	-125.4		
Na^+	10.4	109	$+59.2 = 58log(\frac{109}{10.4})$					
Cl^-	1.5	77.5	$-99.4 = -58log(\frac{77.5}{1.5})$	**88.9**	**82.7**	124.1		
Ca^{2+}	0.021*	2.1	$+58.0 = 29log(\frac{2.5}{0.25})$					
	* used for the published value 4.9							
	Typical mammalian cell T=310 K^o						**6**	11.1
K^+	140	5	$-89.7 = 62log(\frac{5}{140})$	**57.7**	**57.3**	-85.8		
Na^+	15	145	$+61.1 = 62log(\frac{145}{15})$					
Cl^-	4	110	$-89.2 = -62log(\frac{110}{4})$	**57.7**	**56.9**	85.9		
Ca^{2+}	0.04*	5	$+60.8 = 31log(\frac{5}{0.04})$					
	* used for the published value 0.0001							
	Na^+ ions only [62] T=310 K^o						6	10.0
Na^+	12	145	$+66.6 = 62log(\frac{145}{12})$	**60.7**	**66.6**			

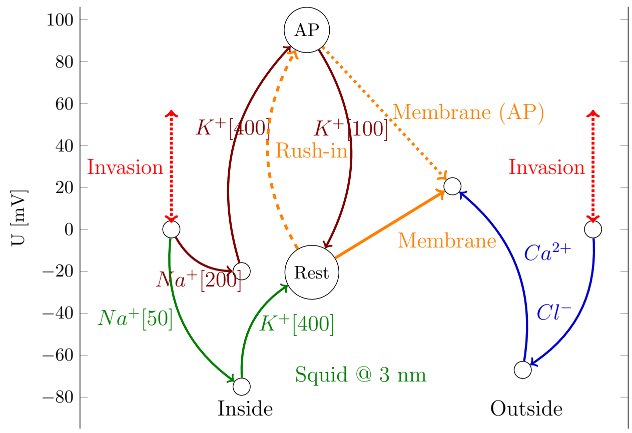


Figure 6.9: The mechanism of producing AP (using numbers referring to the case of squid shown in Table 6.1). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to $+90\ mV$. The neuron must decrease its concentrations to restore the resting potential, mainly by releasing an action potential through the AIS.

electrolyte may also play a role. Dedicated, complete investigations can reveal the details.

Transient state

In a special type of invasion, when suddenly a large amount of Na^+ rushes in (i.e., the trigger is external, but the ions derive from a biological process), the current throughput of the "resting ion channels" is not sufficient anymore: the internal Na^+ concentration in the proximal layer drastically increases, and so does the membrane's potential. At that point, the neuron is out of balance: although most of the Na^+ ions can flow out through the AIS, the system also changes its K^+ concentration through the limited capacity "resting ion channels" until the balanced state is regained. This latter effect must not be confused with the capacitive current, which (owing to its reversed direction) causes the phenomenon known as "hyperpolarization".

Compare the figure to Fig. 7.2 in [2]. Notice the two-sided negative feedback effect: a change in the value of the "command potential" or "command concentration" (the "setpoint" in the language of control theory) changes the other value. In the case of voltage invasion, the ion concentration ratio also changes. In this case, the feedback amplifier is represented by the delicate balance of the electrical and thermodynamic force fields. In the case of clamping, another (electrical) feedback amplifier is introduced into the system, and they work against each other. The neuron attempts to restore its balance without external potential (it cannot distinguish the foreign potential from its membrane potential), and changes its concentrations accordingly. A vital difference is the *speed* of the feedback amplifier. The electrical one operates at electrical speed (not considering the electrolyte electrodes), while the thermodynamic one operates at a speed several orders of magnitude lower. When switching (or changing) the external ("command") voltage, the neuron must first get into its "steady state". It takes time. "The command potential, which is selected by the experimenter and can be of any desired amplitude and waveform" [2]. However, if the measured data readings are not from a steady state, the measurement is wrong: the different amplifier speeds (different current speeds) surely distort the measured value.

Figure 6.9 shows the schematic course of potentials and concentrations during issuing an AP. Follow the green path: the concentration ratio of Na^+ and K^+ ions define a thermodynamic contribution in the intracellular segment. Follow the blue path: the Cl^- and Ca^{2+} ions define a thermodynamic contribution in the extracellular segment. On the left side, two positive ions are in the solution, and the membrane potential (due to charge separation) rises toward the outside segment; the ions cannot go "uphill". Similarly, the Cl^- ions cannot go uphill (due to their negative charge, the same potential is also rising for them). This single membrane potential keeps all ions in their segments. However, the Ca^{2+} ions can travel "downhill". Calcium pumps are needed: although in minimal concentration, Ca^{2+} ions flow into the intracellular segment, and they must be pumped out (BTW: this is likely also the reason why measuring Ca^{2+} concentration is inaccurate).

As discussed above, the concentrations and voltages in the extracellular segment, with their vast amount of ions, remain unchanged in resting and transient states. Follow the red path: Na^+ ions rush into the intracellular segment. They increase the concentration from [50 mM] to (say) [200 mM] (from point (Rest) to point (AP)). This sudden change increases the thermal contribution of Na^+ potential to [-20 mV], but because the total charge on the membrane significantly increases, the resulting voltage on the intracellular side of the membrane increases to [90 mV]. The neuron could decrease its potential to the resting value if it could decrease the K^+ concentration to [100 mM]. However, the ion-delivery capacity of the "resting ion channels" is insufficient: they are sized to maintain the resting state. (Furthermore, the electrical contribution appears instantly, while the thermodynamic one, due to the finite speed of ions, only with some delay [63].) Follow the dashed orange path: the large amount of rush-in Na^+ ions drastically increases the surface concentration on the membrane, so the potential suddenly changes to [90 mV]. Follow the dotted orange path: the electrical field across the membrane suddenly changes its direction from positive to negative (see the slopes of the solid and dotted orange lines) so that the excess positive ions can move "downhill" out of the intracellular space through the high-conductance ion channel array (AIS) at the end of the neuron.

Its internal end is at the high membrane potential, while the external end is at the low potential of the extracellular segment.

The setpoint on the extracellular side remains fixed. On the intracellular side, at the beginning of the AP, the actual voltage is above that of the setpoint on the extracellular side, so at the two ends of the AIS, a positive driving force moves the ions toward the downstream neuron. Given that positive Na^+ ions leave the intracellular segment, its potential continuously decreases. At some point, *the voltage drops below the setpoint of the extracellular side, and the electrical field across the membrane reverses.* The driving force also decreases, and a low-intensity current restores the balanced state to the previous setpoint. When measuring at the AIS, one observes that the potential rises suddenly, then decreases below the setpoint (hyperpolarizes the membrane), then a decreasing current (flowing in the opposite direction due to the changed slope of the resulting potential, the changed electrical field across the membrane, but comprising the originally rushed-in Na^+ ions) slowly restores the resting potential. Here comes the finite speed of ions again. The ions can follow the potential changes with a delay due to their finite speed, causing an apparent delay in the time course of the AP's current.

Models in neuroscience (as reviewed in [89]) almost entirely ignore these aspects. In our physical model, we see that the measurable membrane potential and current change in the function of the ions' speed, the concentration, and its time derivative. Furthermore, all mentioned quantities depend on the effective potential. As Eq. (6.3) shows, the "thermodynamic electrical field" increases as the temperature increases, and raises the setpoint. As Eq.(6.30) shows that the current decreases as the temperature increases, in line with the experimental evidence [72, 90]. The amplitude of the action potential is decreased, given that the setpoint has increased, and its duration is reduced, given that the current decreases.

6.2.2 Controlling neuronal stages

From control theory, it is known that the goal of a controlled system is to govern the application of system inputs to drive the system to a desired state while minimizing delay and overshoot, steady-state error, and ensuring control stability. The neuron implements a controller that monitors the controlled process variable (membrane voltage) and compares it with the reference or setpoint (resting potential). Recall that the geometry and the composition and concentration of electrolytes define the setpoint. The difference between the process variable's actual and desired values, known as the error signal, represents the actual offset potential. It is applied as feedback to generate a control action that brings the controlled process variable to its setpoint.

Neuron as a PID controller

Biology uses a simple Proportional-Integral-Derivative (PID) controller with proportional, integral, and derivative components. In simple words, it means that the output of the system (the output voltage measurable on the AIS) depends on the input voltage (the present: input current if the resistance is constant), the integrated input current(s) (the past), and their derivatives (the future). Notice that calculating the derivatives of the interdependent parameters needs care, as discussed in [91]. The sum of those summands defines the resulting output voltage. The biological case is more complicated than the technical one because biology works with slow ions, and the components have their temporal dependence (applying the laws created for fast currents requires emulation, as discussed in section 6.2). Furthermore, the summands have several constituents, and the neurons handle those current-related constituents autonomously. That is, unlike what is assumed in classical physiology, the output voltage is not necessarily a simple function of the mentioned input variables. Furthermore, it is not sufficient to consider the "present" of the system. The classical models all omit the derivatives' contributions (a direct consequence of using 'clamping': in that "frozen" state, the "future" cannot be interpreted) except that of the capacitive current, which is the passive consequence of the input currents. Furthermore, the 'Integrate and Fire'-type models consider the integrated currents only in the 'charge-up' ('Computing') period of operation, and they entirely omit that different physical processes are going on in different stages of operation. That is, *even the best classical models can not describe neuronal operation completely and correctly.*

The gradients are used to adjust the process variable through their positive and negative contributions (corresponding to rising and falling edges), and the different speeds of the thermodynamic and electrical interactions minimize the delay (i.e., provide the maximum operational speed, vital for survival). The steady-state error is minimized by setting the process variable to the reference point using long-term stable parameters (geometry and overall concentration). The low-intensity current through the always-open resting ion channels provides dynamic stability in the steady state.

We considered that the neuron has a stable base state. On the one side, this resting state must be dynamically stabilized for little perturbations using as little energy as possible (and to provide a mechanism when the cell grows, divides, or ages). On the other side, it must be able to restore the state after rough perturbations as quickly as possible, causing short-time transients (when restoring the membrane's potential after issuing a spike). In both the resting and transient states, the system attempts to return to its balanced state, though the mechanisms required differ.

In its excited state, the system aims to provide an intense output that informs the downstream neurons. This overshoot is initiated by a large number of voltage-gated ion channels (distinct from the non-gated ion channels used to maintain the resting potential) distributed in the neuron's membrane. The overshoot current flows through those persistently open ion channels, which are concentrated in AIS (which has about two orders of magnitude higher channel density than the wall). The intense slow current produces a condenser-like behavior (capacitive current); the phenomena called "polarization" and "hyperpolarization" (not polarization; instead, a movement of completely separated

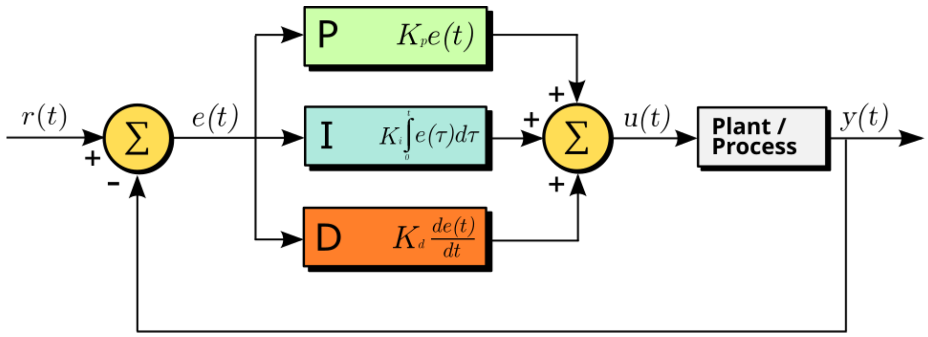


Figure 6.10: A block diagram of a PID controller in a feedback loop. r(t) is the desired process variable (PV) or setpoint (SP), and y(t) is the measured PV. (Wikipedia)

charge in the surface layer of the electrolyte) of the membrane provide the necessary positive and negative error signals for the controller in the transient state by moving the actual potential value of the membrane above and below the resting potential value. The potentials and the electrical field's magnitudes depend on the concentration and the geometry (the finite size of the membrane). The ions' chemical nature comes into play only if the polarizability can differ for different molecules.

The time course of the overall control function $u(t)$ of the general PID controller in function of the error variable $e(t)$ in Fig. 6.10 is described by

$$u(t) = K_p \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right) \tag{6.43}$$

where $T_i = \frac{K_p}{K_i}$ is the time integration constant, $T_d = \frac{K_d}{K_p}$ is the derivative time constant (notice the independent time constants). That is, in the world of a neuron, the "theory of everything" is

$$V_M^{OUT}(t) = K_p \left(\underbrace{\overbrace{V_M^{IN}(t)}^{proportional}}_{clamping,\ synaptic} + \underbrace{\overbrace{\frac{1}{T_i} \int_0^t V_M^{IN}(\tau) d\tau}^{integral\ (parallel\ RC)}}_{membrane's\ wall} + \underbrace{\overbrace{T_d \frac{dV_M^{IN}(t)}{dt}}^{derivative\ (serial\ RC)}}_{AIS,\ synaptic} \right) \tag{6.44}$$

The first term represents the constant effect of the external world (including synaptic inputs, clamping, and excitatory effects on brain tissue). The second term represents the time-averaged contribution, such as charging up the membrane, including clamping current. The voltage is mainly due to current through ion channels in the membrane's wall; it represents a parallel RC circuit. The third term represents the sum of the gradients due to all effects (the external world, the internal processes, and the outflow through the AIS) and constitutes a serial RC circuit. The underbraces of the equation identify which components of the neuron contribute the terms of the PID equation. The presence of some terms depends on the neuron's actual stage; furthermore, the R values in the parallel and serial RC circuit differ (resulting in different time constants).

One can reformulate the HH's famous Eq. (1) in [17] to the form of the PID equation

$$\underbrace{I \times R_{AIS}}_{V_M^{OUT}} = \underbrace{\overbrace{I_i \times R_{AIS}}^{proportional}}_{ionic} + \overbrace{\underbrace{\frac{1}{C_M \times R_{AIS}}}_{T_i} \int_0^t \cancel{(I_{bio} - I_{clamp})(\tau)}\, d\tau}^{integral,\ clamping\ cancels\ this} + \overbrace{\underbrace{C_M \times R_M}_{T_d} \frac{dV_M}{dt}}^{derivative,\ capacitive} \tag{6.45}$$

The proportional term comprises the ionic (and external) currents. At the time when Eq. (6.45) was set up, AIS was not yet known, so HH assumed that in the proportional term, the resistance is identical to the membrane's resistance implemented by the ion channels in the wall (leading to the ideas of "leakage current" and "resting potential", misleading physiological research.) The experimental conditions canceled the integral term entirely: the negative feedback from clamping precisely counterbalances the neuron's internally generated current, so the current integrates to zero. (BTW: the "integrate and fire" type models artificially revive the forgotten integral term.) The derivative term assumes that all currents are constants, so the voltage changes only due to the capacitive current. It entirely forgets that gradients are needed to operate an electrical system (leading to the ideas of constant-voltage batteries and magically operated resistors in the equivalent circuits). Although it was known from the beginning that the APs comprise relatively steep rising and falling edges, it was forgotten that a non-constant current contributes a gradient to the derivative term. HH's formalism describes only the 'present' (as can be expected when freezing the state), has no predictive power, nor can it give an account of neuronal memories.

HH considered only the proportional term, plus in the derivative term, the voltage gradient due to the condenser (but not the capacitive current itself, directly leading to the wrong ad-hoc hypothesis of the presence of K^+ for explaining hyperpolarization). Given that clamping obscures the presence of gradient-like changes in the system, they missed that *the input currents also produce a voltage gradient and that the output current through the AIS also produces a gradient.* Furthermore, the equation shows that external currents (e.g., the rising and falling edges of step functions) also cause significant changes in the output voltage. Similarly, a "foreign" invasion (changing mechanically the positions of charges by pressure, ultrasound, magnetic pulse, changing the concentration in one of the segments, and so on) generates a change in the position of charges on the neuronal membrane. This way, it generates a voltage gradient (along with pressure and other gradients), which may trigger neuronal spikes. As discussed in section 6.3.1, subthreshold excitations provide direct experimental evidence that the description of the resting state includes both serial and parallel contributions.

From a control theoretical point of view, the condenser geometry and the ion concentrations set the 'always the same' value (the setpoint; see also Figure 6.1) and the currents serve to keep or restore that value. After introducing a finite-width membrane into an electrolyte, a potential difference between the two membrane surfaces is created; see Eq. (6.19). When adjusting the membrane's potential, we must consider its ground and excited states separately, given that the perturbations in these two cases are vastly different. Nature employs various mechanisms to maintain the ground state and recover it after generating a spike. Although nature's tools are remarkably similar (channels and pumps; here, for discussing 'downhill' channels, we must consider the charge-up of different components), the significant difference in current intensity warrants a separate discussion.

When comparing Eq. (6.46) describing the operation of a simple serial RC oscillator taken from the theory of electricity, Eq. (6.45) describing HH's differential equation (having PID in mind), and the complete Eq. (6.43) describing PID in general, one can see that they describe the same process in different (experimental and mathematical) approximations. As experience shows, Eq. (6.46) describing the 'net electric' model provides a sufficiently accurate description of the AP, proving that the dominating effect comes from the derivative term and that the thermodynamic term is linearly proportional to the electrical one. Furthermore, as we emphasized by discussing the physical processes and the mathematical terms, to some measure, the "parallel RC circuit" is also present in the process, although its current amplitude is by two orders of magnitude lower than that of the "serial RC circuit". Furthermore, the time constants of the two circuits are also largely different, making the effect of the parallel circuit unnoticeable. The slow current also plays a role here: the ion currents are 'local'. Given that the channels in the membrane's wall are always open, the current on its way towards the AIS may flow out through a nearby ion channel, so only a fragment of the 'resting current' reaches the AIS. Eq. (6.43) provides a high-accuracy description of the process, but for most practical applications using Eq. (6.46) provides sufficiently good results, while Eq. (6.45) is based on an incorrect 'physical model' and requires a series of incorrect ad-hoc hypotheses to describe observations. Note that the forces have both electrical and thermodynamic components, but they are proportional to each other. That is, although the absolute values of the coefficients in the equations are not accurate, their effects are. Surely, the description of the processes is complicated, but appropriate approximations give sufficiently precise and computationally reasonable results.

Robustness of the control circle

An exciting thought experiment is discussed in [62]. "First, we consider a hypothetical membrane that is permeable only to Na^+ ions. Suppose that $[Na^+]_o$, the outside or extracellular sodium concentration, is 145 mM, and $[Na^+]_i$ is 12 mM. Moreover, suppose that initially, there is no membrane potential. The diffusion gradient for Na^+ favors Na^+ entry into the cell, and the initial Na^+ influx carries a charge that builds up on the inside of the cell. This produces a potential (recall that separation of charge produces a potential) across the membrane that impedes further Na^+ ion movement because the positive charges repel the positively charged Na^+ ion."

The paper correctly explains that in the case of Na^+ only, the potential is set up to 66.6 mV. (We quietly add that plus a layer of the corresponding Cl^- ions on the other segment, in a similar concentration ratio, builds up; so also, the attraction between positive and negative charges contributes to the effect.) In the absence of our argumentation, it cannot explain *why* that potential builds up; only why that concentration ratio establishes: the Nernst law alone does not explain the effect. Furthermore, the statement is valid only for a well-defined semipermeable membrane. We can add that the same concentrations develop if initially identical concentrations are present on both sides. The finite width of the membrane creates an electrical potential, a consequence of charge separation, as stated. If there are no ions initially inside, then no charge separation occurs, and therefore, no ion transport begins due to the membrane's electrical potential. However, the thermal driving force exists, and the started ion transport quickly builds the electrical layer, which effectively helps to establish the resting potential.

This robustness also explains why, during replication (cell division), cells also inherit their operating ability and resting potential; the existing membrane cell continues to grow in the same way. Its voltage does not depend on its size, so it remains constant, and the voltage defined in this way adjusts the concentrations accordingly. Moreover, it explains why, during evolution, cells first formed: a couple of lipids came together to form an imperfect membrane with holes, establishing cellular electricity. As Fig. 6.8 depicts, the concentrations may adapt to the living conditions (whether living in seawater), but the general operating principles remain the same. The seawater environment enables the use of thinner membranes and lower membrane voltages; however, the same electrical field is used in all cases.

Given that neurons share the external (global) concentrations (defined by the vast amount of ions in the bulk), they provide a firmly fixed offset value to the always-the-same electrical potential. This way, the operations of the individual neurons do not interfere. Furthermore, the internal (local) concentrations can adjust themselves even following a very rough charge perturbation while issuing an AP; moreover, when the living organism is growing.

6.2.3 Goldman-Hodgkin-Katz potential

It has already been stated, based on experimental evidence, that "the membrane permeability to the ions has nothing to do with the potential generation and the ions' adsorption on the membrane surface generates the membrane potential"; for a review, see [57]. The idea itself is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; the resting potential has nothing to do with either ions' mobility or permeability or with ion absorption [60]. Furthermore, there is no idea in GHK about whether the 'setpoint' (why that specific concentration or potential difference) is present and why the same potential is reset after rough perturbations such as issuing an AP or replicating a cell. The causality is reversed: the potential is a static concept and is set electrically, and the two gradients form the experienced concentrations from the available solvent molecules. For the time course of gradient formation, mobility and permeability play a role, but not in determining concentrations or the resting potential.

In our clear physical picture, the thermodynamic forces on one side and on the other side of the membrane are summed. They counterbalance each other; furthermore, jointly the effect of the neuronal condenser, see Fig. 6.8. As we emphasized, the concentration gradients are ion-specific. Furthermore, as discussed in connection with Eq.(6.2), the Nernst equation comprises a per-ion indefinite constant (a potential difference). To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is adding absolute concentrations on the different sides of the membrane, or changing the base of the logarithm used in a calculation to match the experimental value. It is not more than number magic. The potential is described by coupled equations as discussed in connection with Eq. (6.42).

The Ca^{2+} ions do not fit into the GHK picture. One of the reasons why GHK cannot be good is that $[Ca^{2+}]$ is omitted. Biologically, it is hard to believe that Ca^{2+} does not participate in the game of life (especially since biology sees the need for Ca^{2+} pumps). Physically, a single concentration on one side of the membrane cannot maintain balance; as the different ion-exchange processes shown in Table 6.1 and Fig. 6.8 demonstrate. When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}).

6.3 The Action Potential

As discussed, a smaller number of ion channels is distributed over the surface of the membrane, and the overwhelming majority of ion channels is concentrated in the AIS.

6.3.1 The physical process

In the resting state, we have a condenser that has some low-conductance resistors (the resting ion channels) and a very low intensity (nearly constant) distributed current flows, under the effect of a slightly varying (nearly constant) membrane voltage. Due to the membrane potential, a current through the always-open channels flows on the dendritic surface and the resting channels' conductivity represents a drain with sufficient transmission. The resting current practically does not reach the AIS: the ion channels along the current's path practically "shunt" the current. This is a static state that classical neuroscience assumes: some static current in and some static (leakage) current out; no significant gradient. *In this state*, an almost correct model is a condenser with a *parallelly* switched resistor; an integrator-type RC circuit, see Table 6.2 (although when approaching the threshold voltage, the AIS plays some role. The smaller gradient changes, such as subthreshold excitations, produce "mini-AP waves" [81], providing direct experimental proof that in this state the parallel RC circuit is not correct). Some perturbations must be counterbalanced, but their amount does not exceed a predefined threshold.

In the transient (perturbed) state, the process variable (the membrane's potential) exceeds the threshold. That event acts as a fast trigger signal. The amount of charged particles (and so: the membrane's voltage) suddenly increases, the created gradient drives a current. When Na^+ ions rush into the intracellular layer, they roughly increase the overall concentration and the potential in that thin layer. All nearby ions feel a driving force. The targeted membrane potential is set electrically, and the driving gradients may behave in unexpectedly during the transient period. The actual voltage gradient may temporarily reverse the direction of the chemical gradients.

Our results align with the observation (see caption of 11.22 in [45]), that *the significant processes occur in a thin layer of the electrolyte proximal to the membrane surface*. The amount of unbalanced ions is in the range of 10^7, and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush into the low-concentration side and cause a significant change in the membrane potential (and concentration). Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions in that layer. However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or in a few atomic layers on top of each other, depending on the concentration.

Suppose that at the beginning of an AP a large amount of Na^+ ions are transferred from the extracellular to the intracellular side. In that case, the K^+/Na^+ concentrations change from 145/15 to 45/115 and the corresponding thermodynamic voltage contribution changes from $+61$ $[mV]$ to -25 $[mV]$ (i.e., altogether a 86 $[mV]$ sudden increase in membrane potential). For the Na^+ ions, the resultant potential changes from $+4$ $[mV]$ to -83 $[mV]$. That means when the AP begins, the "Na-K pump" stops (if the resulting potential provides the driving force for the exchange pump). The only way for the neuron to remove the excess Na^+ ions is to generate a current toward the AIS (where the other end of the ion channels remained at the extracellular potential), until the Na^+-specific driving force disappears. The K^+-specific driving force, changes from -90 $[mV]$ to -176 $[mV]$, i.e., the K^+ intake gets more intensive, misleading researchers into believing that it causes the observed hyperpolarization. However, as Fig. 3 in [81] demonstrates, it occurs instantly, not with a delay; furthermore, the "leakage current" and the stimulus current are negligible. The low extracellular K^+ concentration plus the low number of channels in the membrane's wall do not enable a significant increase in the intracellular K^+. (Given that complex changes occur, including changes in the electrical potential that also change the concentrations, different waves start; the statement is not strictly valid.)

As the current flows, the thermodynamic force decreases, and at some point the resultant force changes its sign; so does the current direction. This is seen as the "condenser current" (and mis-identified as an intense K^+ current). The ions move slowly, and due to the enormous electrostatic repulsion, their "electrostatic fluid" must be continuous. A special "damped wave" is formed (with all secondary mechanical, optical, etc. effects) and the condenser current reverses. Electricity interprets the phenomenon as "capacitive current" in the equivalent circuit, and thermodynamics interprets it as changed chemical concentration (or the negative amplitude of the damped elastic oscillation). Our discussion adds that electrical repulsion generates a pressure wave, or shock wave, within the neuron's closed volume.

The neuronal AP generation can be entirely described by the 3-D line in Fig. 6.11. The three dashed diagram lines are drawn in three perpendicular planes; see the figure's caption and the discussion below for the details.

6.3.2 The oscillator model

Despite the early warning that '*it was not possible to separate the change into resistance and capacity components*' [75], a commonly accepted truism was that neurons, in some sense, behave as electrical oscillators. HH introduced the idea explicitly that the electrically equivalent circuit of a neuron is an RC oscillator. We must also add the invention from about three decades later: "Neurons ensure the directional propagation of signals throughout the nervous system. The functional asymmetry of neurons is supported by cellular compartmentation: the cell body and dendrites (somatodendritic compartment) receive synaptic inputs, and the axon propagates the action potentials that trigger synaptic release toward target cells. *Between the cell body and the axon sits a unique compartment called the axon initial segment*. The AIS was first described 50 years ago [i.e., nearly two decades after HH published their study], and its molecular composition and organization have been progressively elucidated during the following decades. .... Recent years have also brought crucial insights into the functions of the AIS: how *ion channels at its surface generate and shape the action potential*." [79] We provide the physics and mathematics of how AIS shapes the action potential (or more precisely, we show what an important role it plays in forming AP).

Inventing AIS changed the viewpoint of neuroscience [79], a half century later, after setting up the commonly accepted model by HH. "The AIS is located at the proximal axon and is the site of action potential initiation. This reflects the high density of ion channels found at the AIS. ... The summation of synaptic inputs gives rise to action potentials at the AIS, a 20–60 μm long domain located at the proximal axon/soma interface that has a high density of voltage-gated ion channels." As discussed in [80], see also their Figure 1, the structure of the AIS is known to the most minor details. As the illuminating investigations in 2008 [47] revealed, the AIS has very dense ion channels. That is, from an electrical point of view, those parallelized channels can be abstracted as a *discrete conductance* (or resistance) between the membrane and the axon. The

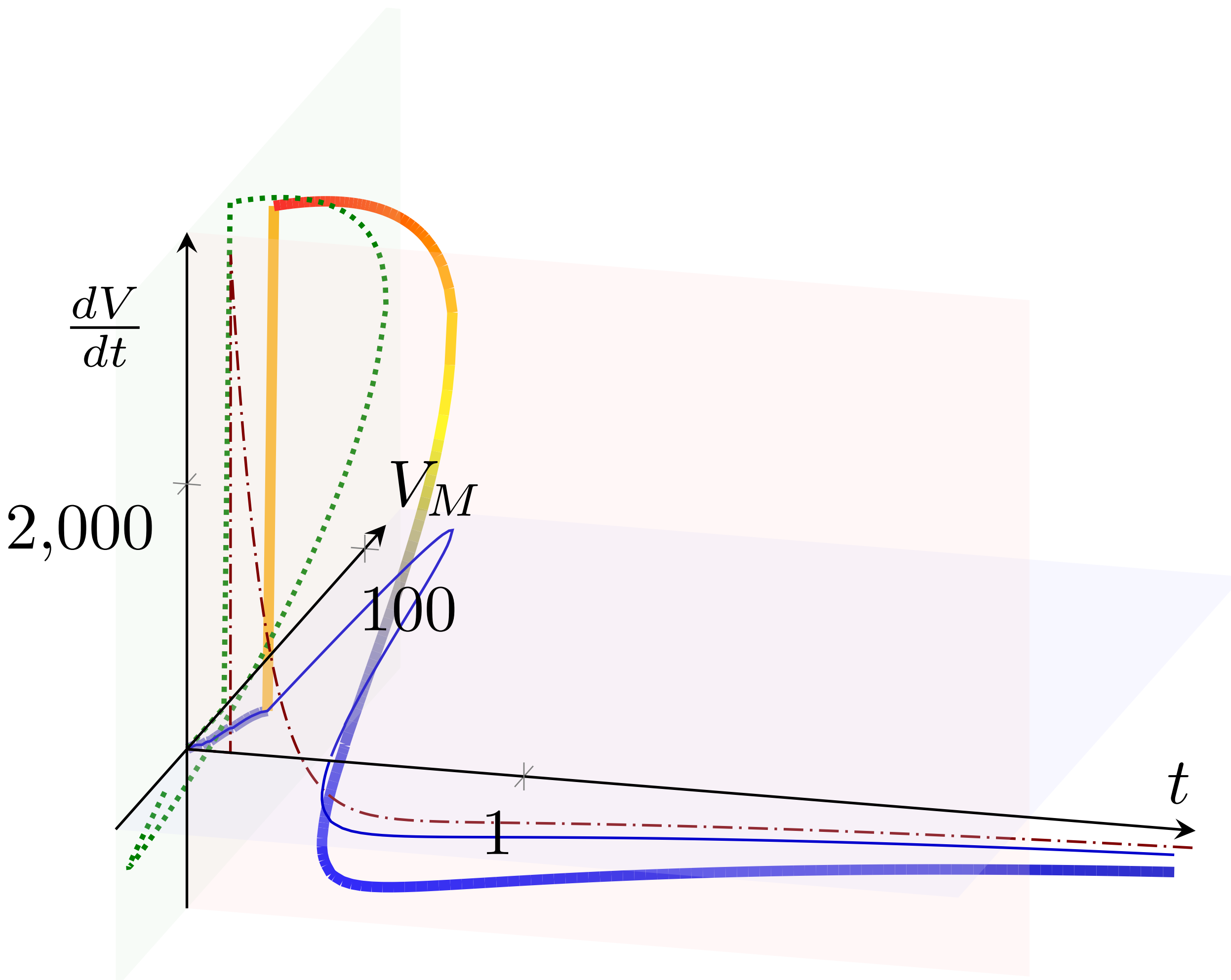
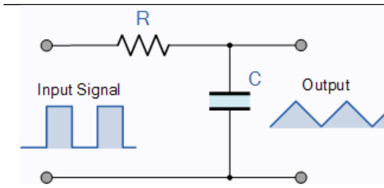
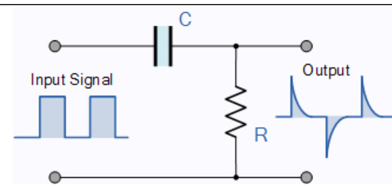


Figure 6.11: A summary of producing neuronal AP as a 3-D figure. The thick, solid 3-D line shows how the color-coded voltage gradient drives the membrane potential over time. The three dashed diagram lines are the projections of the spatial curve to the boundary planes of the cube. The physiology delivers them as 2-D plots; the figure explains how those parameters are interrelated. The red line shows the voltage gradient produced by the rush-in of the Na^+ ions, see also the middle inset in Fig. 6.12. The blue diagram line shows the AP, see also the bottom inset in Fig. 6.12. The interdependence of the gradient and the voltage is called a phase plot; see, for example [92].

Table 6.2: RC circuit types

The RC Integrator	The RC differentiator
$V_{out}^{Integrator} = \frac{1}{RC}\int_0^t V_{in}dt$	$V_{out}^{Differentiator} = RC\frac{dV_{in}}{dt}$
Low Pass Filter	**High Pass Filter**
Resting state	**Transient state**
	
(Mostly) resting state	(Mostly) transient state
Ion channels in the membrane	Ion channels in the AIS

membrane itself can be abstracted as a *distributed condenser* with no resistance. This model is in contrast with the viewpoint of biophysics, that the membrane plus AIS is considered a distributed element, where the capacitor and condenser cannot be separated. The current model is the result of projecting the physical processes in the resting state to the transient state. Notice the important point: "Neurons are also anatomically polarized, as they can be subdivided into a somatodendritic input domain and an axonal output domain" [77]; providing direct evidence that (unlike in HH's model) *the input and output currents (and voltage time derivatives) are independent.*

In our model, the AIS gets part of the membrane, and this leads to crucial changes. (BTW, the name is slightly misleading: the AIS is part of the neuronal oscillator, and it forwards a traveling potential wave to the axon instead of belonging to it.) 'Although by definition a neuron must have an axon to assemble an AIS, the relationship between AIS assembly and axon specification in vivo has not been determined yet' [77]. Anyhow, a neuron can be abstracted (in the disciplinary view of electricity) as comprising a resistor-like and a capacitor-like component. However, in the resting and the transient states, they are combined in different ways.

In the resting state, we can model neurons as a *parallel RC* circuit (see the "integrator" in Table 6.2), where "random" currents flow in and out. Practically, the output voltage is constant (the resting potential): the ion channels and pumps are distributed over the membrane's surface, and due to the low speed of currents, at low local voltage gradients, the AIS is not involved. Only the synaptic inputs cause a measurable gradient on the AIS. In this native state, there is no gradient; in this sense, it is similar to the case of clamped operation, where the gradient is (apparently) suppressed by the feedback. HH did not see any structural elements on the membrane, so logically, they modeled it as a distributed resistor and capacitor, which really has resemblance with a *parallel*

RC oscillator. However, they made a wrong choice of the circuit type, and their choice (likely due to inertia) was repeated in good textbooks such as ([6] Figure 3.1 or [66] Figure 1.1), and it is a commonly accepted fallacy even today [4].

In the transient state, the high-transmission ion channels in the AIS (high conductance R) replace the low-transmission ion channels (low conductance R) in the membrane's wall, and the rush-in current flows out entirely on the AIS (for the model, in this state, we omit the low-transmission ion drain). Those two elements form a *serial RC* circuit (see the "differentiator" in Table 6.2) that acts as a damped oscillator and produces a current pulse, known as AP. Neuroscience mistakenly claims it is a parallel RC circuit; that statement is valid only in the resting state. However, the number and intensity of the 'resting' ion channels in the membrane's wall are insignificant for producing AP, so we use the approximation that only the 'transient ion channels' in the AIS are active when generating an AP.

In the electrical view, the process can be modeled as follows: a sudden voltage gradient (see Figure 6.12 middle inset) appears on the membrane (as a discrete capacitor), and a current flows out (see Figure 6.12 bottom inset) through the serially connected ion channel array (as a discrete resistor). The neuron forms a simple *serially* (*not parallelly*, as assumed by Hodgkin and Huxley [17] and mistakenly claimed by neurophysiology) connected RC oscillator. For the physical and mathematical description, see section 6.3.3, furthermore, section 4 in [83]; for its algorithmic specifics [84], for further details [82]. Notice that the difference between the rush-in and the resultant gradients emphasizes the role of the finite resources: without the AIS, no turn-back of the resultant gradient would occur.

The fundamental difference between the parallel and serial RC circuits is that *the parallel circuit cannot produce output voltage with opposite sign*. As discussed, the capacitive current, which by definition changes its direction and so generates an opposite voltage on the resistor represented by the AIS, perfectly describes the so-called "hyperpolarization". It is not the effect of a K^+ current: the resting ion channels do not have sufficient conductance. Not knowing about the AIS leads to assuming that the resting ion channels work also as transient channels (in other words, assuming a parallel RC circuit instead of the serial one, furthermore, that the output current flows through the membrane instead of the axon). Some K^+ current certainly exists; for the magnitude of currents during AP, see [81].

The shape of the output waveform depends on the ratio of the pulse width to the RC time constant. When RC is much larger than the pulse width, the output waveform resembles the input signal, even with a square wave input. (In the case of the neuronal oscillator, the shape of the front side of the spike is similar to the one derived for the rush-in current, while the back side is very prolonged.)

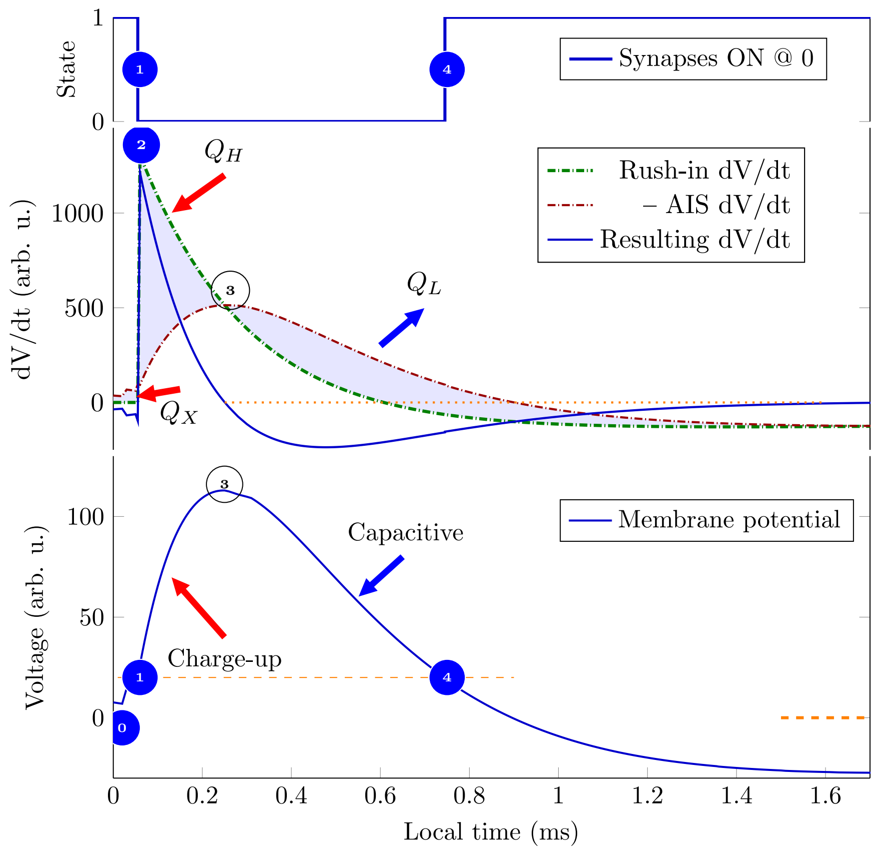


Figure 6.12: The physical processes describing the membrane's operation. The rush-in Na^+ ions instantly increase the membrane's charge, and the membrane's capacity discharges (producing an exponentially decaying voltage derivative). The ions are created at different positions on the membrane, so they take different times to reach the AIS, where the current produces a peak in the voltage derivative. The resulting voltage derivative, the sum of the two derivatives (the AIS current is outward), drives the oscillator. Its integration produces the membrane potential. When the membrane potential crosses the threshold, it switches the synaptic currents on or off.

6.3.3 Mathematics of the ActionPotential

Although the implementation of a neuronal circuit works with ionic currents and has a charge transmission mechanism drastically different from the one based on free electrons used in metals, the operation can align (to some measure) with the instant currents of physical electronics (as we suggest, the effect of the slow current and finite size can be reasonably imitated with current generators generating a special current shape). The input currents in a biological circuit are a large rush-in current through the membrane in an extremely short time, and much smaller gated synaptic currents arriving at different times through the synapses. Those currents generate voltage gradients, and those gradients direct the operation of the RC oscillator (Notice that due to the multi-physical effects, the parameters R and C may change in the function of time and the actual voltage). As it is well known from the theory of electrical circuits, the output voltage measured on the output resistor is as follows:

$$V_{out}^{Differentiator} = RC\frac{dV_{in}}{dt} \tag{6.46}$$

where the input voltage is as follows:

$$\frac{d}{dt}V_{in} = \sum \frac{d}{dt}V_{IN}^{Component} - \frac{d}{dt}V_{OUT}^{AIS} \tag{6.47}$$

that is, the (temporally gated) sum of the input gradient that the currents generate, plus the gradient of the output current through the AIS [80, 79]. The latter term can be described as follows (see the discussion around also Eq.(6.43):

$$\frac{d}{dt}V_{OUT}^{AIS} = \frac{1}{C}\frac{V_{internal} - V_{external}}{R_{AIS}} \tag{6.48}$$

The input currents are slow (we typically imitate them using a current generator with a special form). In the simplest case, the resulting voltage derivative comprises only the rush-in contribution and the AIS contribution described by Equation (6.48). The gating implements a dynamically changing temporal window and dynamically changing synaptic weights. The appearance of the first arriving synaptic gradient starts the “Computing” stage. The rush-in gradient starts the “Delivering” stage [84]. The charge collected in that temporal window is as follows:

$$Q(t) = \int_{t_{0,i}}^{t_{thr}} dt \underbrace{\sum_i}_{t_{0,i} \leq t \leq t_{thr}} I_{syn,i}(t) \tag{6.49}$$

The total charge is integrated in a mathematically entirely unusual way: this is *NOT* a simple "integrate and fire": the collected charge defines the upper bound of the integration time. Through adjusting the integration time window, the neuron selects which one(s) out of its upstream neurons can contribute to the result; the individual contributions are the result of a bargain between the sender and the receiver. The upstream neuron decides when the time window begins, but the neuron decides when a contributing current terminates. *The beginning of the time window is set by the arrival of a spike from one of the upstream neurons, and it is closed by the neuron when the charge integration results in a voltage exceeding the threshold.* The speculations [91] about the spatial length of the spike are correct, but its "payload length" is defined by the receiving neuron. The computation (a weighted summation) is performed on the fly, excluding the obsolete operands (the idea of Analog In-Memory Computing (AIMC) [93] almost precisely reproduces the charge summation except for timing). The only operation that a neuron can perform is that unique integration. Its result is the reciprocal of the weighted sum, and it is represented by the time when a spike is sent out (*when* that charge on the membrane's fixed capacity leads to reaching the threshold). Notice that the time is defined on the neuron's local time scale and is interpreted similarly by the downstream neurons: the data transmission time matters.

Chapter 7

Thermodynamics for neurons

Chapter 8

Transmission

Quotation:

"Transient electrical signals are particularly important for carrying *time-sensitive information* rapidly and over long distances. These transient electrical signals—receptor potentials, synaptic potentials, and action potentials—are all produced by *temporary changes* in the electric current into and out of the cell, changes that drive the electrical potential across the cell membrane away from its resting value." [6]

List of Tables

List of Figures

Index

Glossary

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